

KDA Accumulator on VDA16 — Full Analysis Atlas

kda-vda16-seed0: psychometrics, frozen probes, and the complete map battery

run vlca020uvsqtgh (checkpoint d35781fa...)

2026-08-29/30

1 Provenance and methods

Model. `KDAConvMemoryModel`: conv stem $\rightarrow$ KDA visual accumulator (gated delta rule; per-pixel matrix state $S \leftarrow \text{Diag}(\alpha_t)S + \beta_t k (v - \tilde{S}^\top k)^\top$, 4 heads $\times$ 32 channels, L2-normalized k/q , RMS-normalized sigmoid-gated readout) $\rightarrow$ vision conv-attention (input widened to $[X_t \parallel \text{acc}]$) $\rightarrow$ H1/H2 memory conv-attention $\rightarrow$ per-pixel JEPA head + change classifier. $R = [H1 \parallel H2 \parallel Z \parallel \text{att_vis}]$, 1,200,630 parameters.

Training. VDA16 (4 $\times$ 4 grid, 100 $\times$ 100 px, $T=7$, change at $t = 5$), 500,736 trials, JEPA (1.0) + change CE (1.0), fixed teacher EMA 0.996, shrinking- θ curriculum ($65^\circ \rightarrow -3^\circ$ at > 0.85 per-collection accuracy, floor 8°), H1 memory noise $\sigma = 0.05$, AMP, seed 0. Terminal state: collection 488/489 complete, change accuracy plateau 0.80–0.82, curriculum stalled at $\theta = 29^\circ$.

Analysis. All evaluation episodes are fresh from the training generator ($T=7$, change at $t=5$, sensory noise 5.0, $\theta=65^\circ$ env setting with exact signed $\Delta\theta$ overrides). Probes: 6,000 trials, frozen $R@last$, 80/20 split. Maps: 100 trials per condition, gates and $A \odot V$ energies recomputed exactly; accumulator $\alpha/\beta/\text{surprise}$ recomputed from the accumulator’s own convs. $t = 0$ is the blank reset frame (zero stem output), so all maps are uniformly zero there.

2 Psychometrics

2.1 Validity gradient (16 cue proportions)

2.2 The cueing effect: valid vs invalid at 100% validity

The false-alarm floor ($\theta = 0^\circ$, change trials with zero applied change) is 10

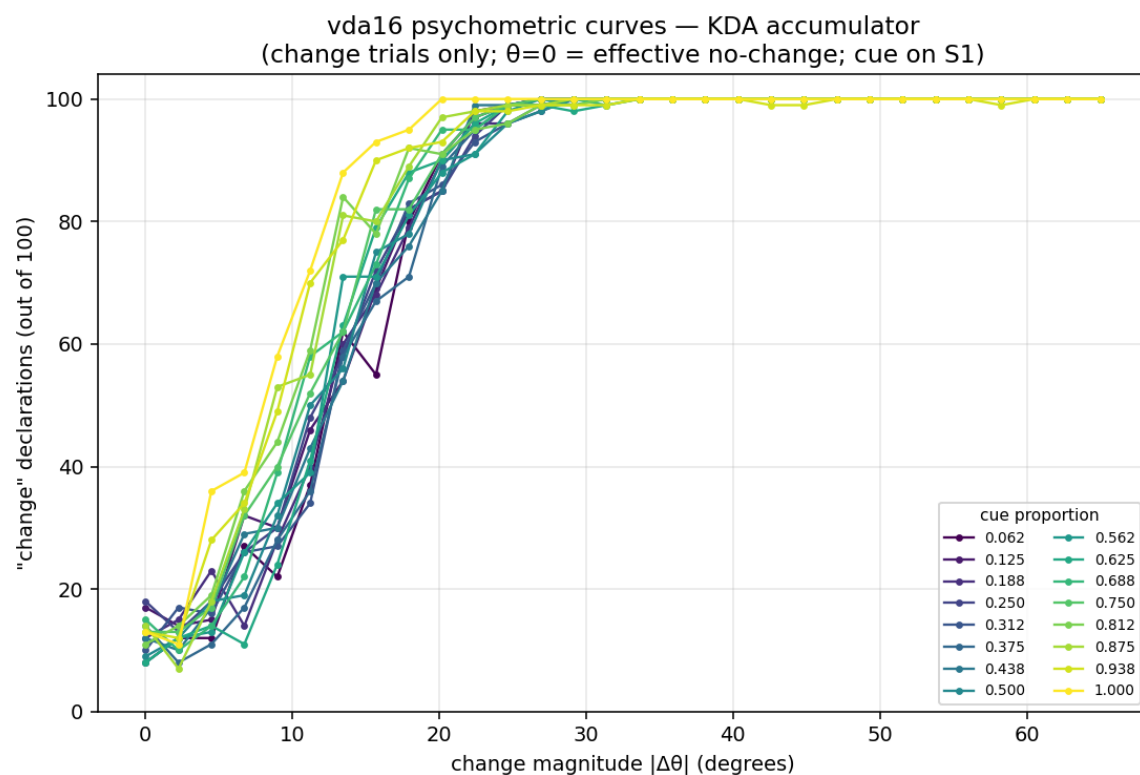


Figure 1: Change declarations vs change magnitude for each cue-proportion level ($k/16$, $k = 1 \dots 16$; 100 trials per point, cue on S1, natural change placement).

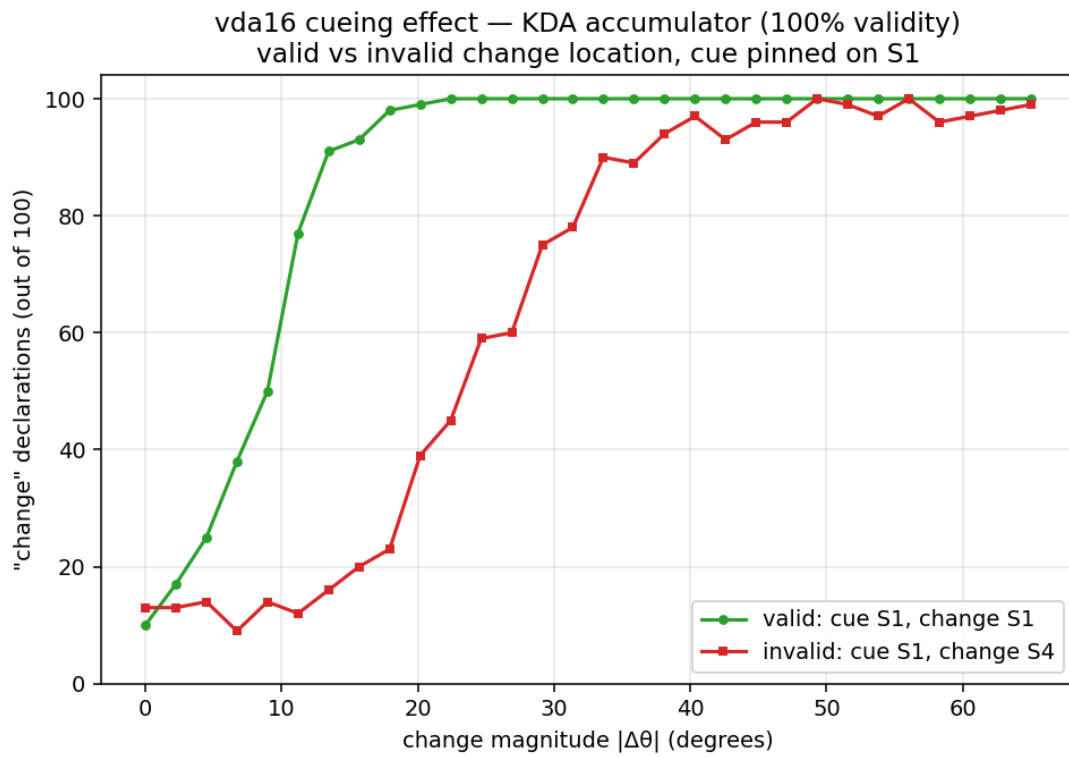


Figure 2: Cue pinned on S1, proportion 1.0; change pinned at S1 (valid, green) vs S4 (invalid, red). 100 trials per point.

3 Map battery: 48 conditions

Each subsection shows three figures for one condition: the *vision layer* (gates A_X/A_H and energies $\|A_X V_X\|/\|A_H V_H\|$), the *memory layer* (gates A_Z/A_{H1} and energies $\|A_Z V_Z\|/\|A_{H1} V_{H1}\|$), and the *KDA accumulator* (α_t decay, β_t write gate, and per-pixel surprise $\|v - \tilde{S}^\top k\|$). All maps are 16×16 with 4×4 cell crosshairs; columns are timesteps $t = 0 \dots 6$; the change lands at $t = 5$.

3.1 cueS1_1of16_nochange

Cue fixed on cell S1 (red), ring completeness $1/16$ ($\text{p}(\text{change at cued cell} \mid \text{change}) = 1/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.55\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.02\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.

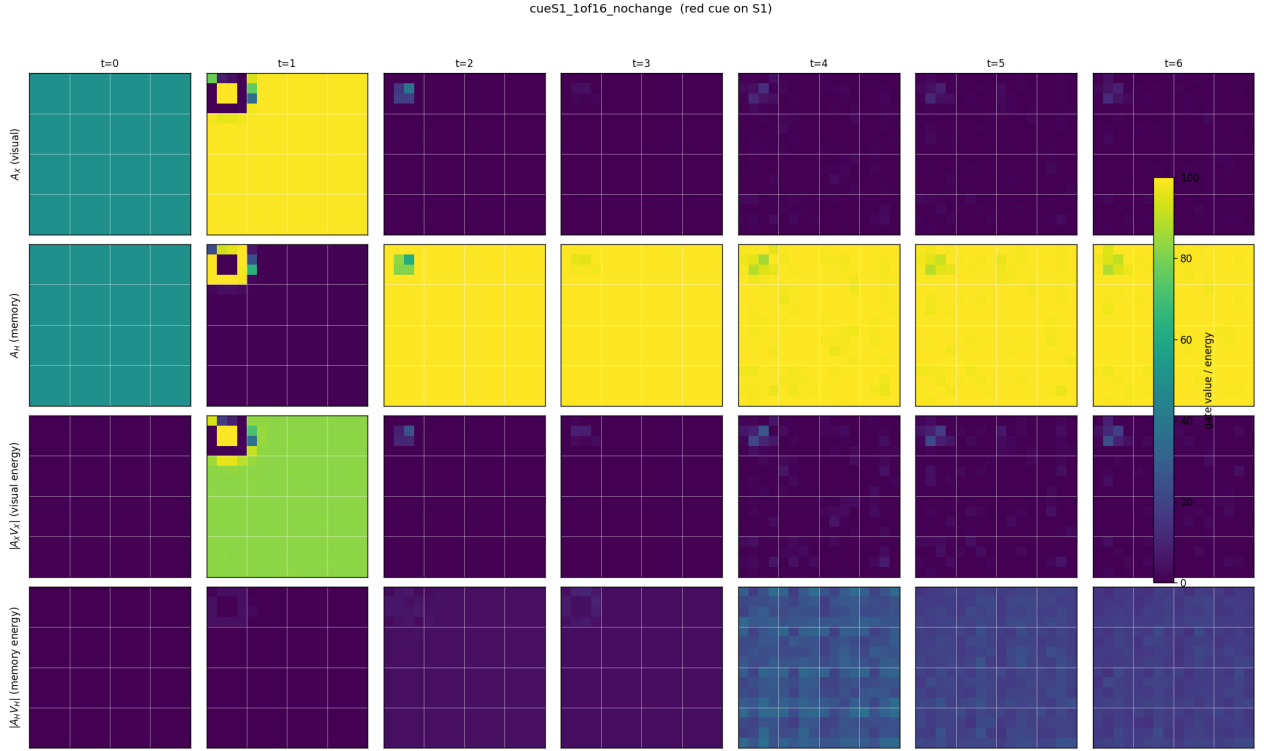


Figure 3: cueS1_1of16_nochange — vis layer.

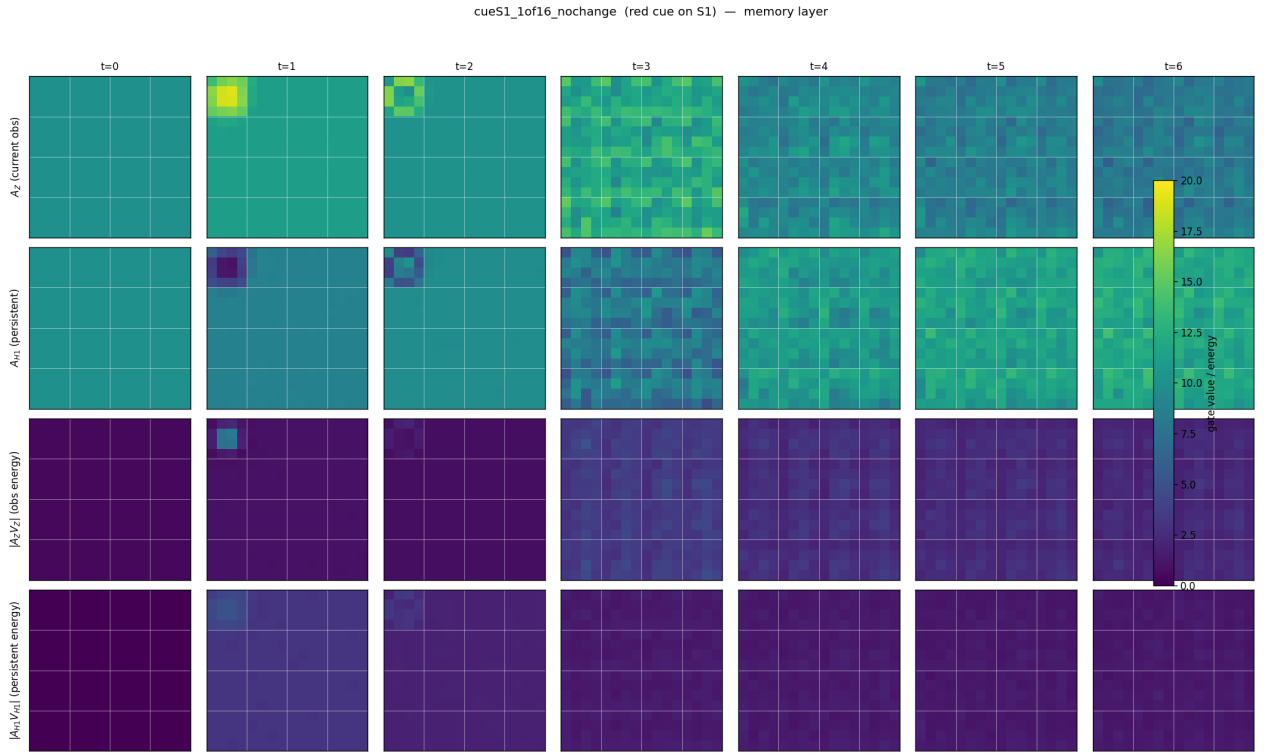


Figure 4: cueS1_1of16_nochange — mem layer.

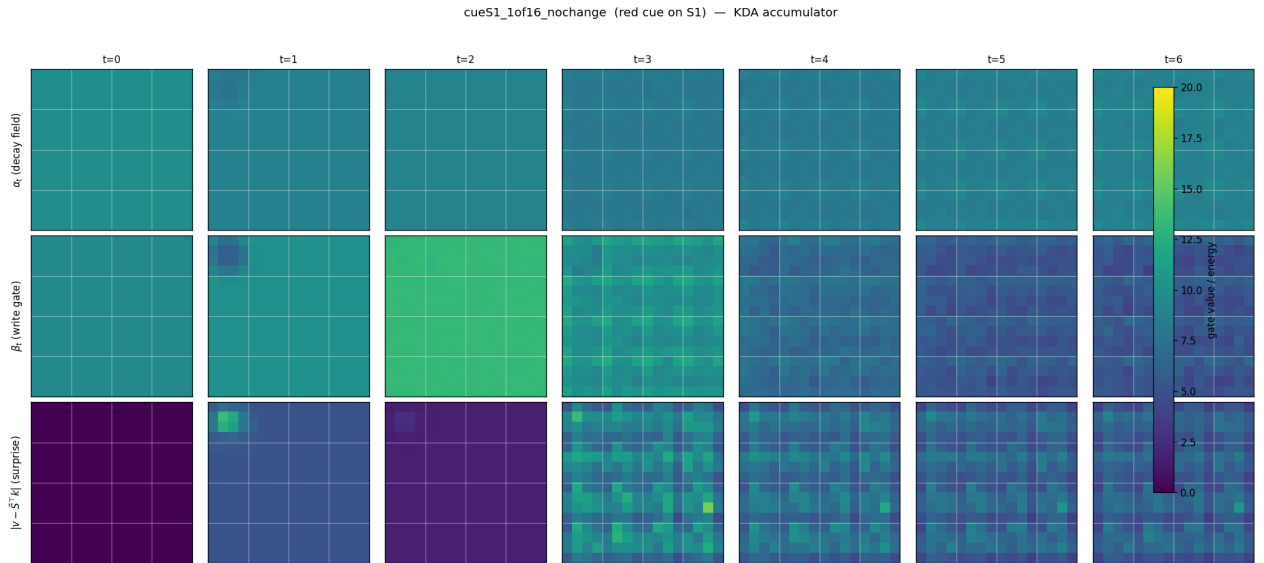


Figure 5: cueS1_1of16_nochange — acc layer.

3.2 cueS1_1of16_changeS1

Cue fixed on cell S1 (red), ring completeness 1/16 ($p(\text{change at cued cell} \mid \text{change}) = 1/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.55\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.23\times$ the global mean ($t = 6$: $1.22\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.44 vs A_{H1} (persistence) 0.56. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 6: cueS1_1of16_changeS1 — vis layer.

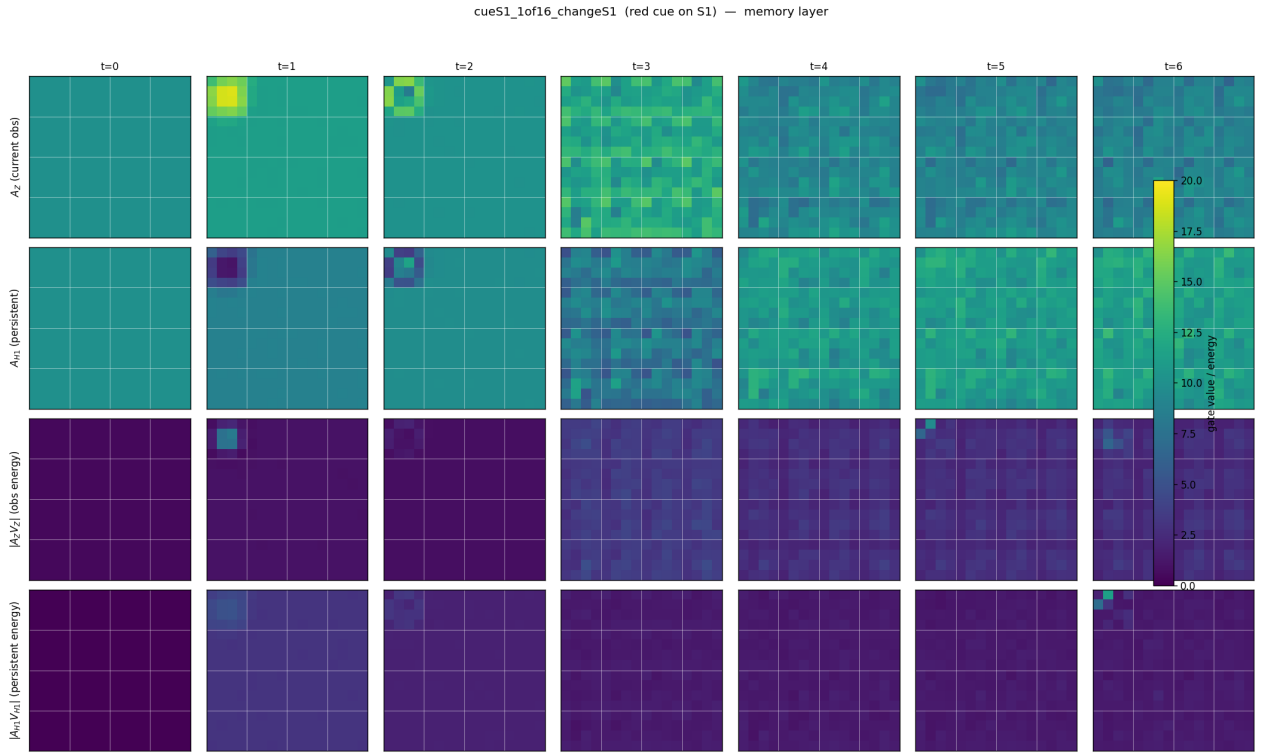


Figure 7: cueS1_1of16_changeS1 — mem layer.

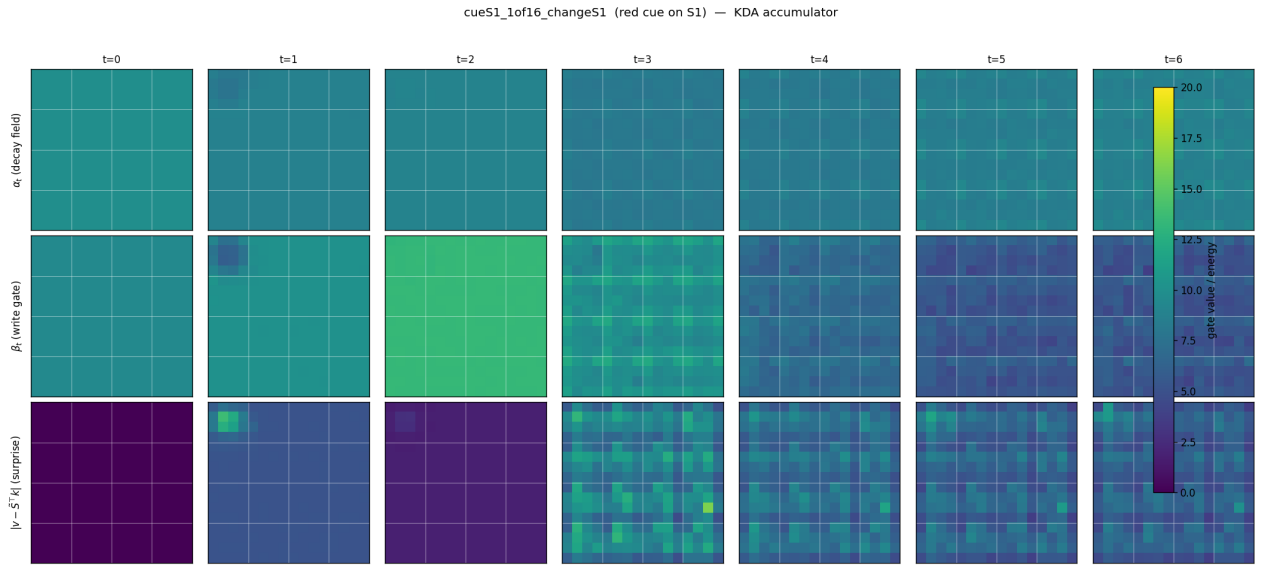


Figure 8: cueS1_1of16_changeS1 — acc layer.

3.3 cueS1_1of16_changeS4

Cue fixed on cell S1 (red), ring completeness 1/16 ($p(\text{change at cued cell} \mid \text{change}) = 1/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.55\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.01\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

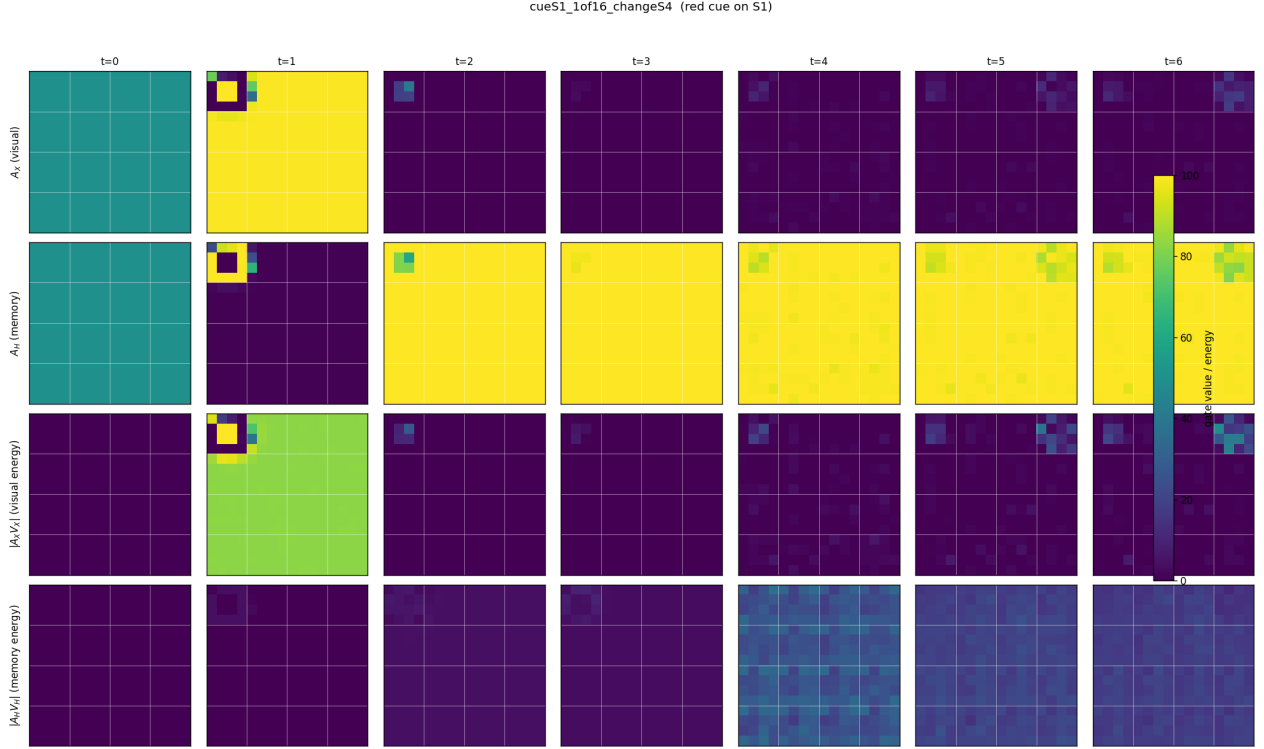


Figure 9: cueS1_1of16_changeS4 — vis layer.

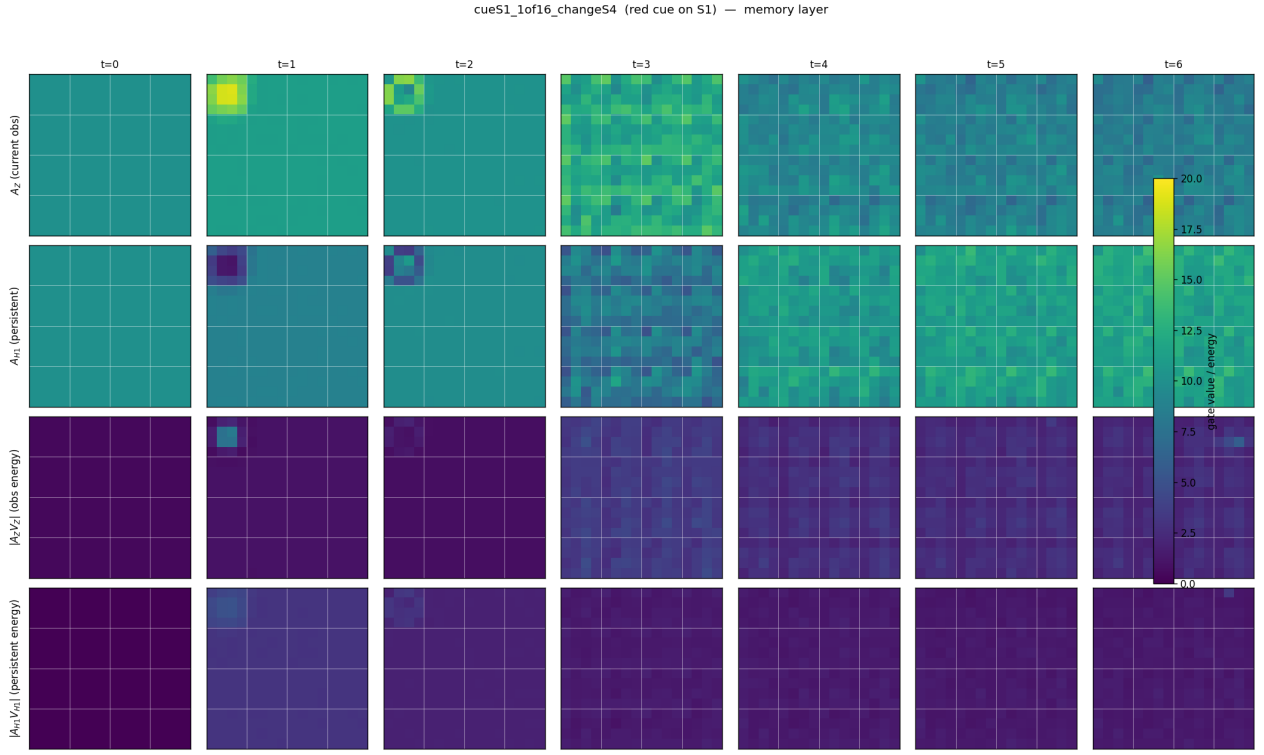


Figure 10: cueS1_1of16_changeS4 — mem layer.

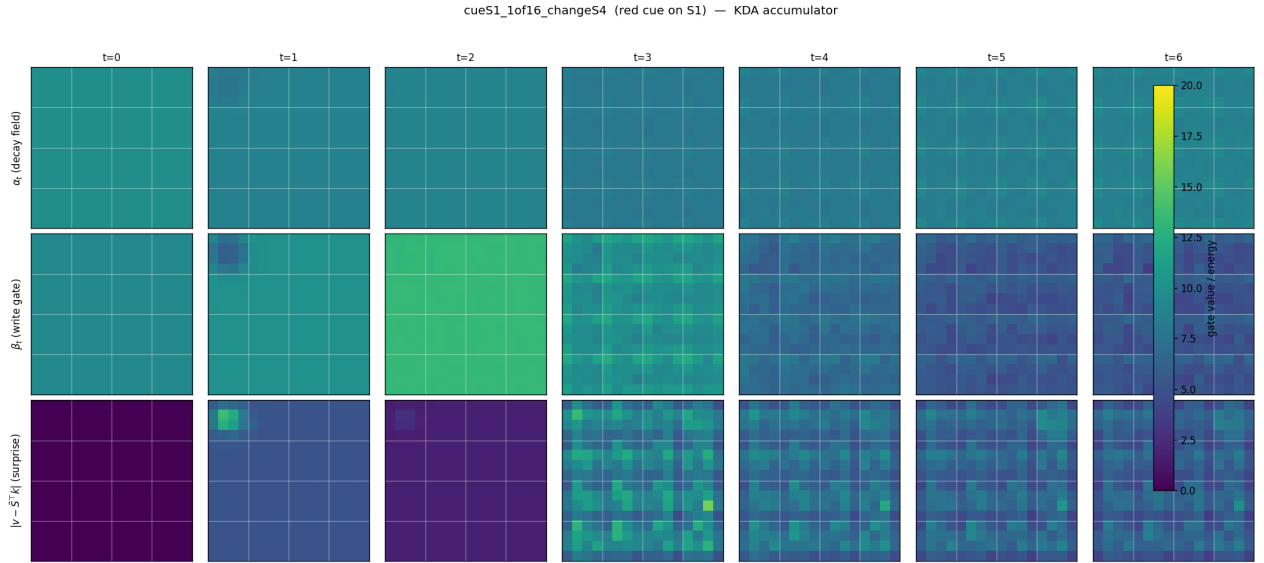


Figure 11: cueS1_1of16_changeS4 — acc layer.

3.4 cueS1_2of16_nochange

Cue fixed on cell S1 (red), ring completeness 2/16 ($p(\text{change at cued cell} \mid \text{change}) = 2/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.56\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.03\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.00/1.00 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.

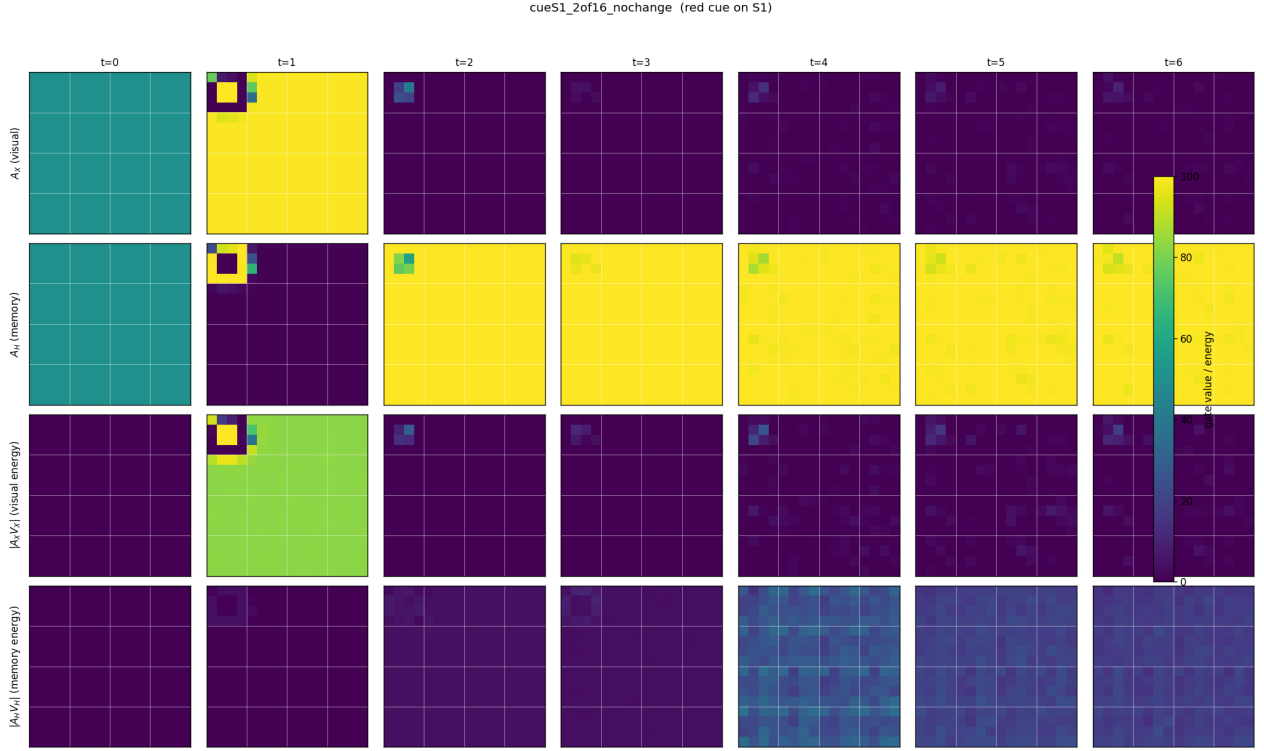


Figure 12: cueS1_2of16_nochange — vis layer.

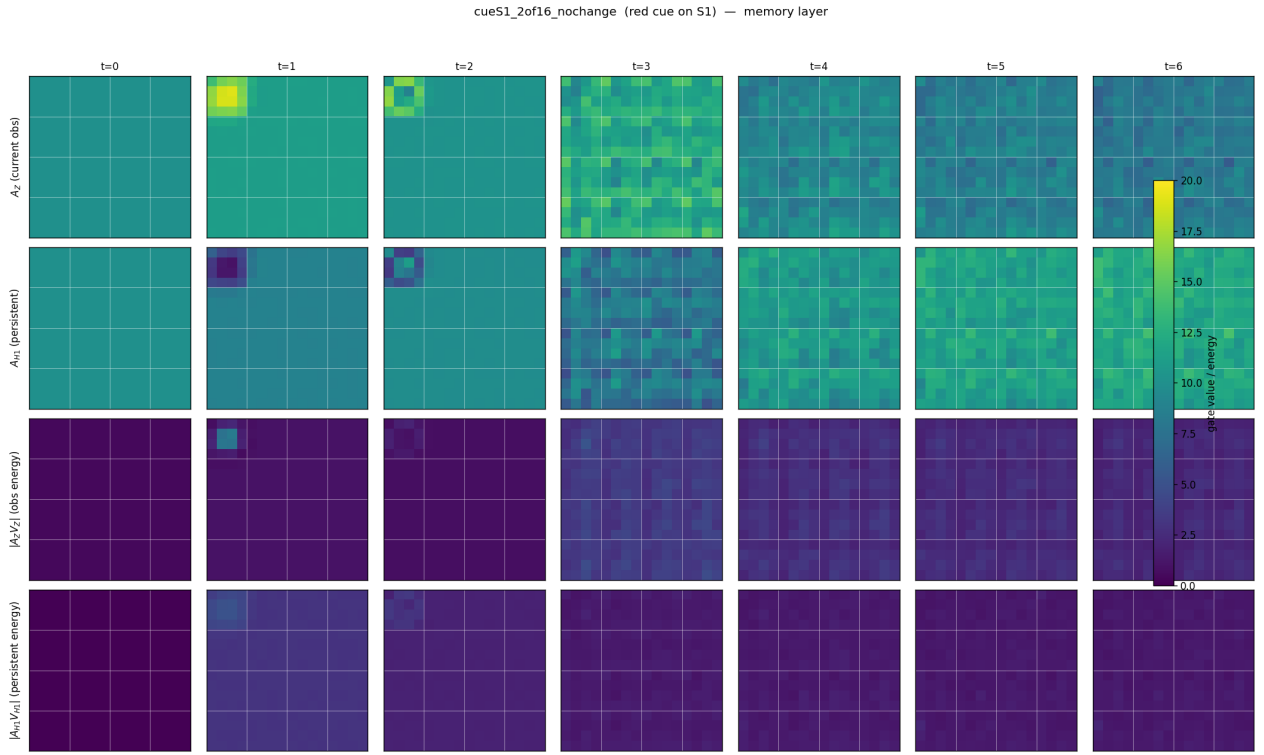


Figure 13: cueS1_2of16_nochange — mem layer.



Figure 14: cueS1_2of16_nochange — acc layer.

3.5 cueS1_2of16_changeS1

Cue fixed on cell S1 (red), ring completeness 2/16 ($p(\text{change at cued cell} \mid \text{change}) = 2/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.56\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.21\times$ the global mean ($t = 6$: $1.23\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.45 vs A_{H1} (persistence) 0.55. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.

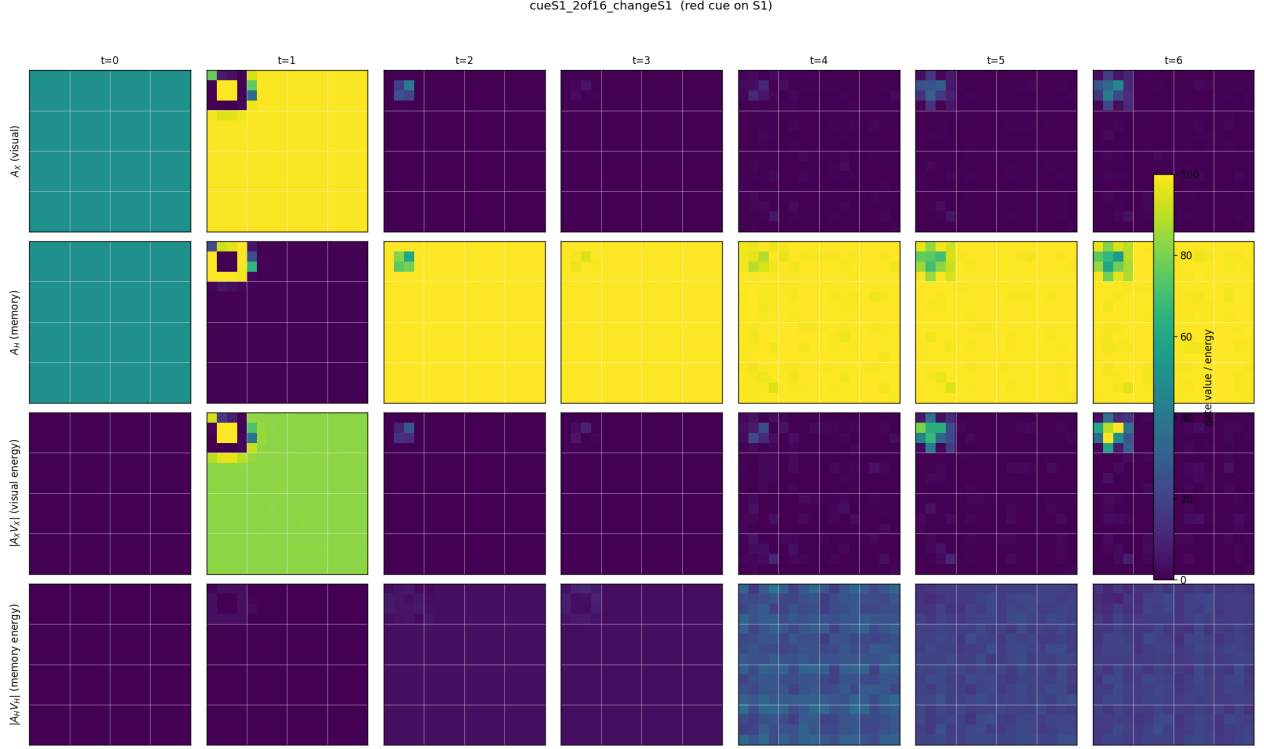


Figure 15: cueS1_2of16_changeS1 — vis layer.

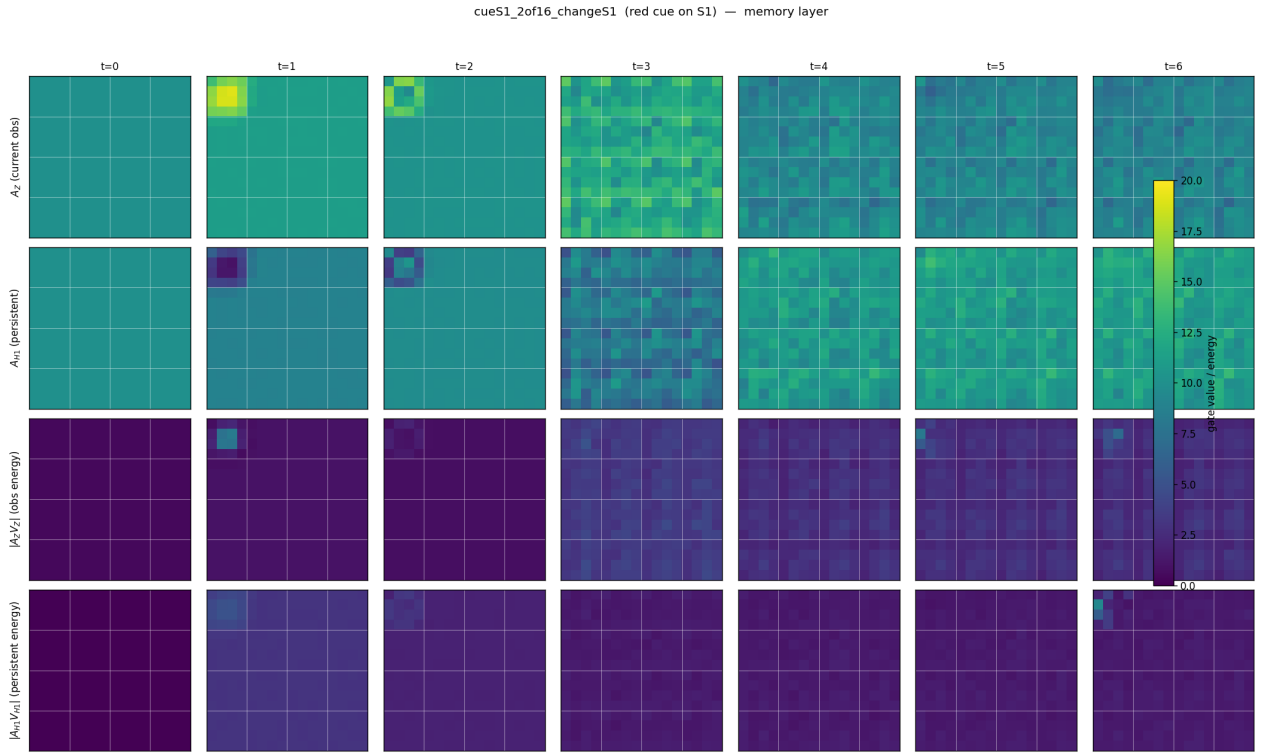


Figure 16: cueS1_2of16_changeS1 — mem layer.



Figure 17: cueS1_2of16_changeS1 — acc layer.

3.6 cueS1_2of16_changeS4

Cue fixed on cell S1 (red), ring completeness 2/16 ($p(\text{change at cued cell} \mid \text{change}) = 2/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.56\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $0.99\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.43 vs A_{H1} (persistence) 0.57 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

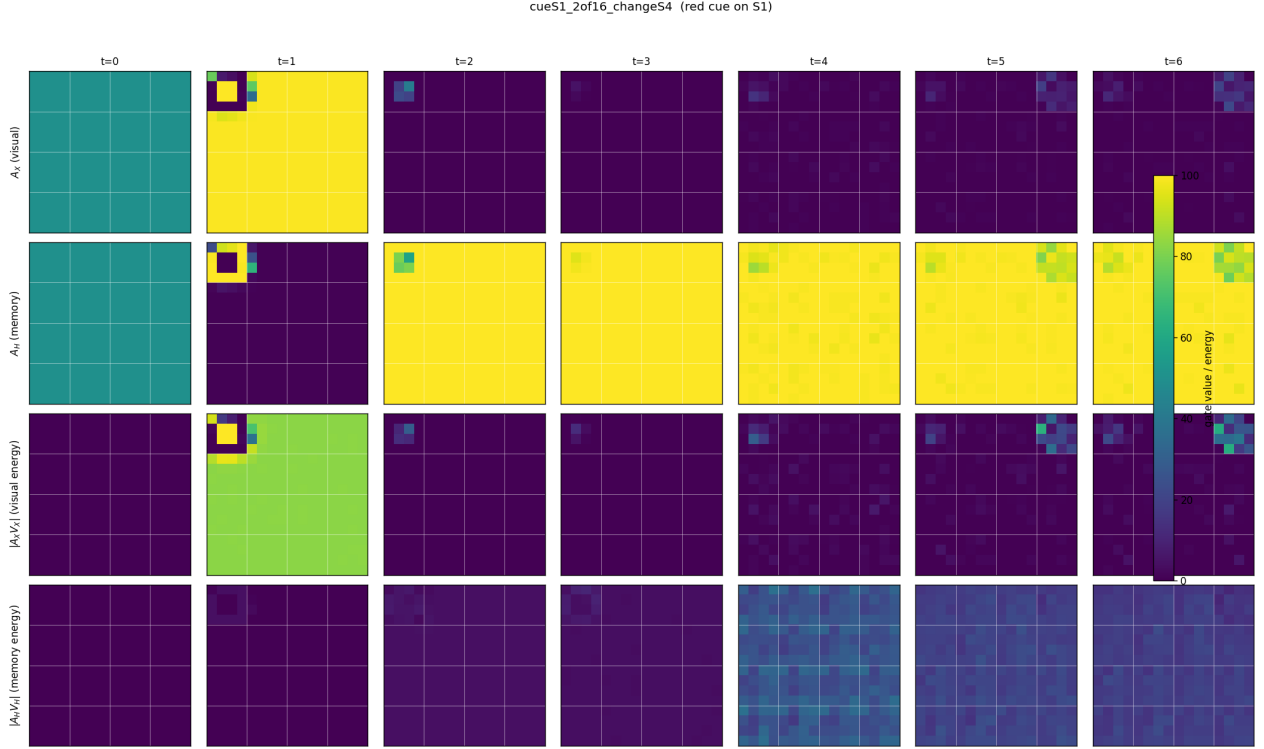


Figure 18: cueS1_2of16_changeS4 — vis layer.

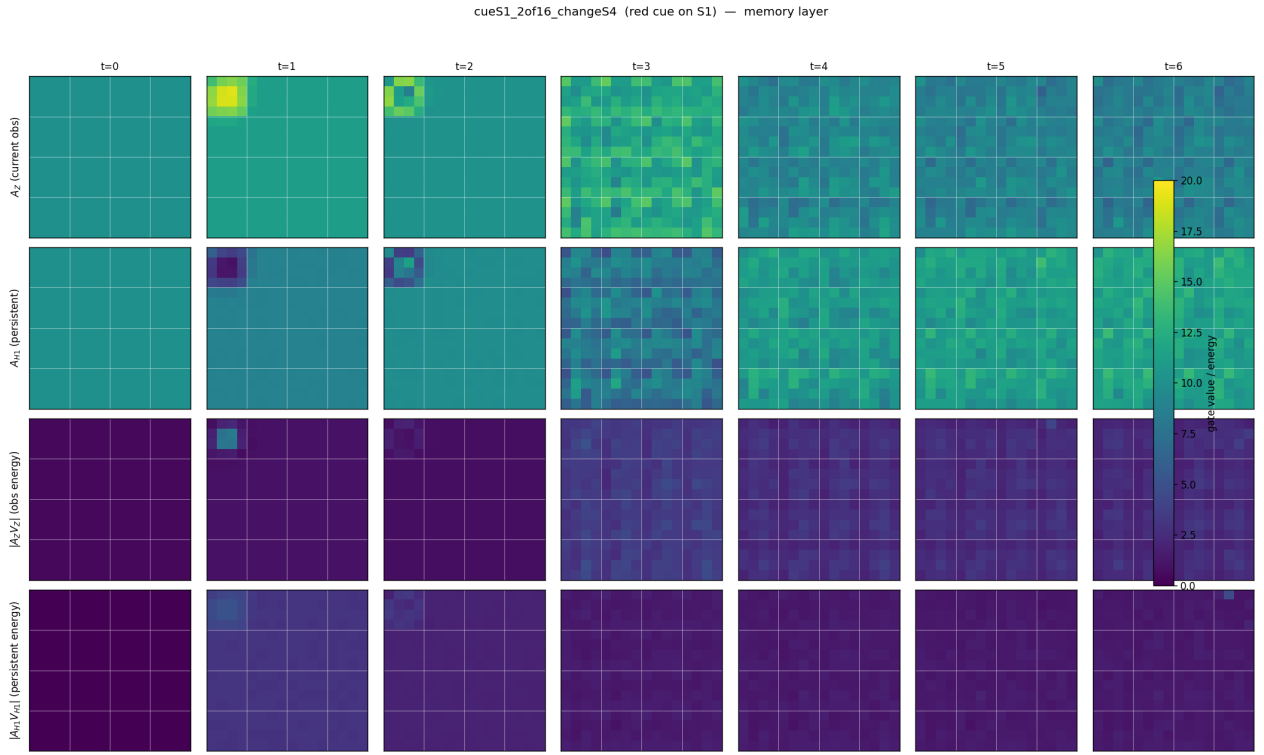


Figure 19: cueS1_2of16_changeS4 — mem layer.



Figure 20: cueS1_2of16_changeS4 — acc layer.

3.7 cueS1_3of16_nochange

Cue fixed on cell S1 (red), ring completeness 3/16 ($p(\text{change at cued cell} \mid \text{change}) = 3/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.57\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.02\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.26 ; spatially flat — localization lives in the error term, not the gates.

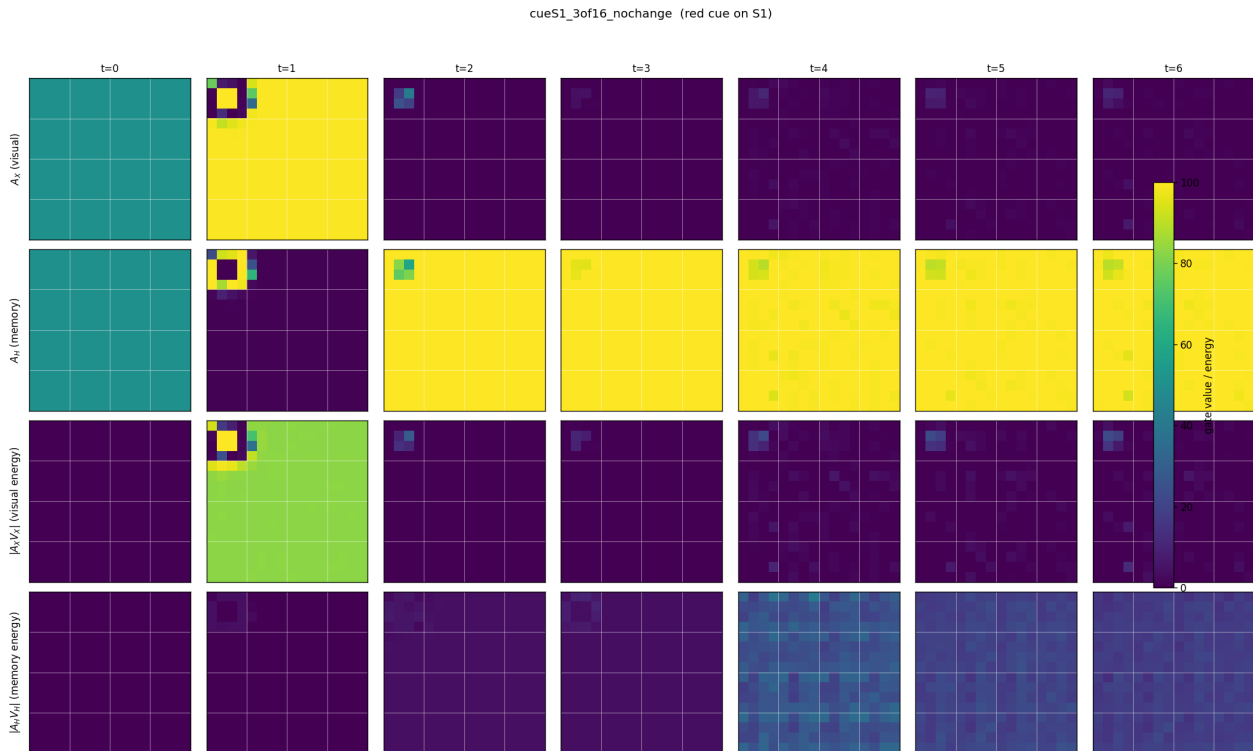


Figure 21: cueS1_3of16_nochange — vis layer.

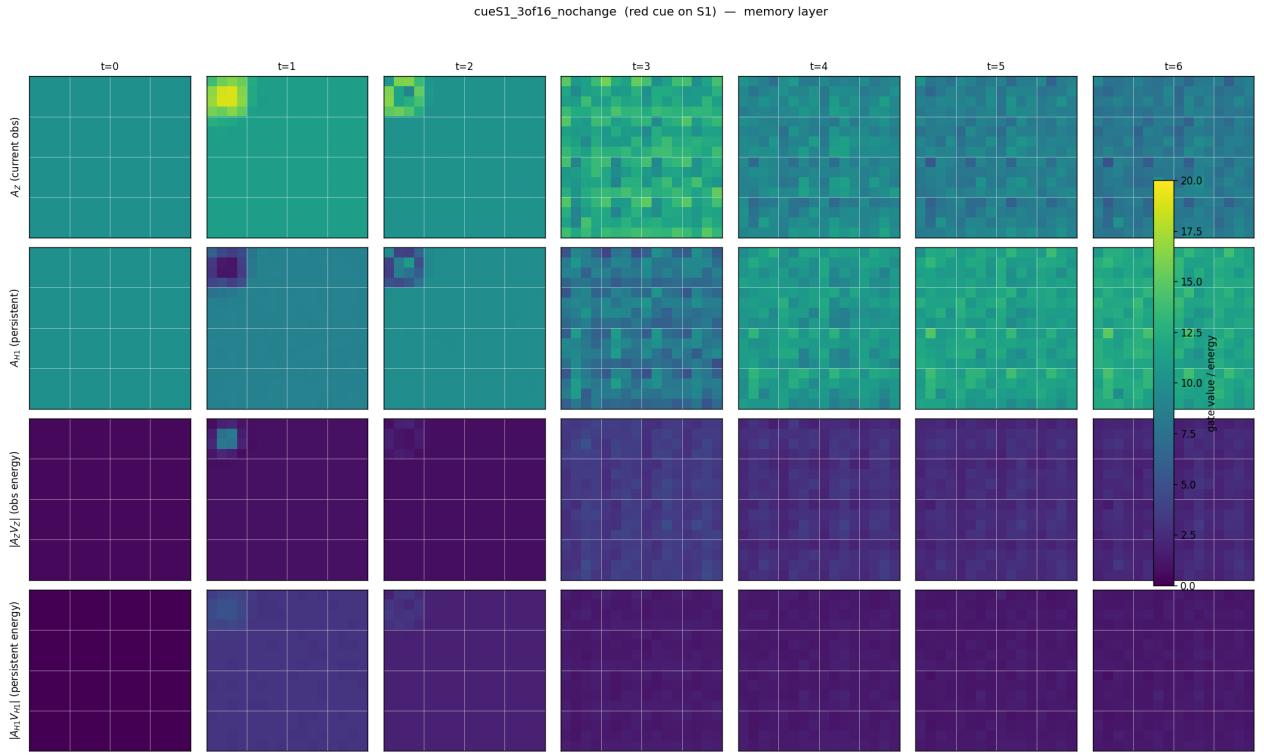


Figure 22: cueS1_3of16_nochange — mem layer.



Figure 23: cueS1_3of16_nochange — acc layer.

3.8 cueS1_3of16_changeS1

Cue fixed on cell S1 (red), ring completeness 3/16 ($p(\text{change at cued cell} \mid \text{change}) = 3/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.57\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.22\times$ the global mean ($t = 6$: $1.21\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.44 vs A_{H1} (persistence) 0.56. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 24: cueS1_3of16_changeS1 — vis layer.

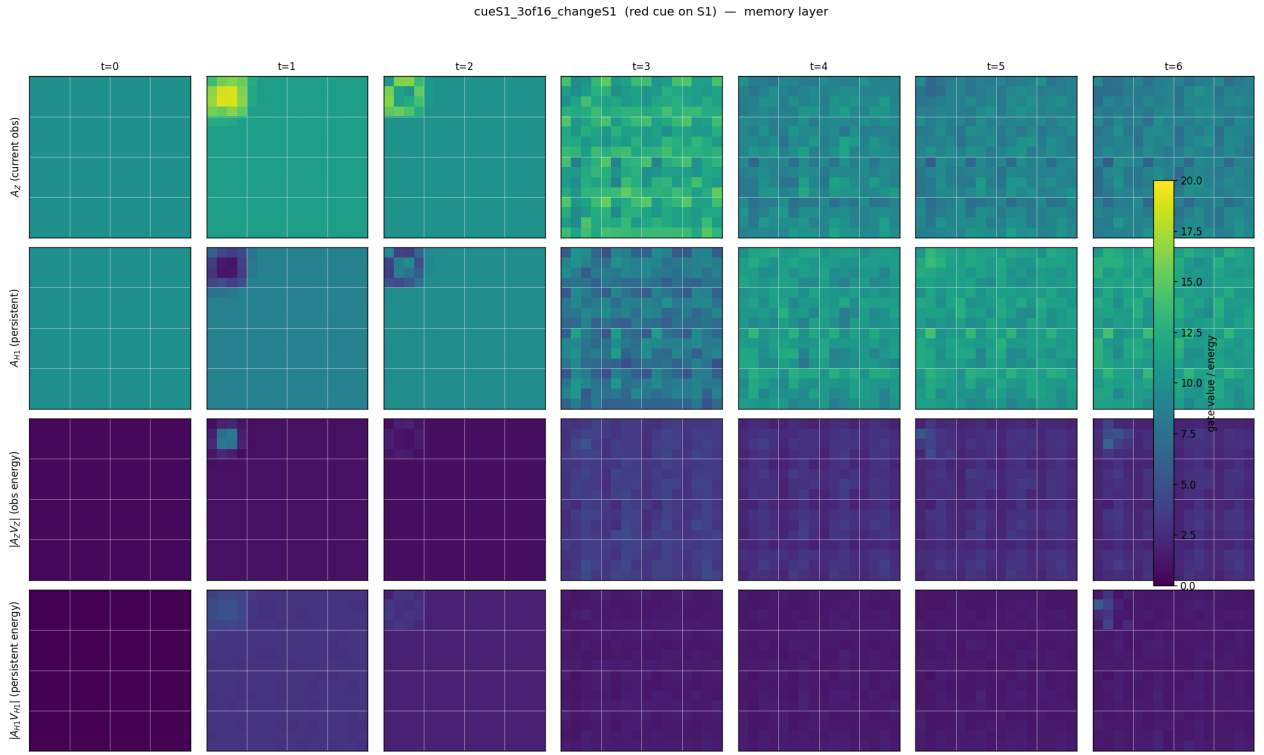


Figure 25: cueS1_3of16_changeS1 — mem layer.

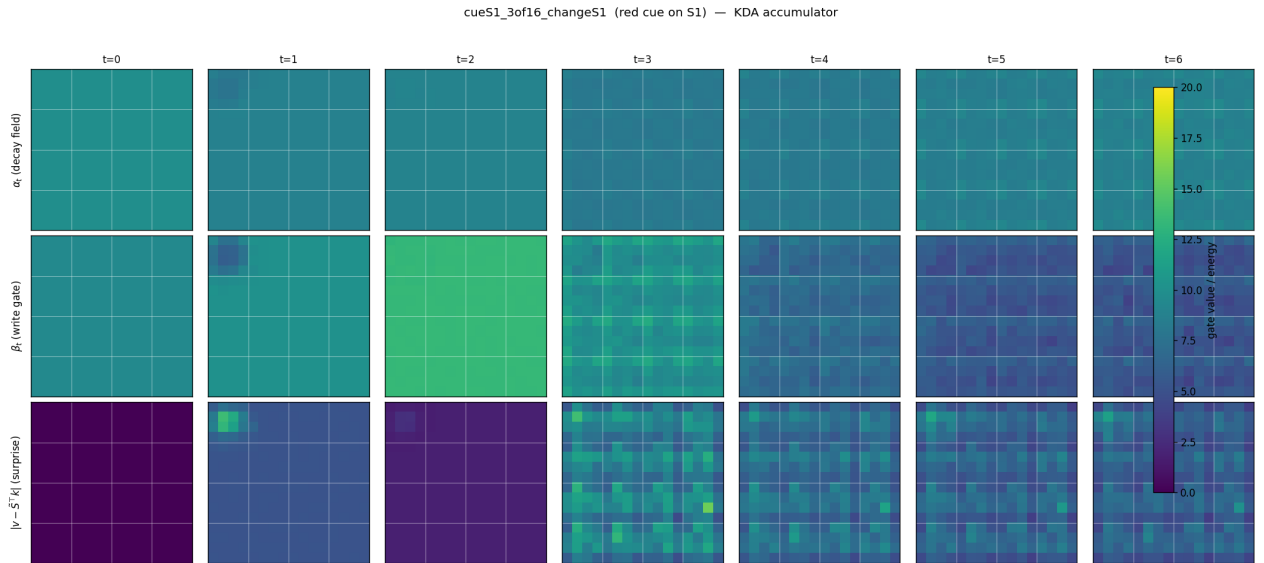


Figure 26: cueS1_3of16_changeS1 — acc layer.

3.9 cueS1_3of16_changeS4

Cue fixed on cell S1 (red), ring completeness 3/16 ($p(\text{change at cued cell} \mid \text{change}) = 3/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.57\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $0.99\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.43 vs A_{H1} (persistence) 0.57 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

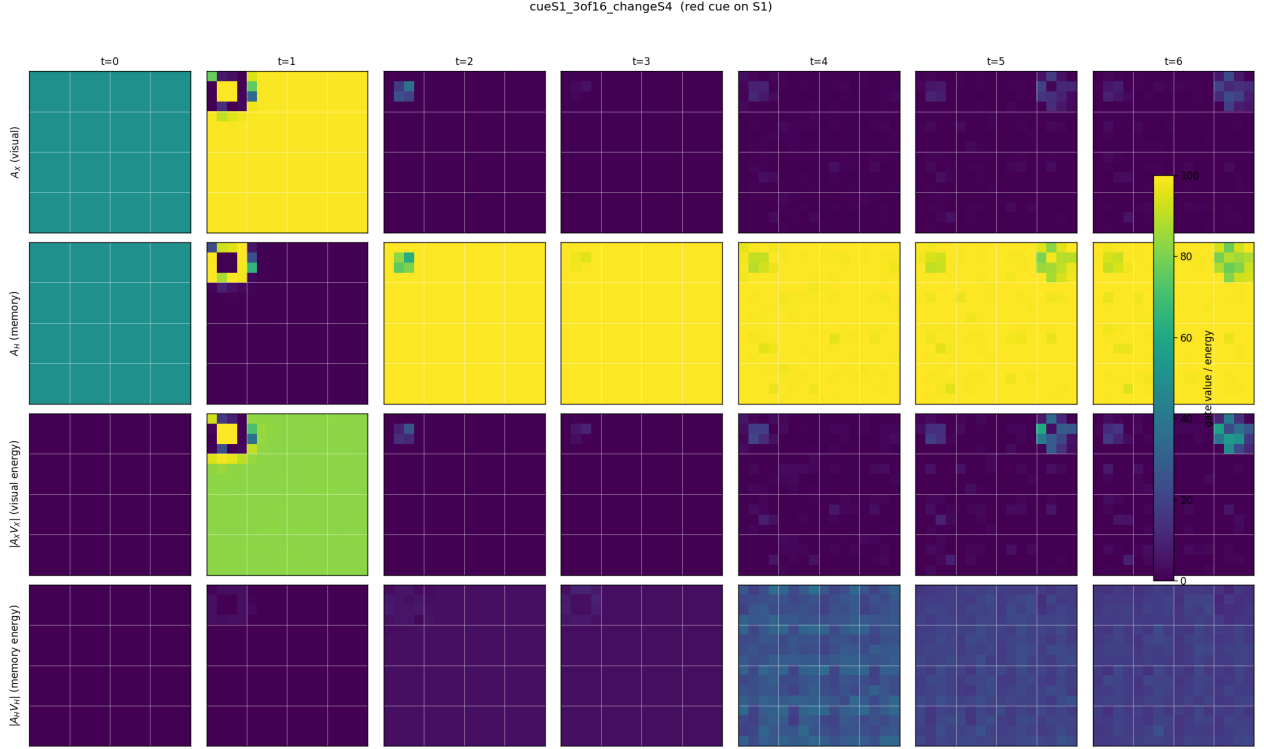


Figure 27: cueS1_3of16_changeS4 — vis layer.

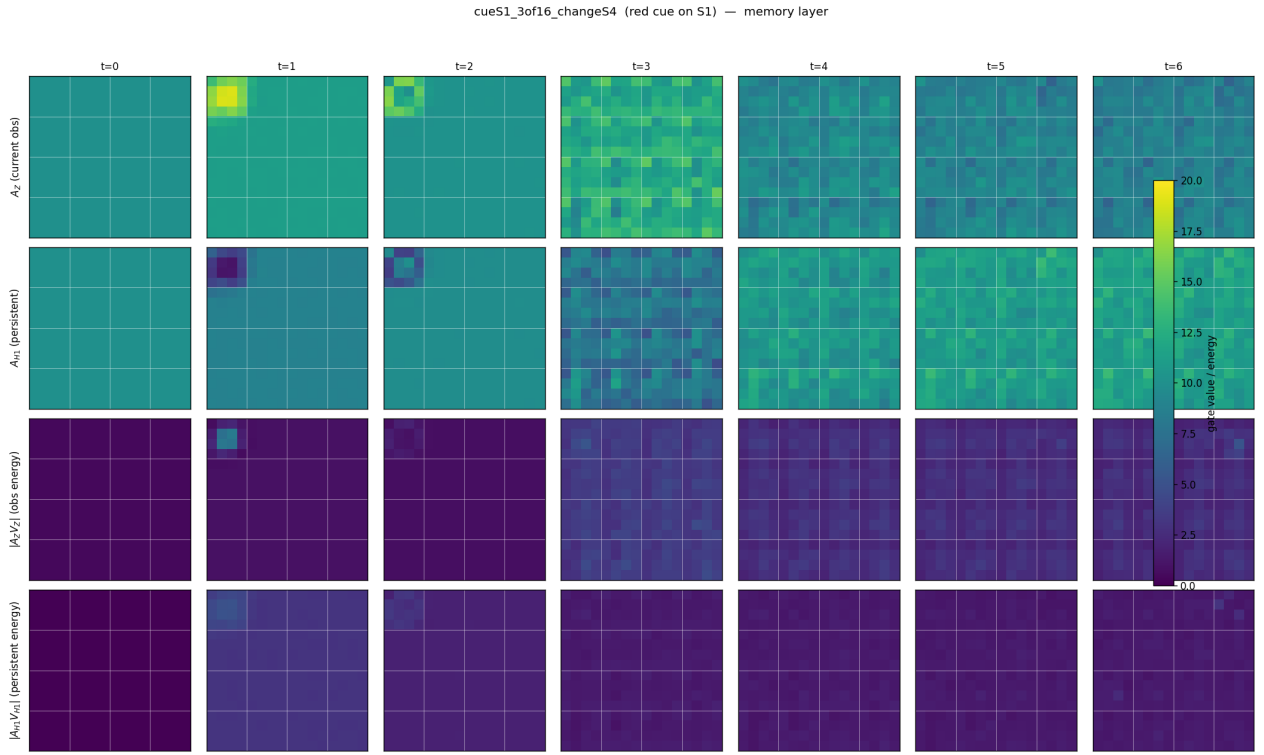


Figure 28: cueS1_3of16_changeS4 — mem layer.



Figure 29: cueS1_3of16_changeS4 — acc layer.

3.10 cueS1_4of16_nochange

Cue fixed on cell S1 (red), ring completeness 4/16 ($p(\text{change at cued cell} \mid \text{change}) = 4/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.60\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.02\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.

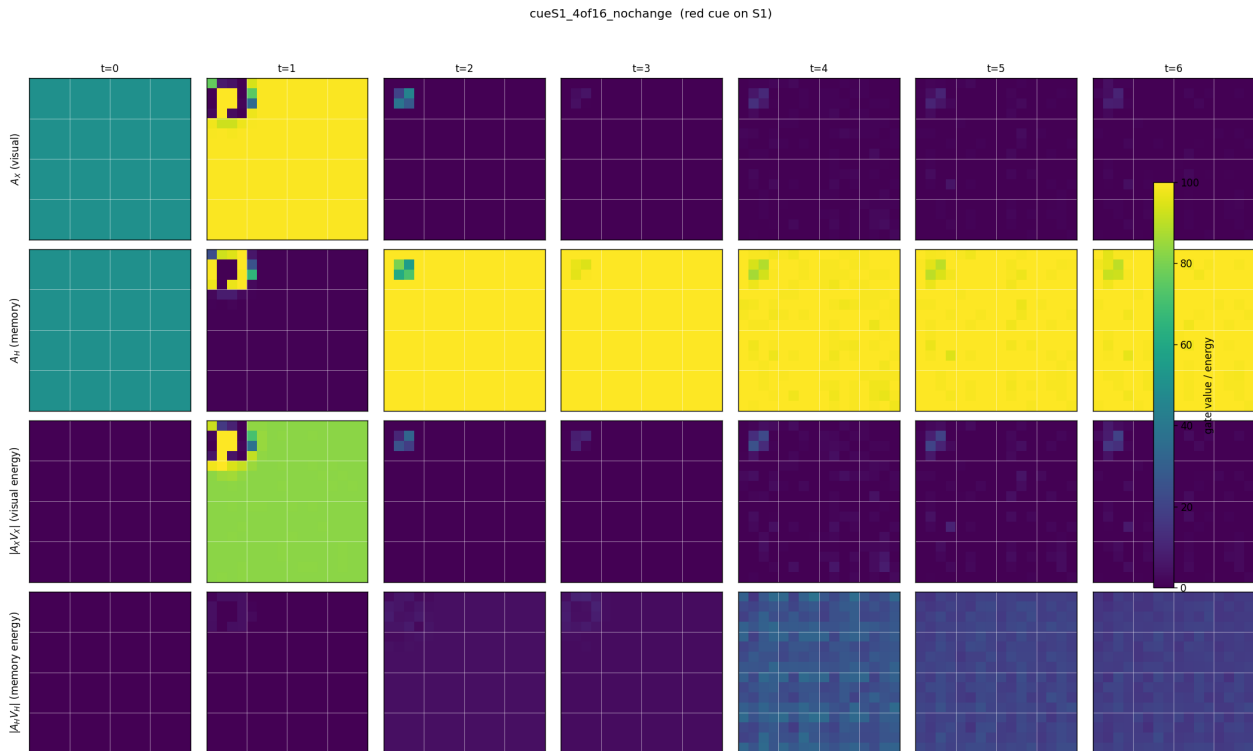


Figure 30: cueS1_4of16_nochange — vis layer.

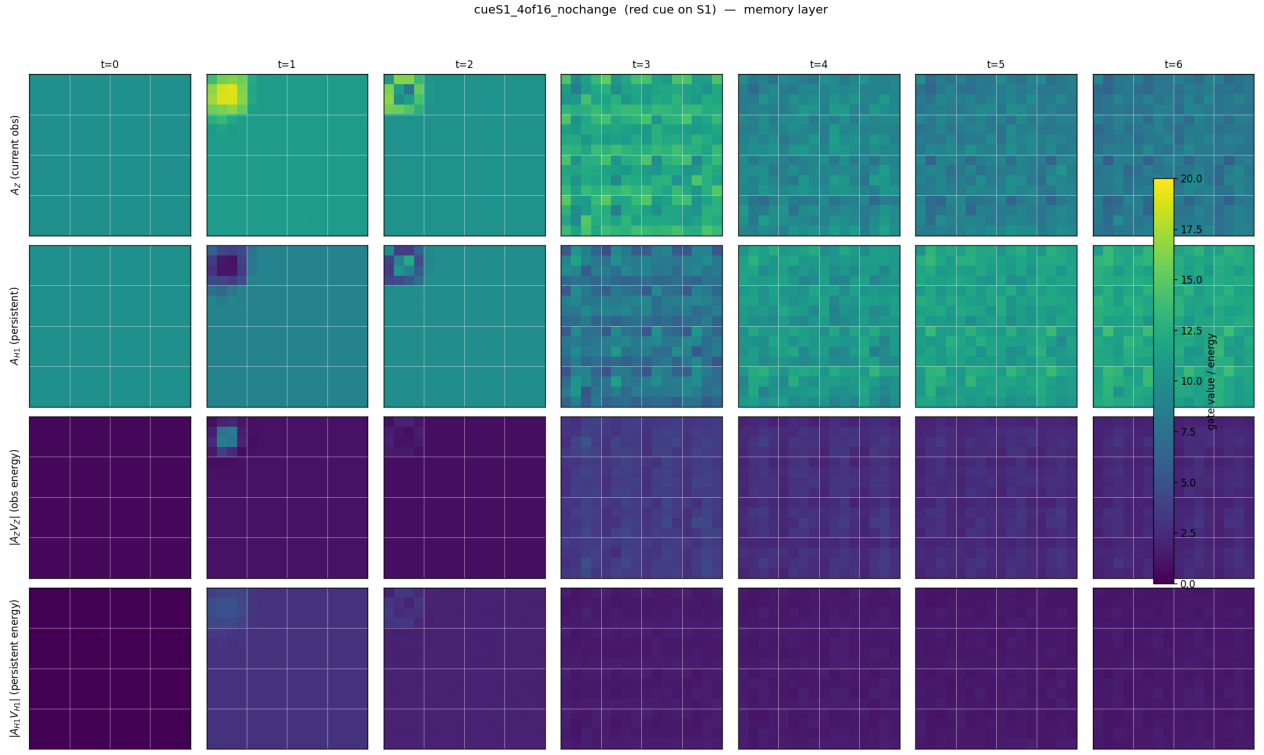


Figure 31: cueS1_4of16_nochange — mem layer.

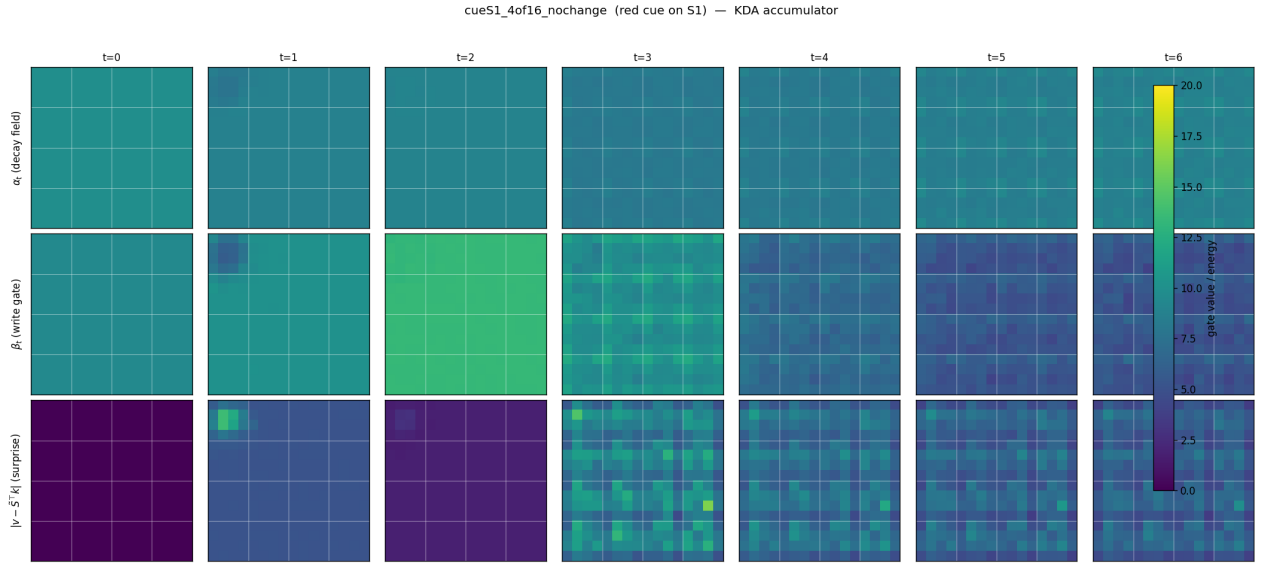


Figure 32: cueS1_4of16_nochange — acc layer.

3.11 cueS1_4of16_changeS1

Cue fixed on cell S1 (red), ring completeness 4/16 ($p(\text{change at cued cell} \mid \text{change}) = 4/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.60\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.22\times$ the global mean ($t = 6$: $1.24\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.44 vs A_{H1} (persistence) 0.56. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 33: cueS1_4of16_changeS1 — vis layer.

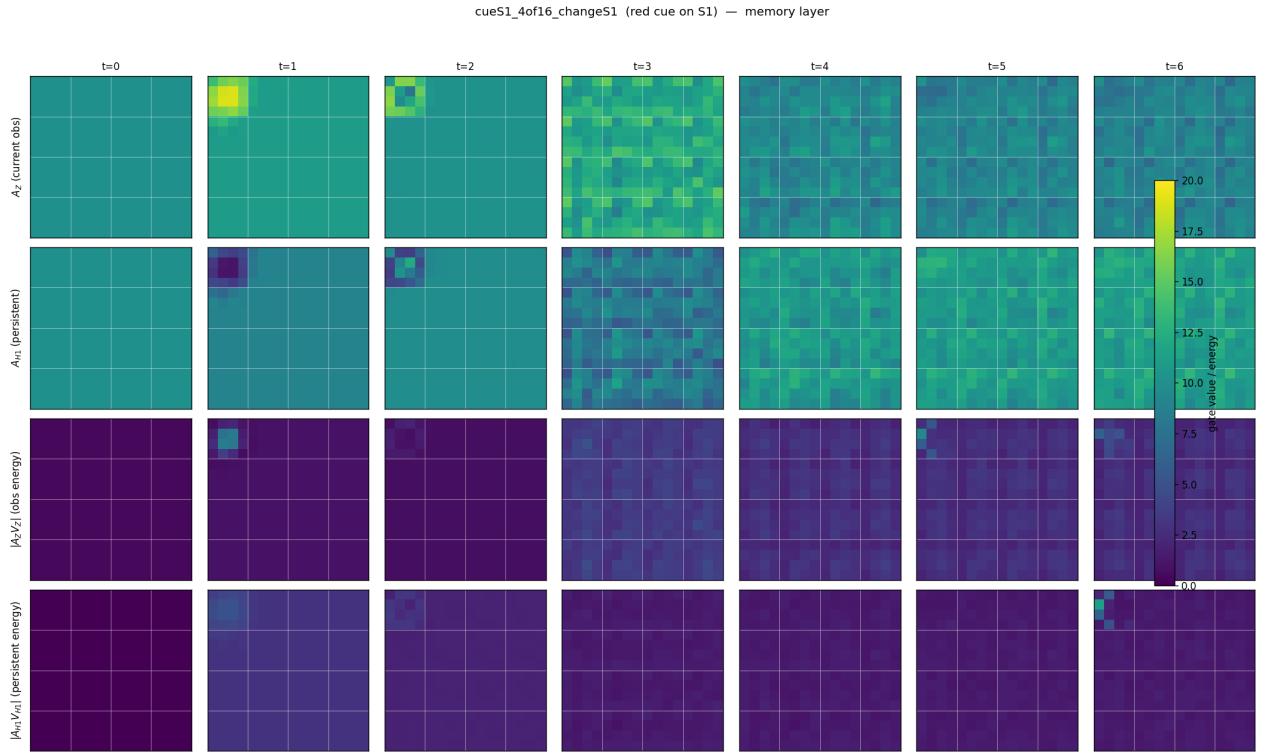


Figure 34: cueS1_4of16_changeS1 — mem layer.



Figure 35: cueS1_4of16_changeS1 — acc layer.

3.12 cueS1_4of16_changeS4

Cue fixed on cell S1 (red), ring completeness 4/16 ($p(\text{change at cued cell} \mid \text{change}) = 4/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.60\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $0.98\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

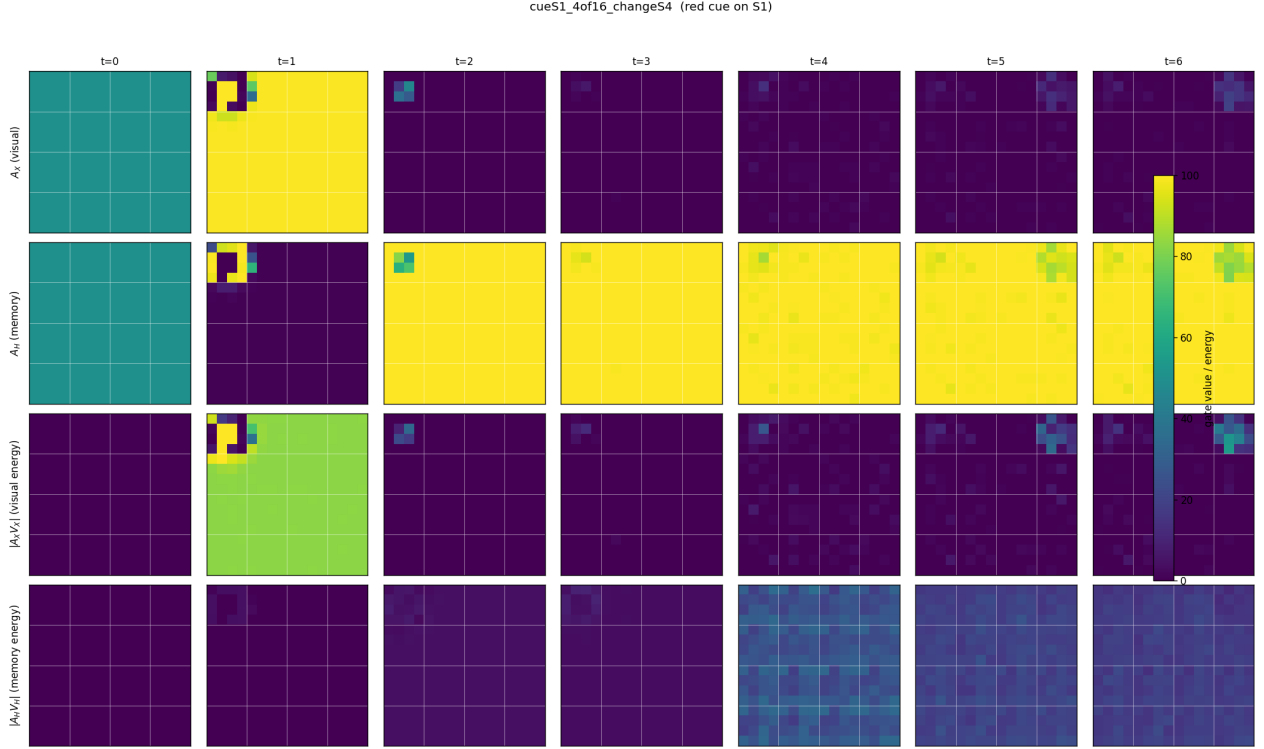


Figure 36: cueS1_4of16_changeS4 — vis layer.

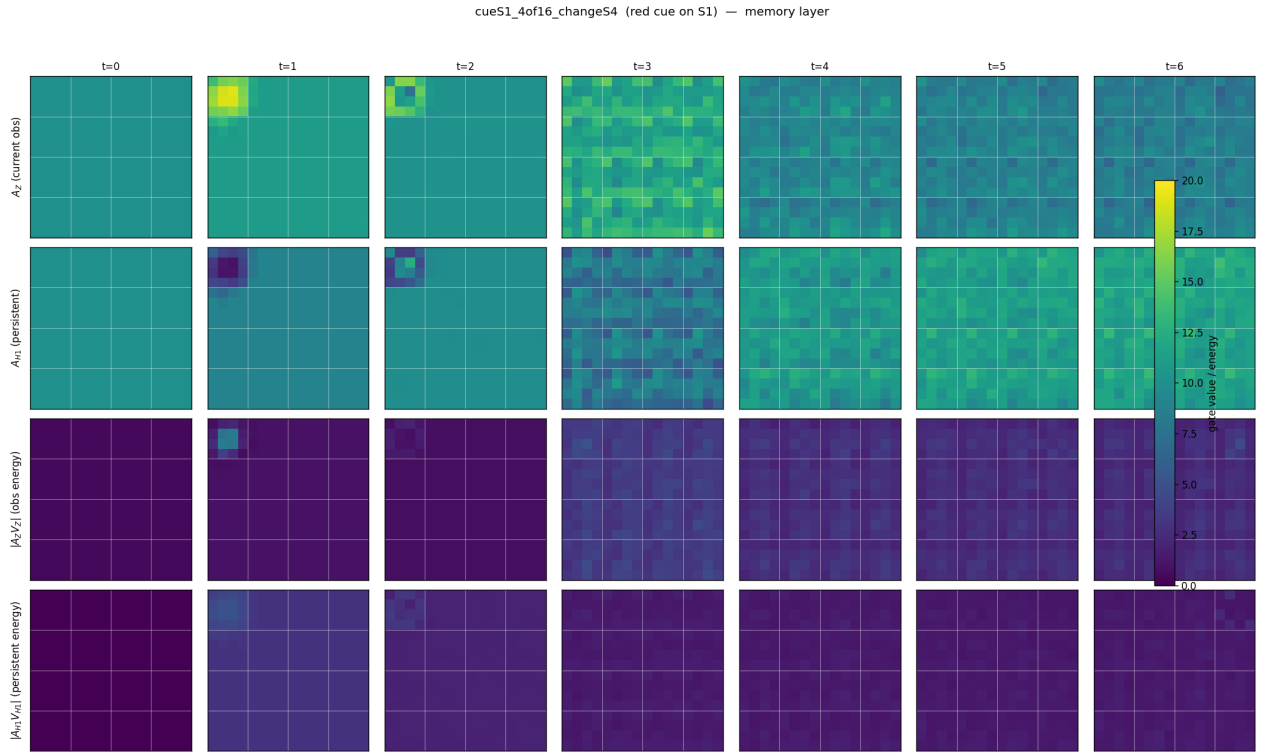


Figure 37: cueS1_4of16_changeS4 — mem layer.



Figure 38: cueS1_4of16_changeS4 — acc layer.

3.13 cueS1_5of16_nochange

Cue fixed on cell S1 (red), ring completeness 5/16 ($p(\text{change at cued cell} \mid \text{change}) = 5/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.62\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.00\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.

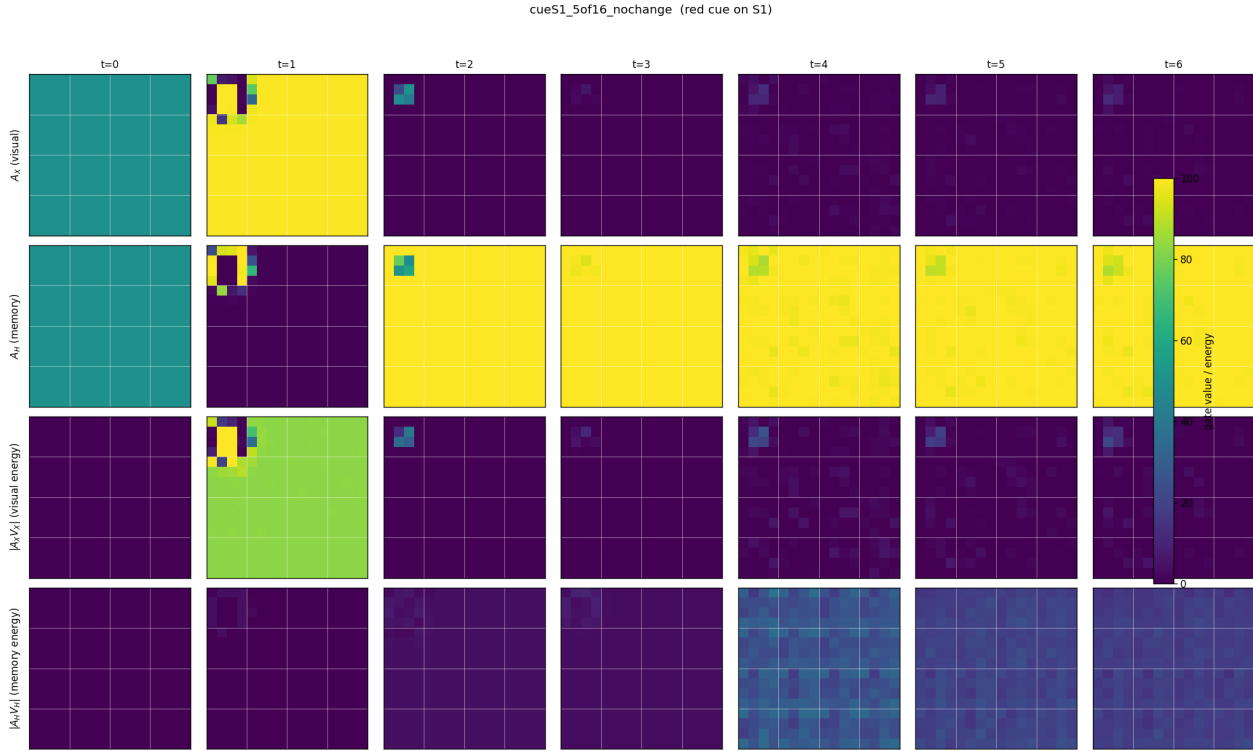


Figure 39: cueS1_5of16_nochange — vis layer.

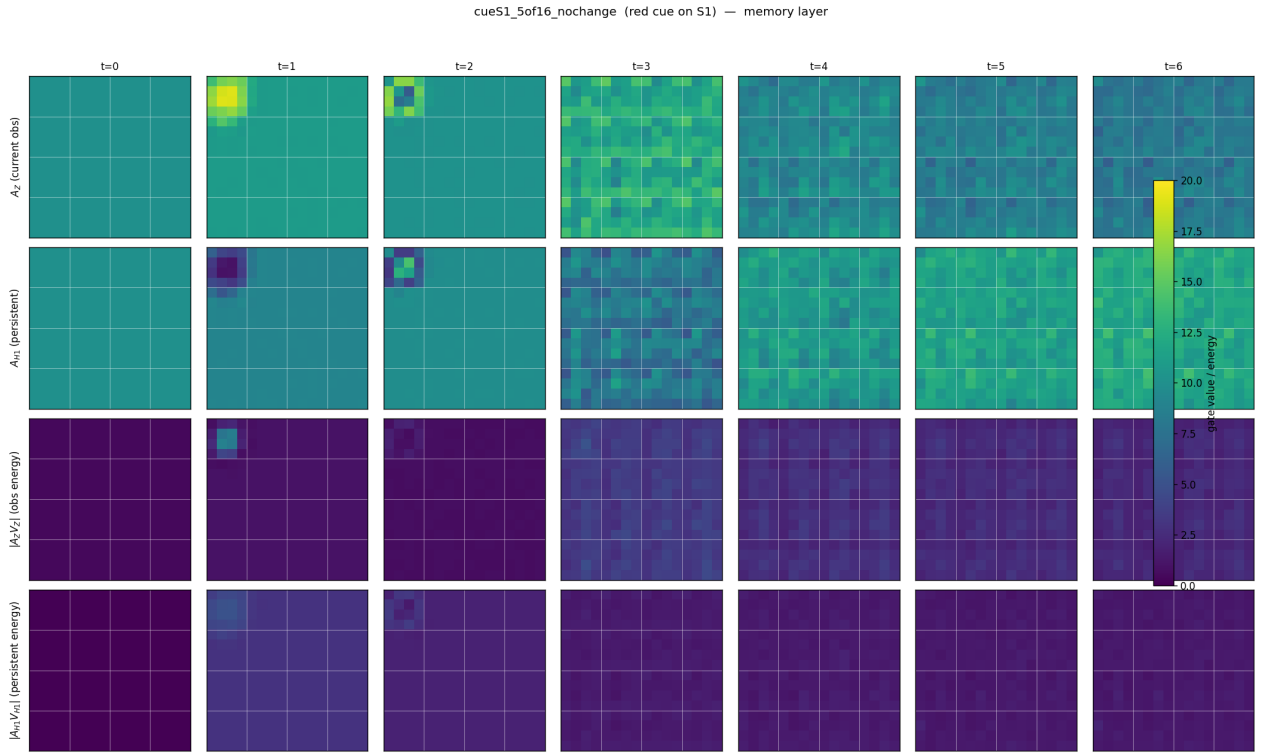


Figure 40: cueS1_5of16_nochange — mem layer.



Figure 41: cueS1_5of16_nochange — acc layer.

3.14 cueS1_5of16_changeS1

Cue fixed on cell S1 (red), ring completeness 5/16 ($p(\text{change at cued cell} \mid \text{change}) = 5/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.62\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.23\times$ the global mean ($t = 6$: $1.21\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.45 vs A_{H1} (persistence) 0.55. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 42: cueS1_5of16_changeS1 — vis layer.

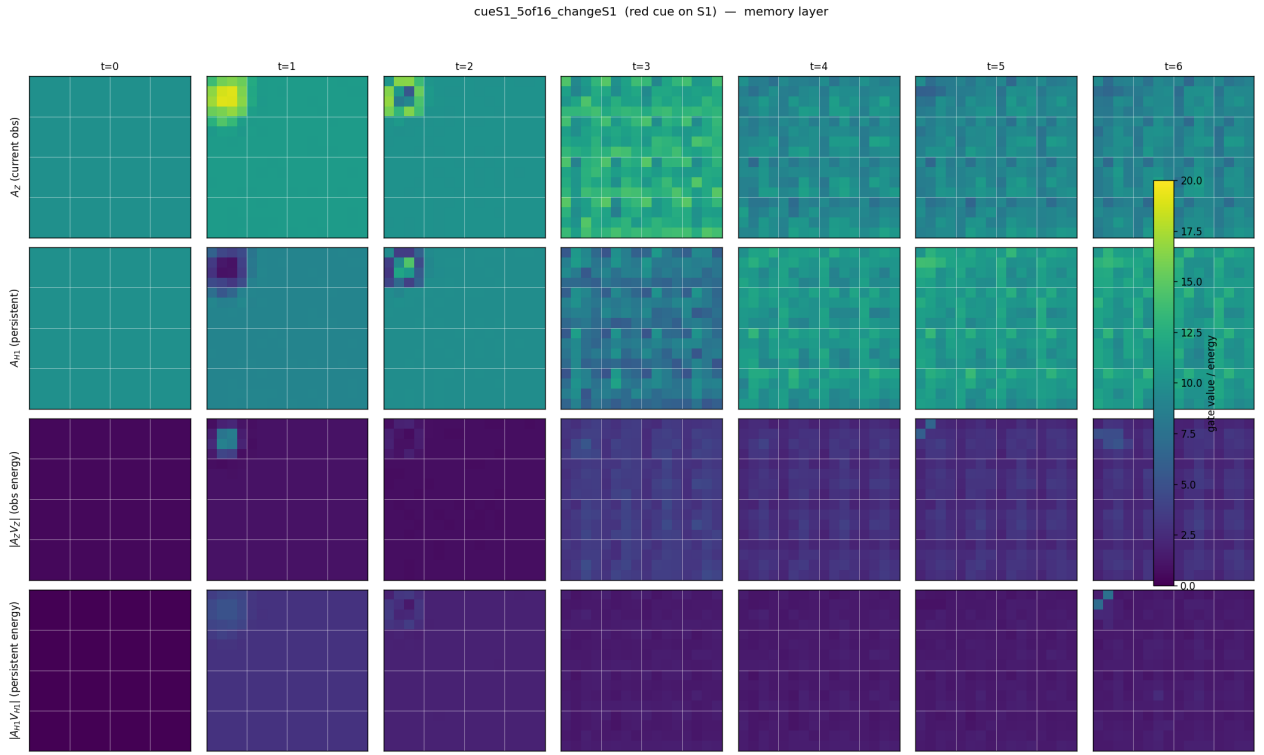


Figure 43: cueS1_5of16_changeS1 — mem layer.

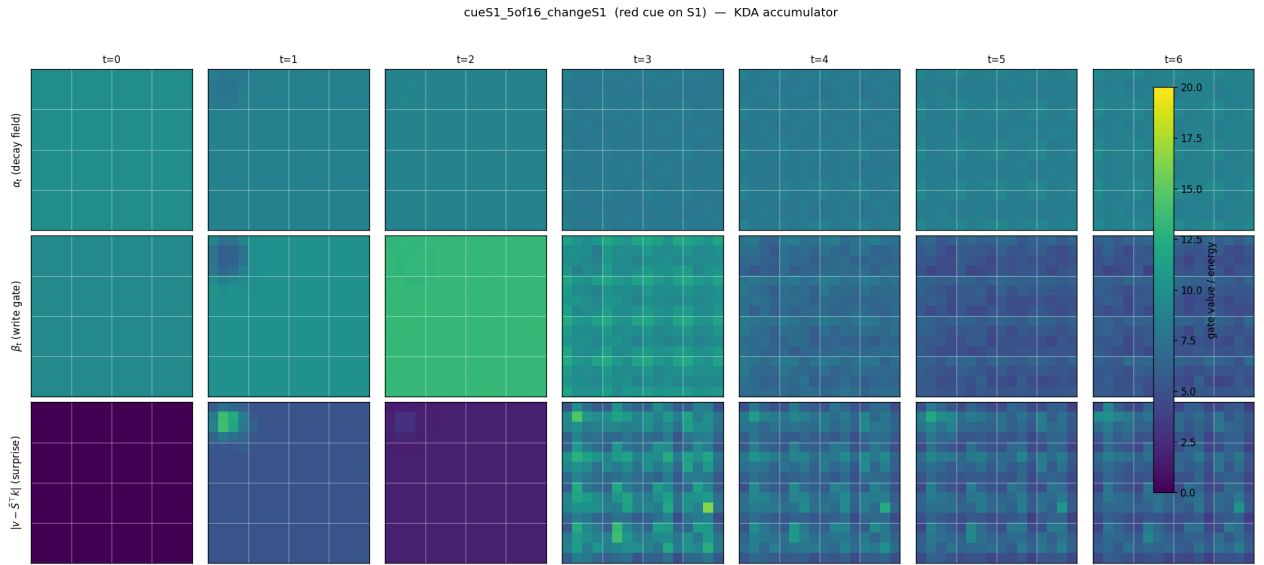


Figure 44: cueS1_5of16_changeS1 — acc layer.

3.15 cueS1_5of16_changeS4

Cue fixed on cell S1 (red), ring completeness 5/16 ($p(\text{change at cued cell} \mid \text{change}) = 5/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.62\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.00\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

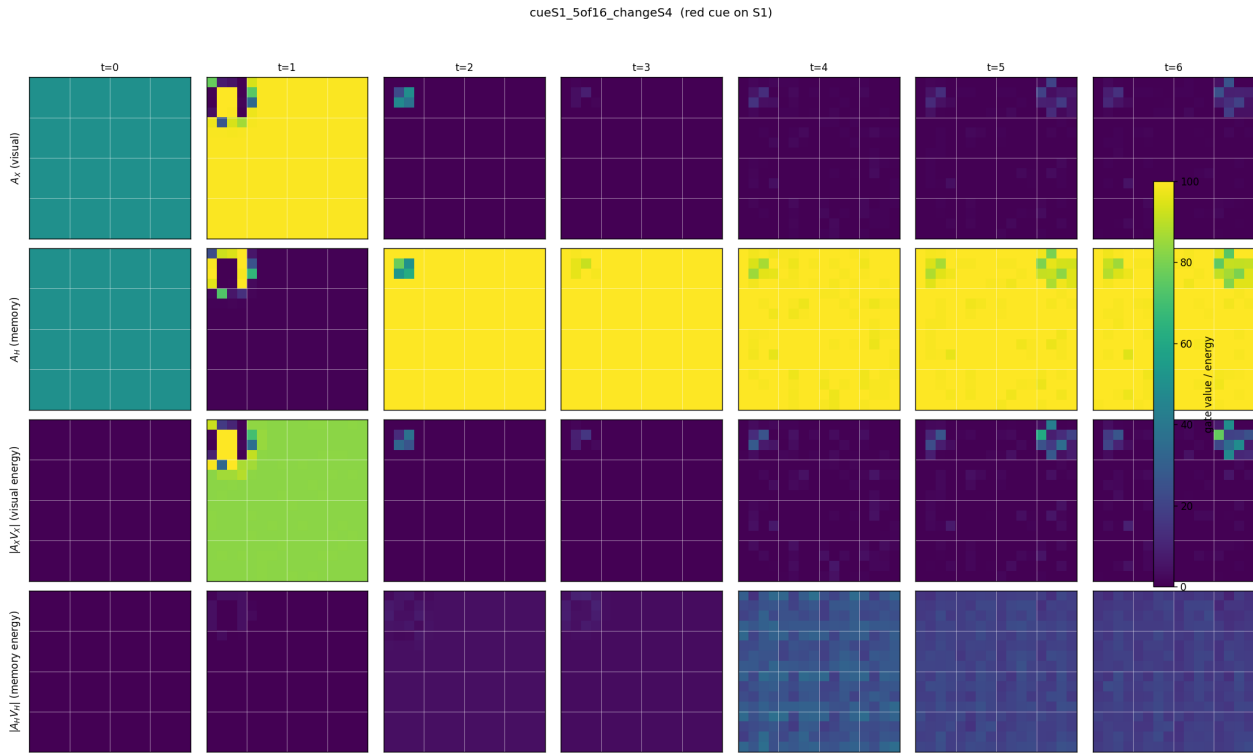


Figure 45: cueS1_5of16_changeS4 — vis layer.

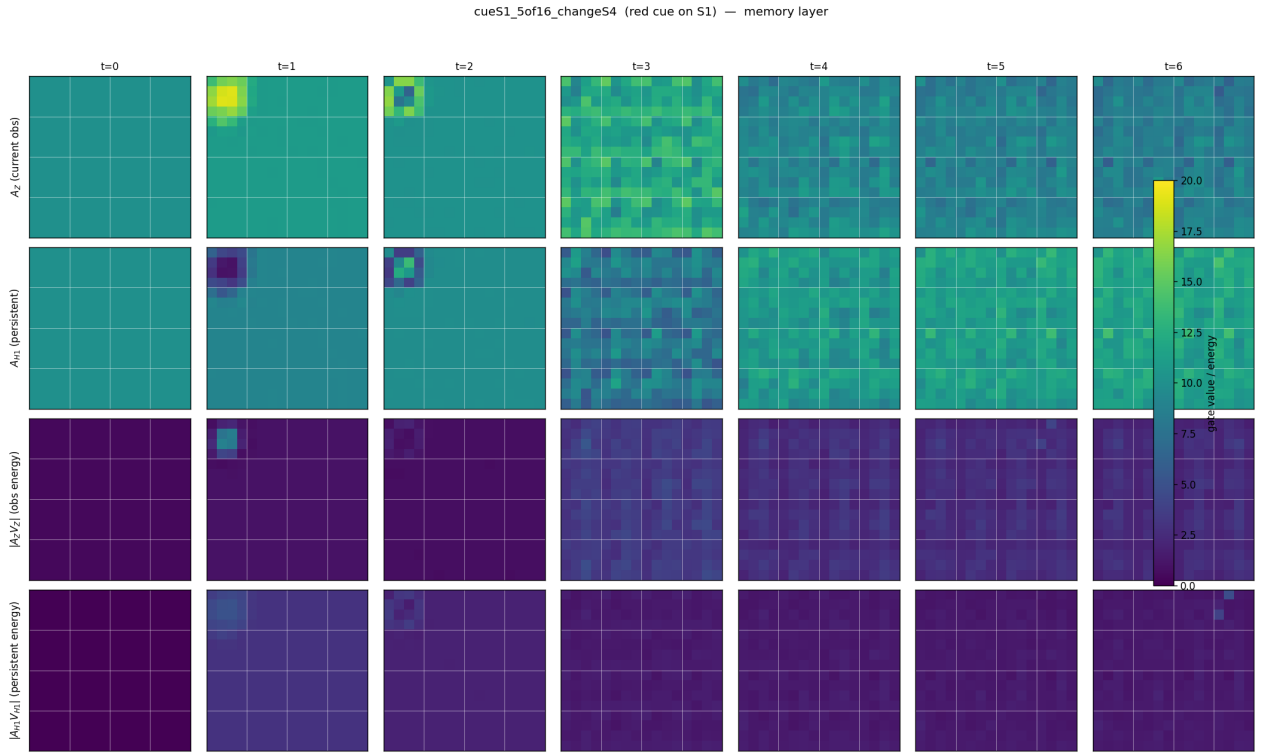


Figure 46: cueS1_5of16_changeS4 — mem layer.

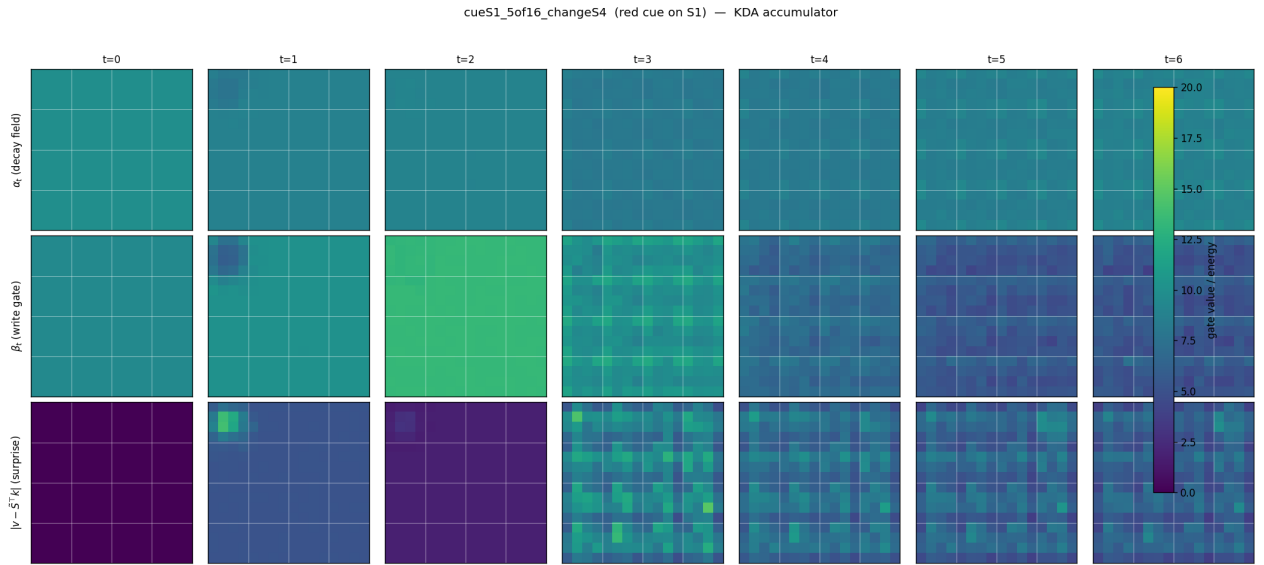


Figure 47: cueS1_5of16_changeS4 — acc layer.

3.16 cueS1_6of16_nochange

Cue fixed on cell S1 (red), ring completeness 6/16 ($p(\text{change at cued cell} \mid \text{change}) = 6/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.64\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.01\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.26 ; spatially flat — localization lives in the error term, not the gates.

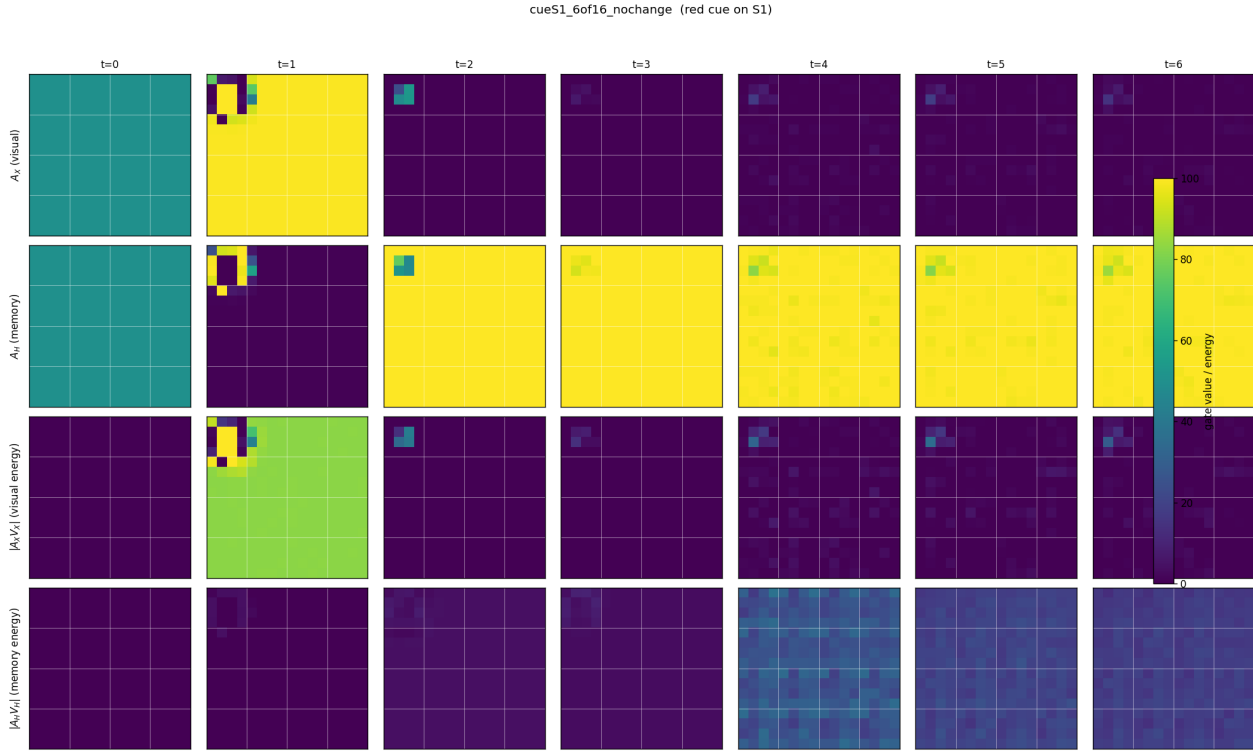


Figure 48: cueS1_6of16_nochange — vis layer.

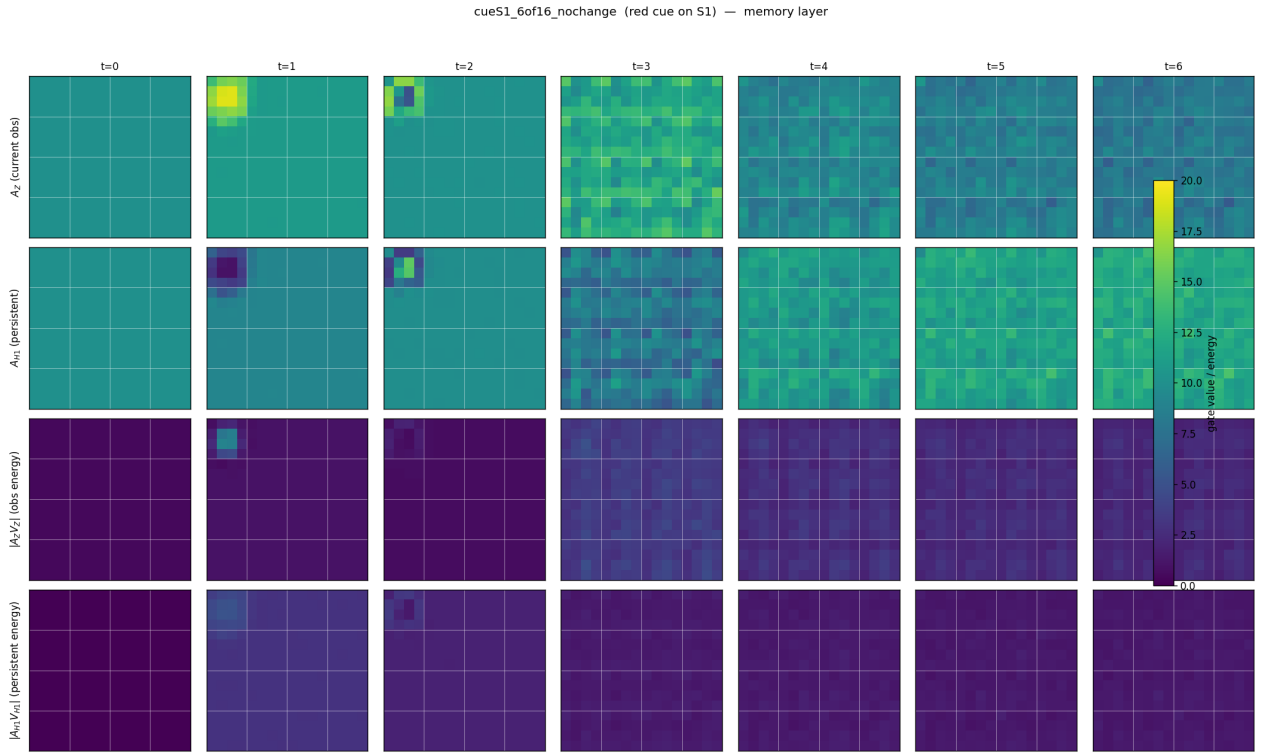


Figure 49: cueS1_6of16_nochange — mem layer.



Figure 50: cueS1_6of16_nochange — acc layer.

3.17 cueS1_6of16_changeS1

Cue fixed on cell S1 (red), ring completeness 6/16 ($p(\text{change at cued cell} \mid \text{change}) = 6/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.64\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.24\times$ the global mean ($t = 6$: $1.24\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.44 vs A_{H1} (persistence) 0.56. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 51: cueS1_6of16_changeS1 — vis layer.

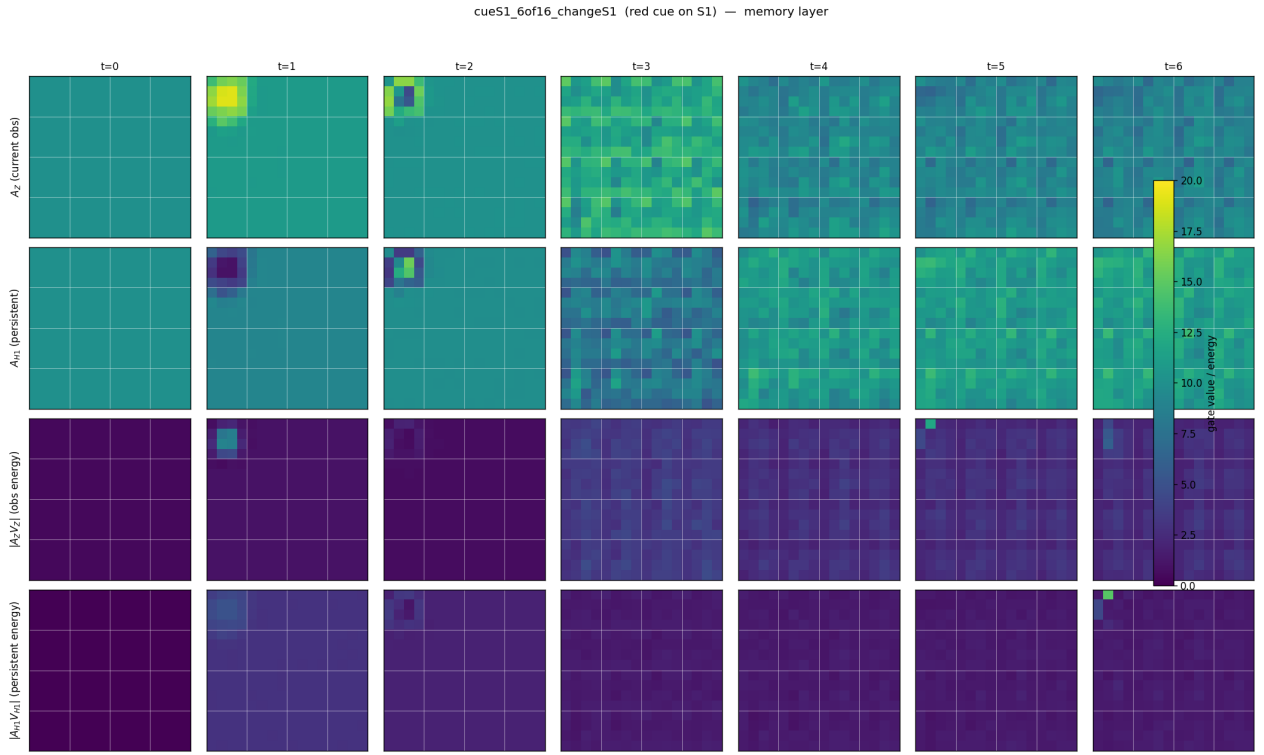


Figure 52: cueS1_6of16_changeS1 — mem layer.



Figure 53: cueS1_6of16_changeS1 — acc layer.

3.18 cueS1_6of16_changeS4

Cue fixed on cell S1 (red), ring completeness 6/16 ($p(\text{change at cued cell} \mid \text{change}) = 6/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.64\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.01\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

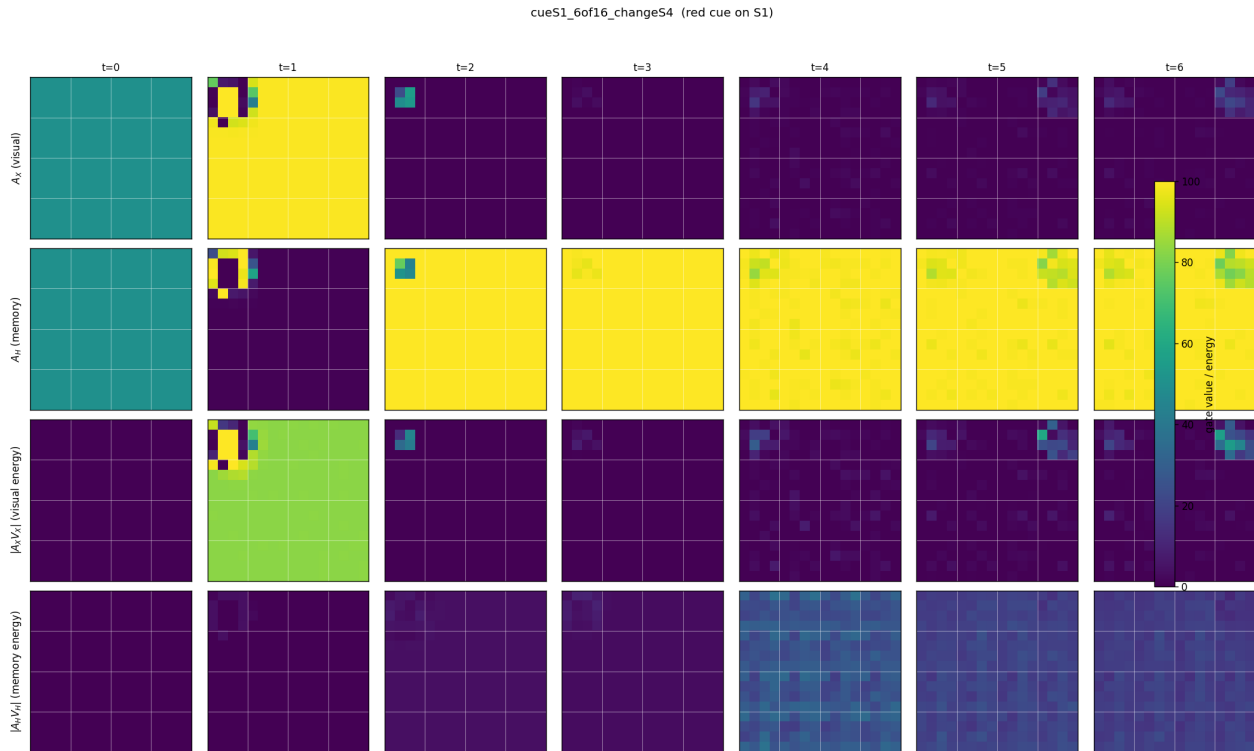


Figure 54: cueS1_6of16_changeS4 — vis layer.

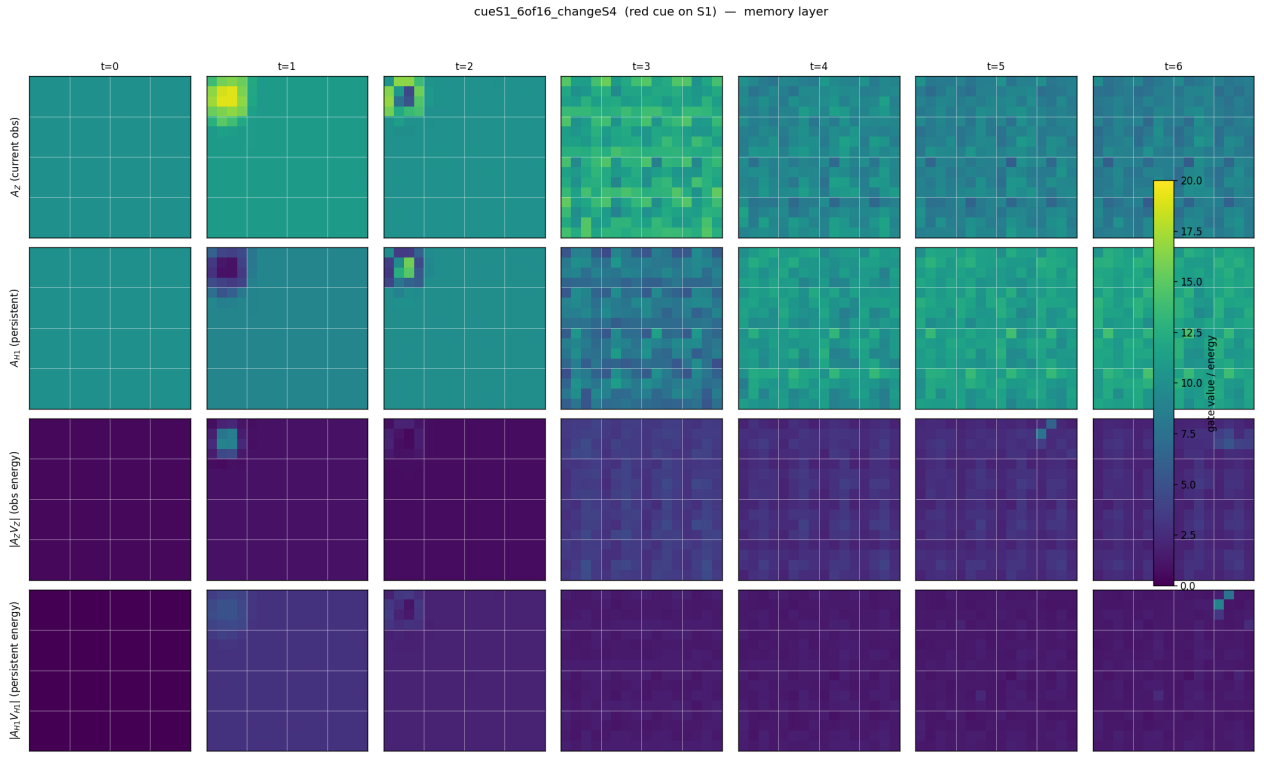


Figure 55: cueS1_6of16_changeS4 — mem layer.

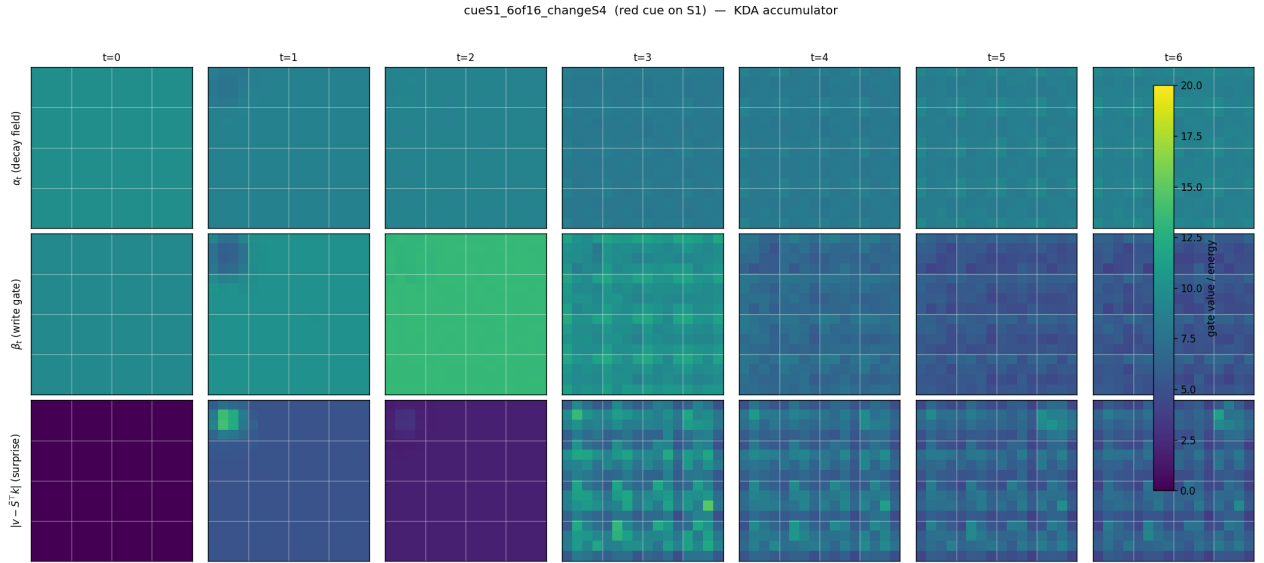


Figure 56: cueS1_6of16_changeS4 — acc layer.

3.19 cueS1_7of16_nochange

Cue fixed on cell S1 (red), ring completeness 7/16 ($p(\text{change at cued cell} \mid \text{change}) = 7/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.64\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.00\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.26 ; spatially flat — localization lives in the error term, not the gates.

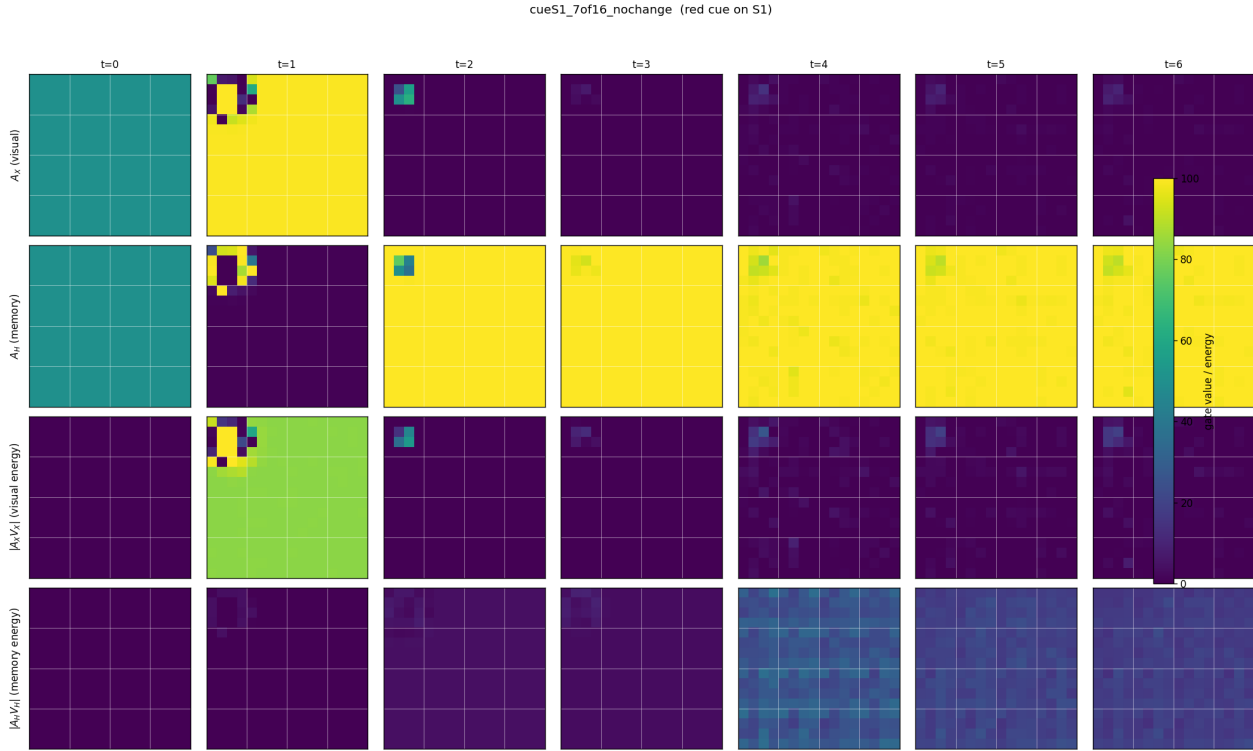


Figure 57: cueS1_7of16_nochange — vis layer.

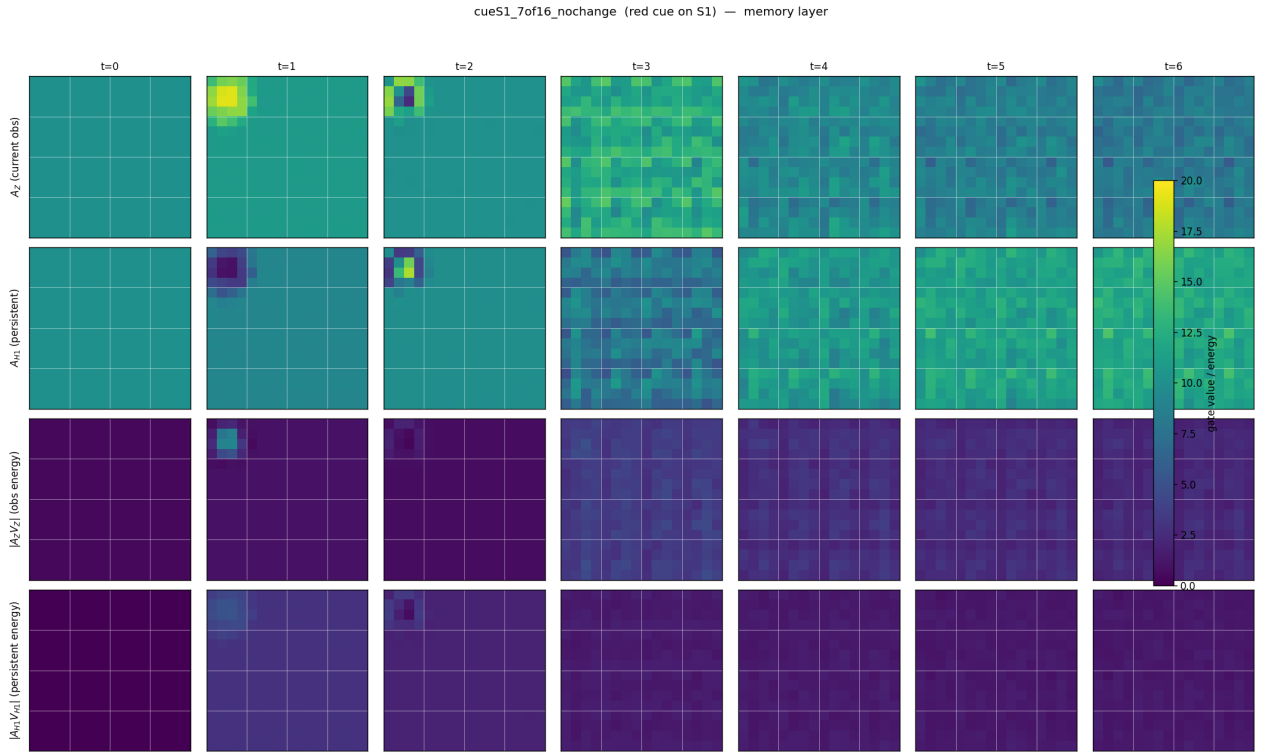


Figure 58: cueS1_7of16_nochange — mem layer.

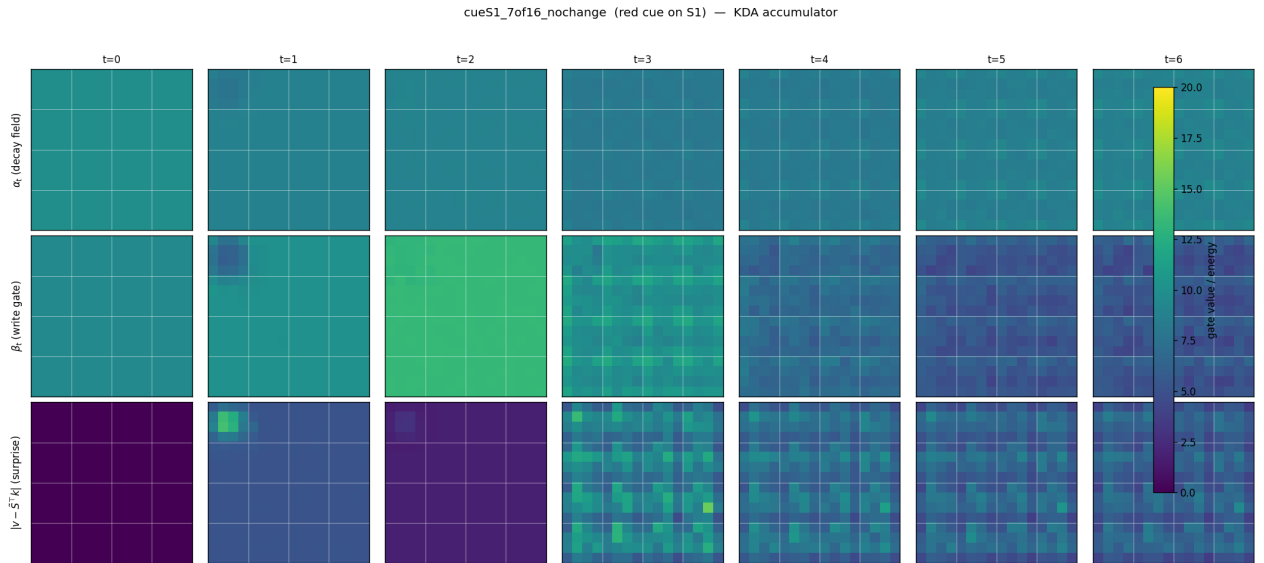


Figure 59: cueS1_7of16_nochange — acc layer.

3.20 cueS1_7of16_changeS1

Cue fixed on cell S1 (red), ring completeness 7/16 ($p(\text{change at cued cell} \mid \text{change}) = 7/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.64\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.18\times$ the global mean ($t = 6$: $1.18\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.44 vs A_{H1} (persistence) 0.56. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 60: cueS1_7of16_changeS1 — vis layer.

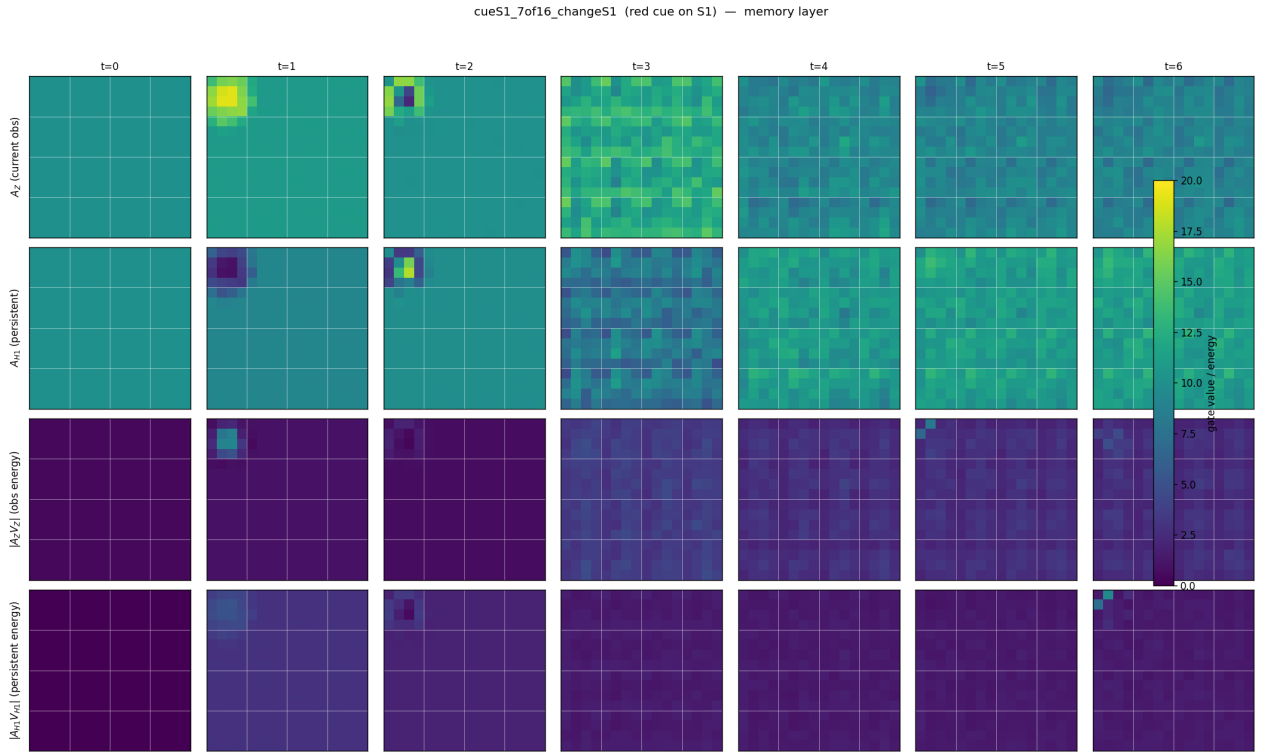


Figure 61: cueS1_7of16_changeS1 — mem layer.

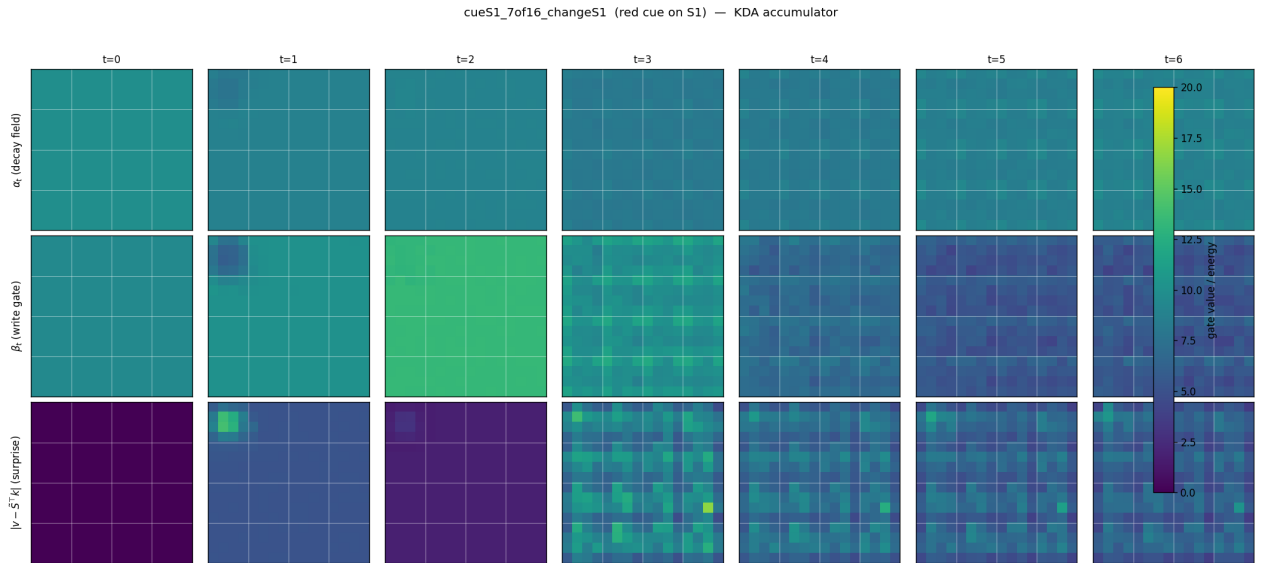


Figure 62: cueS1_7of16_changeS1 — acc layer.

3.21 cueS1_7of16_changeS4

Cue fixed on cell S1 (red), ring completeness 7/16 ($p(\text{change at cued cell} \mid \text{change}) = 7/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.64\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.00\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.43 vs A_{H1} (persistence) 0.57. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

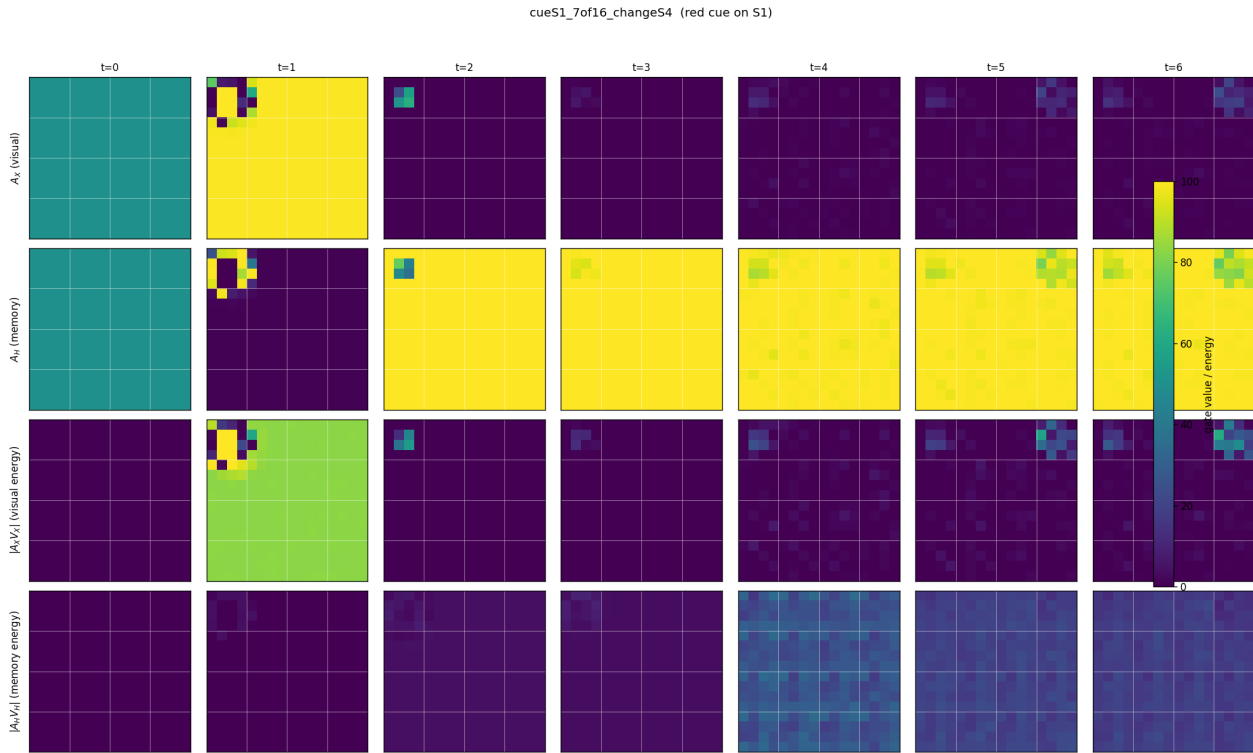


Figure 63: cueS1_7of16_changeS4 — vis layer.

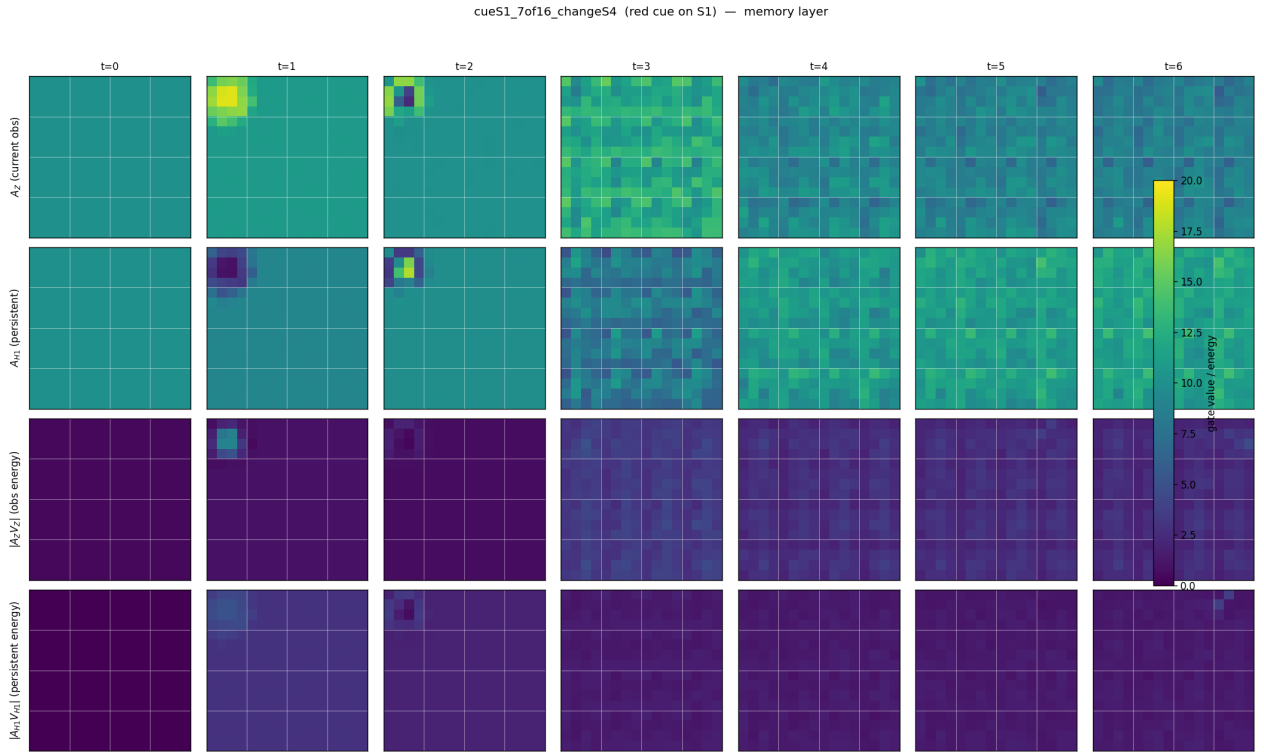


Figure 64: cueS1_7of16_changeS4 — mem layer.

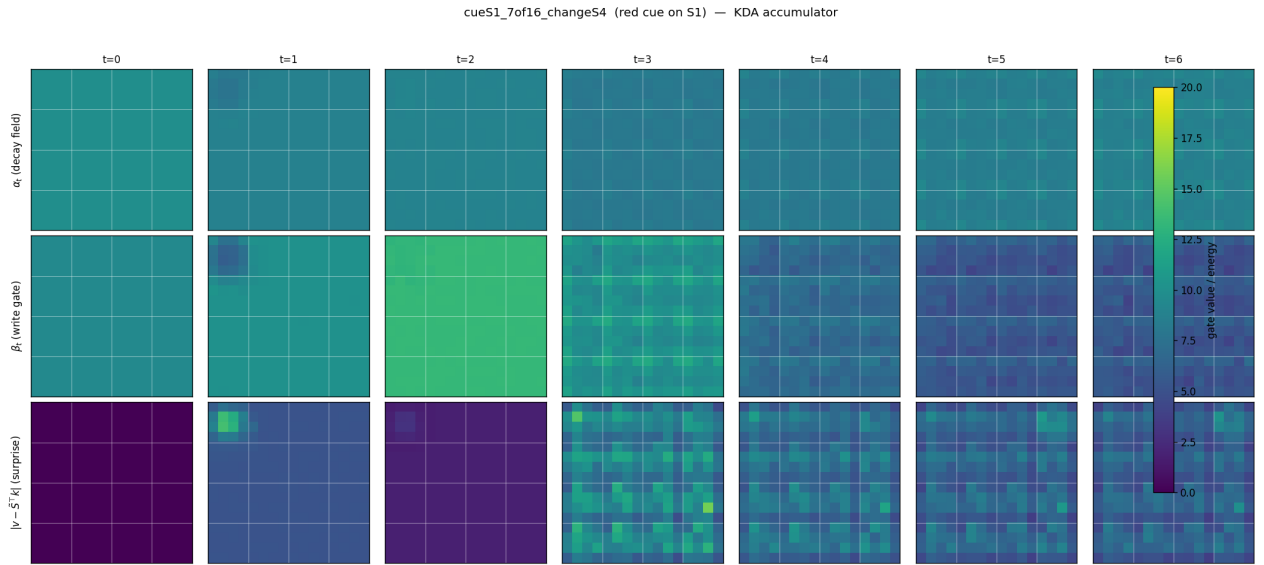


Figure 65: cueS1_7of16_changeS4 — acc layer.

3.22 cueS1_8of16_nochange

Cue fixed on cell S1 (red), ring completeness 8/16 ($p(\text{change at cued cell} \mid \text{change}) = 8/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.65\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.00\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.

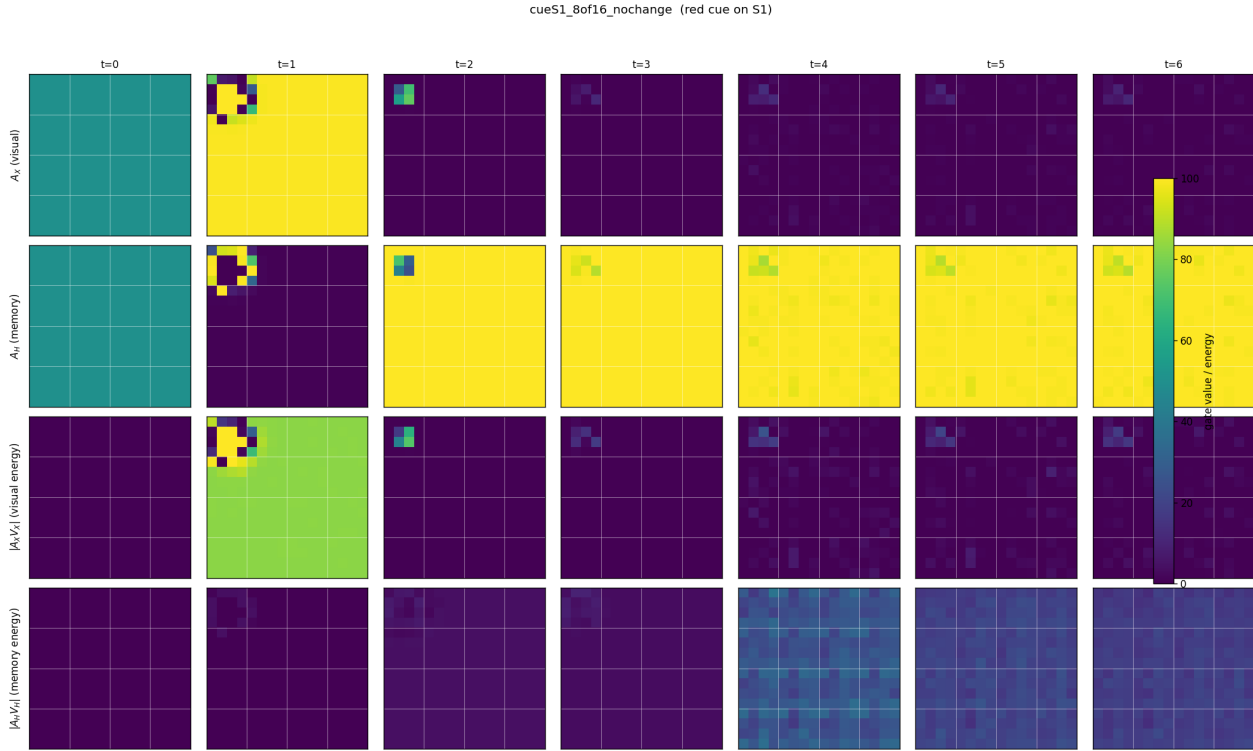


Figure 66: cueS1_8of16_nochange — vis layer.

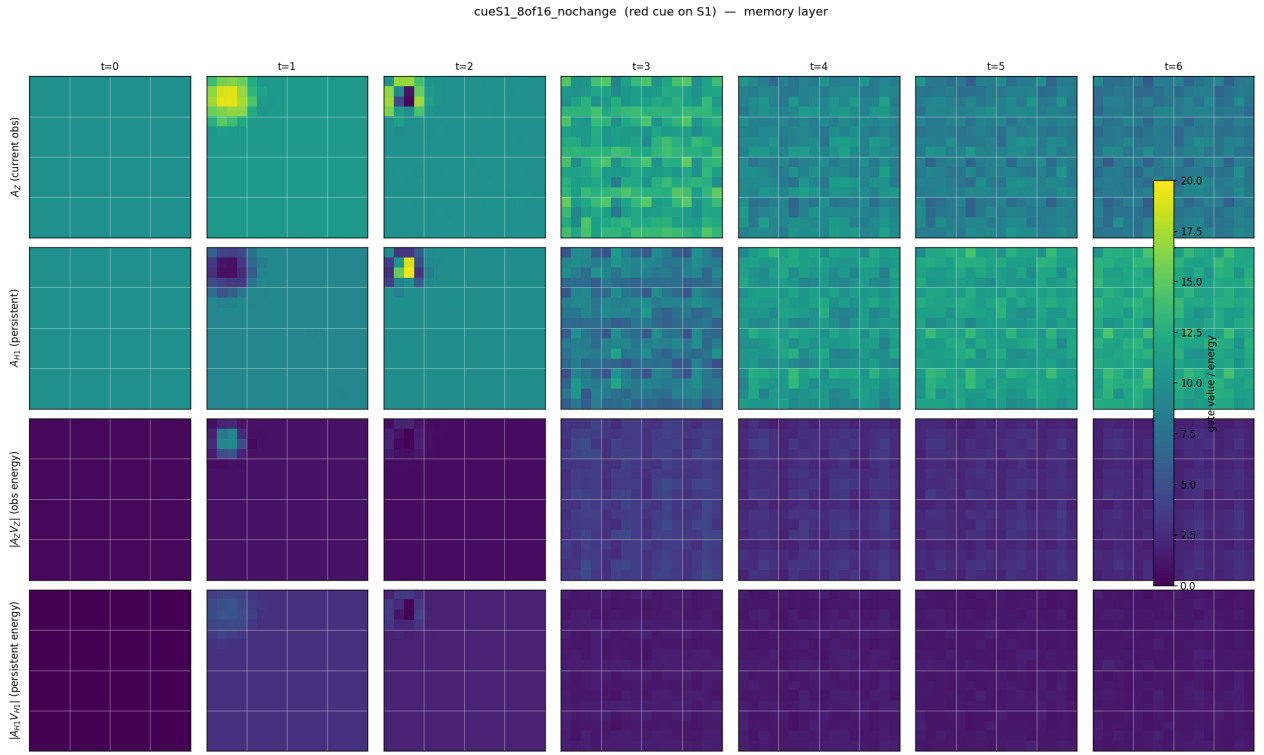


Figure 67: cueS1_8of16_nochange — mem layer.



Figure 68: cueS1_8of16_nochange — acc layer.

3.23 cueS1_8of16_changeS1

Cue fixed on cell S1 (red), ring completeness 8/16 ($p(\text{change at cued cell} \mid \text{change}) = 8/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.65\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.22\times$ the global mean ($t = 6$: $1.22\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.45 vs A_{H1} (persistence) 0.55. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.

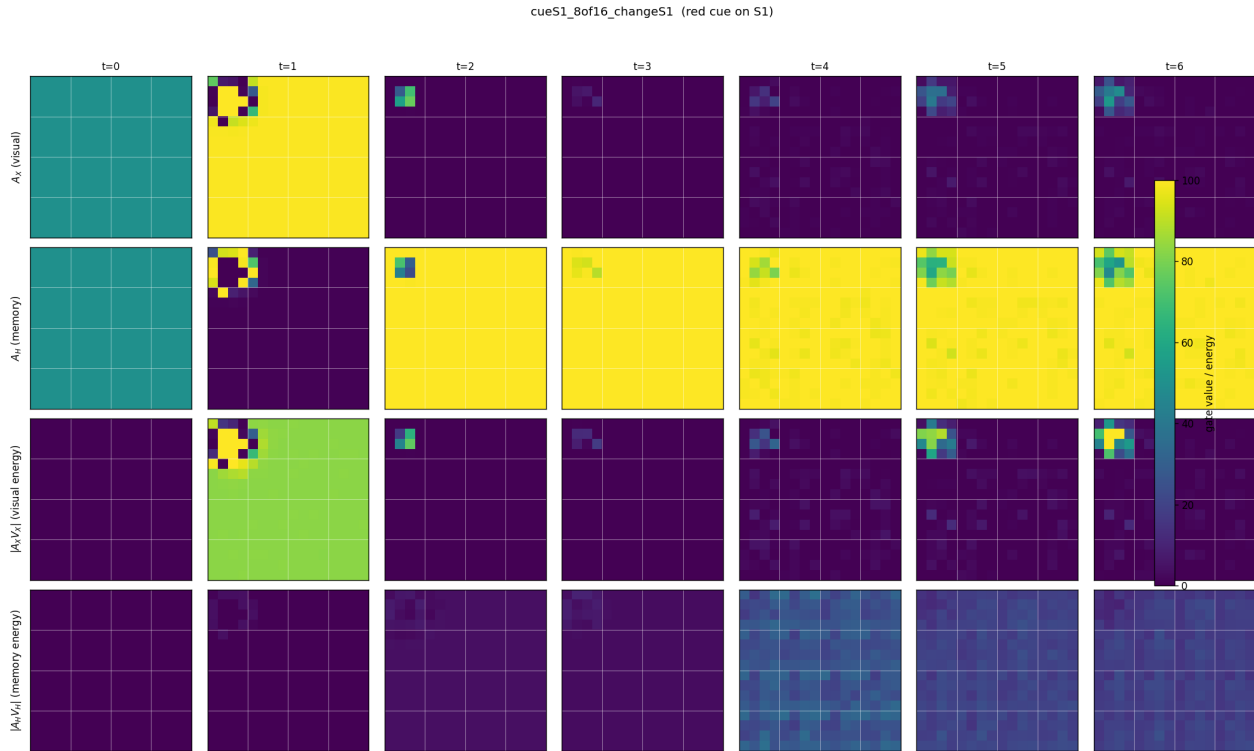


Figure 69: cueS1_8of16_changeS1 — vis layer.

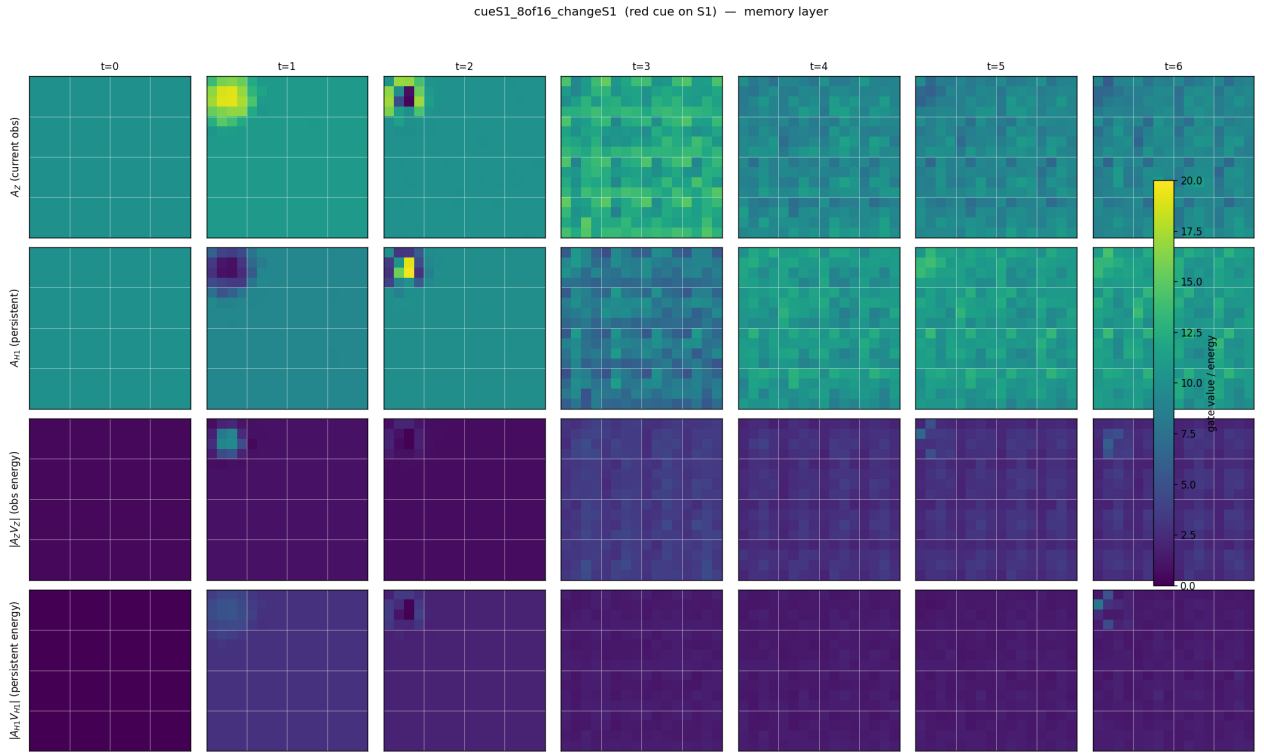


Figure 70: cueS1_8of16_changeS1 — mem layer.



Figure 71: cueS1_8of16_changeS1 — acc layer.

3.24 cueS1_8of16_changeS4

Cue fixed on cell S1 (red), ring completeness 8/16 ($p(\text{change at cued cell} \mid \text{change}) = 8/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.65\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.01\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.



Figure 72: cueS1_8of16_changeS4 — vis layer.

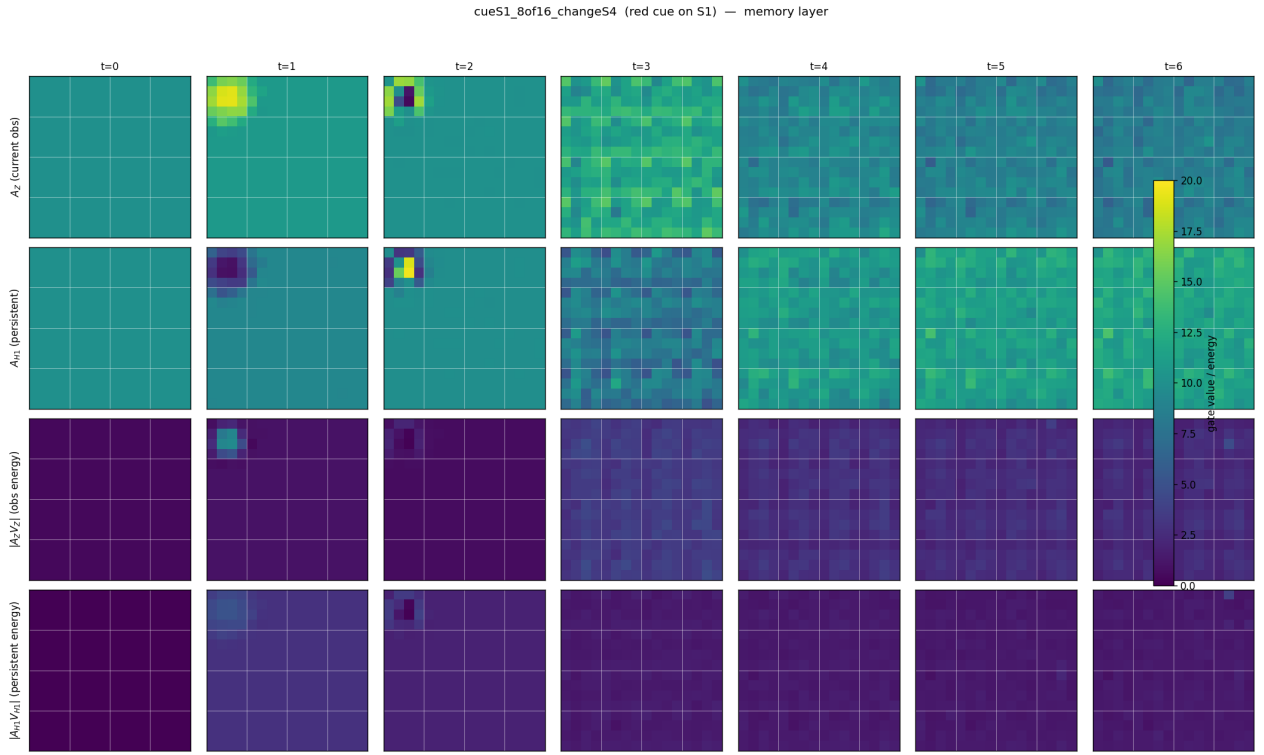


Figure 73: cueS1_8of16_changeS4 — mem layer.

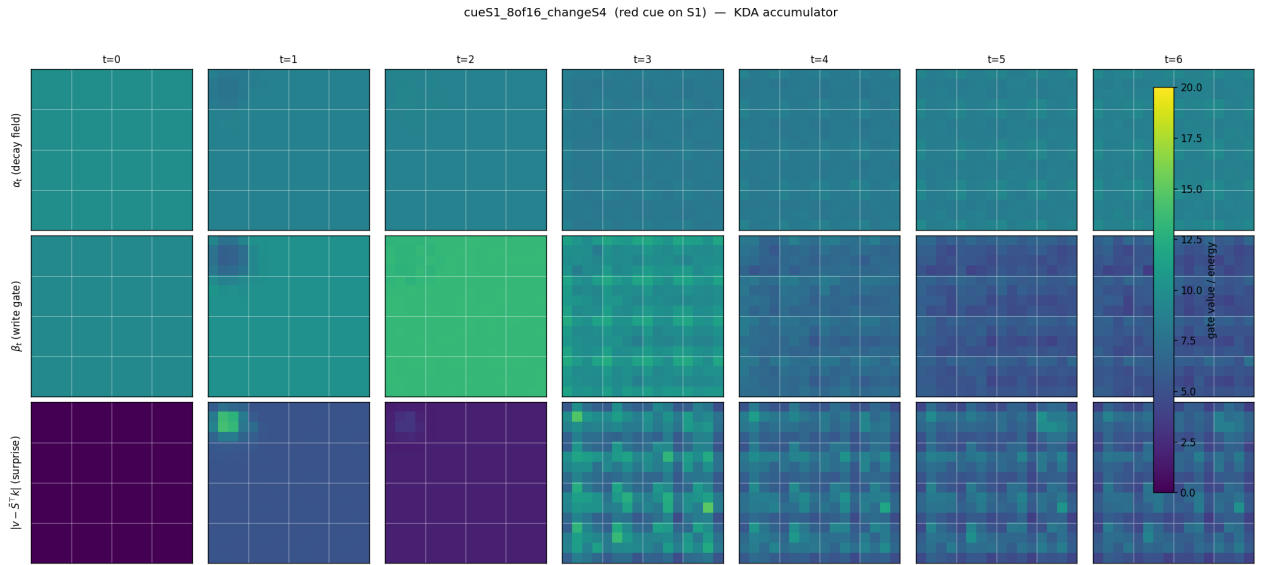


Figure 74: cueS1_8of16_changeS4 — acc layer.

3.25 cueS1_9of16_nochange

Cue fixed on cell S1 (red), ring completeness 9/16 ($p(\text{change at cued cell} \mid \text{change}) = 9/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.65\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.00\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.



Figure 75: cueS1_9of16_nochange — vis layer.

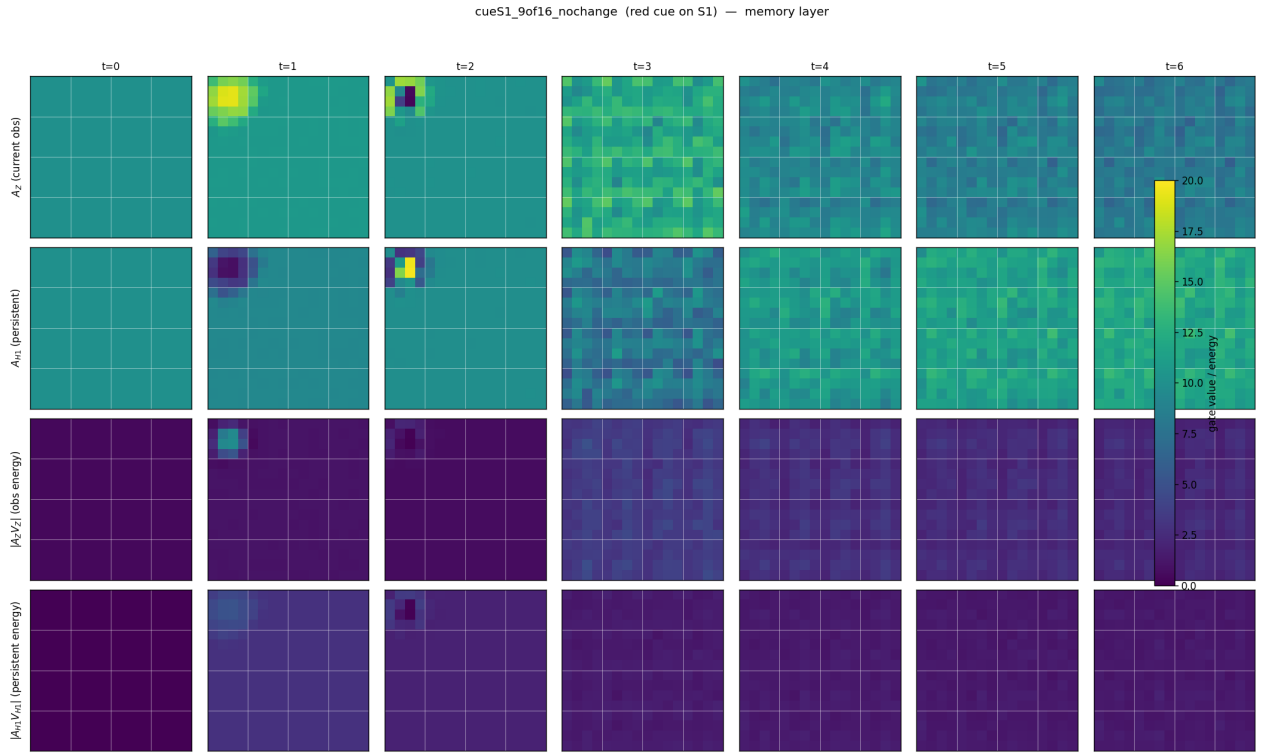


Figure 76: cueS1_9of16_nochange — mem layer.



Figure 77: cueS1_9of16_nochange — acc layer.

3.26 cueS1_9of16_changeS1

Cue fixed on cell S1 (red), ring completeness 9/16 ($p(\text{change at cued cell} \mid \text{change}) = 9/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.65\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.19\times$ the global mean ($t = 6$: $1.23\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.45 vs A_{H1} (persistence) 0.55. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.

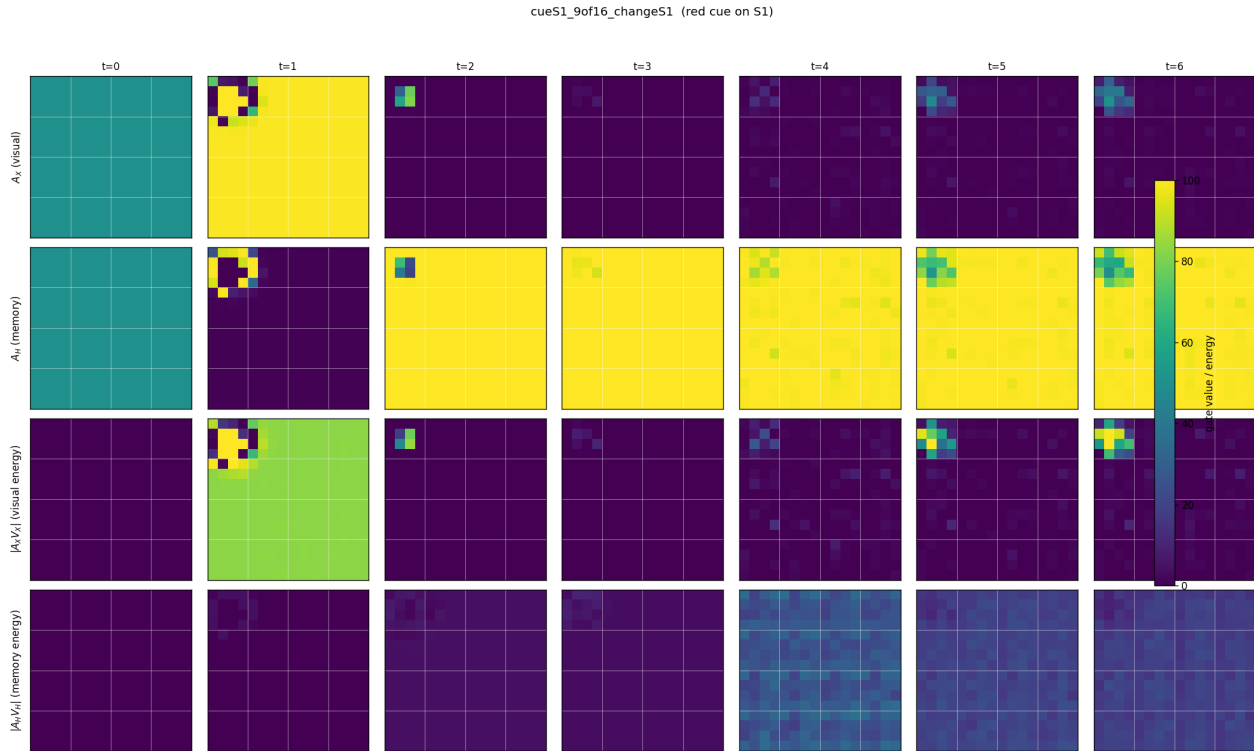


Figure 78: cueS1_9of16_changeS1 — vis layer.

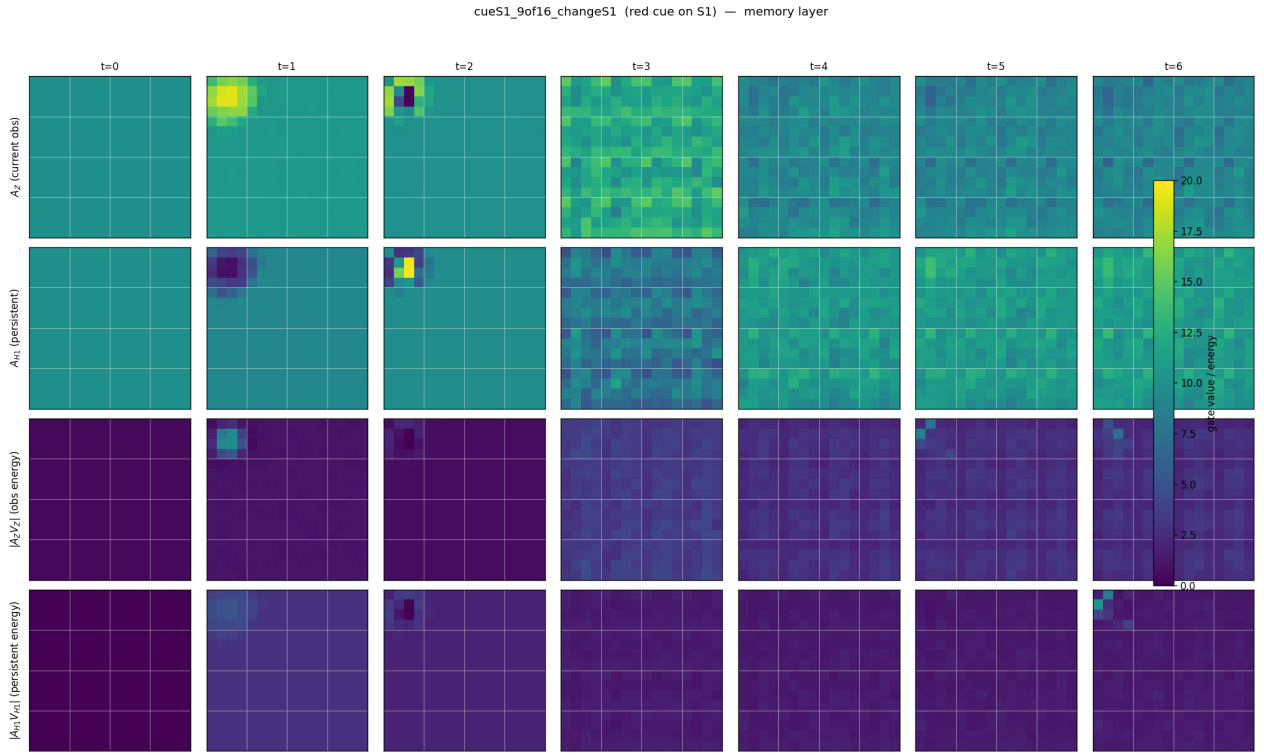


Figure 79: cueS1_9of16_changeS1 — mem layer.



Figure 80: cueS1_9of16_changeS1 — acc layer.

3.27 cueS1_9of16_changeS4

Cue fixed on cell S1 (red), ring completeness 9/16 ($p(\text{change at cued cell} \mid \text{change}) = 9/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.65\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.00\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.43 vs A_{H1} (persistence) 0.57. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

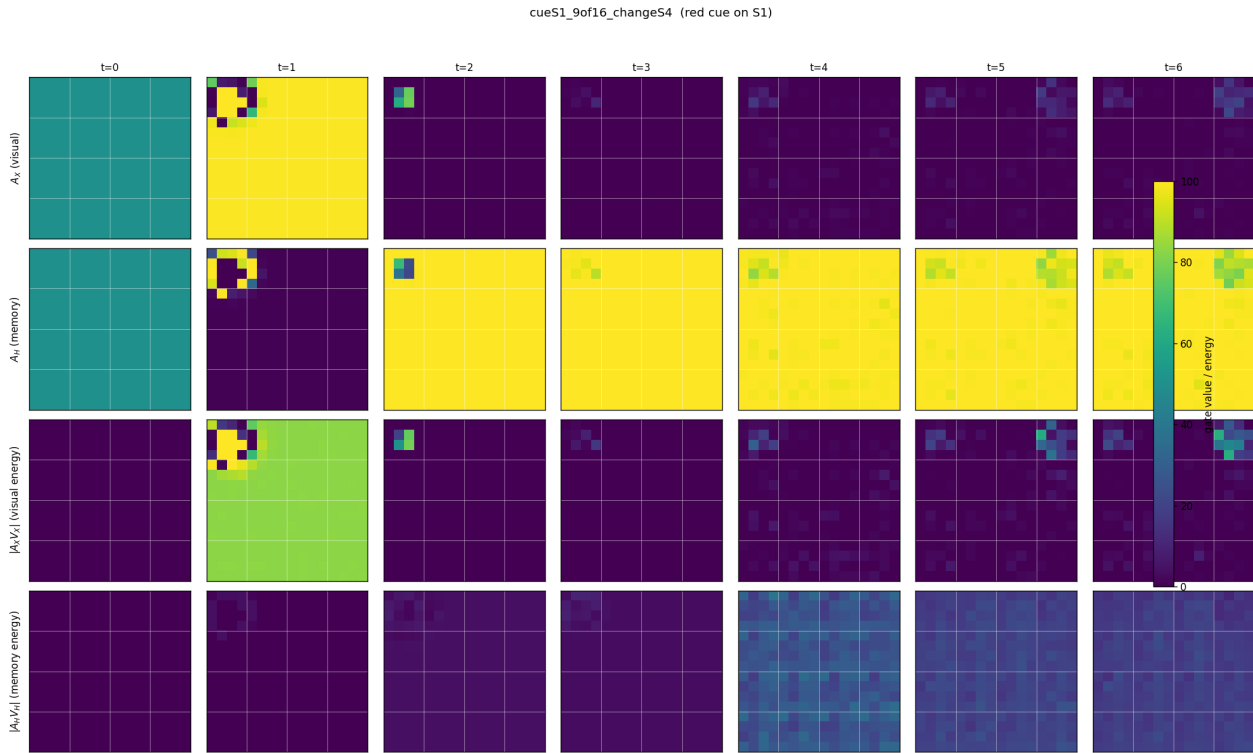


Figure 81: cueS1_9of16_changeS4 — vis layer.

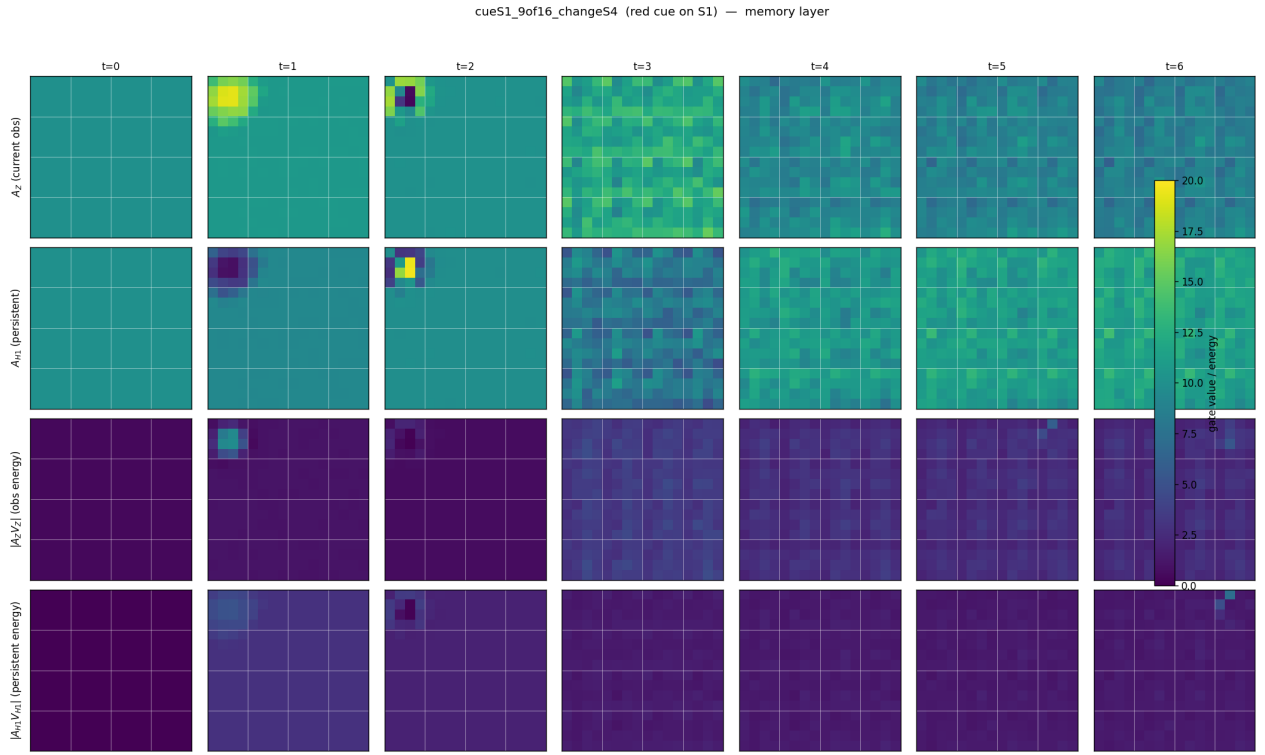


Figure 82: cueS1_9of16_changeS4 — mem layer.



Figure 83: cueS1_9of16_changeS4 — acc layer.

3.28 cueS1_10of16_nochange

Cue fixed on cell S1 (red), ring completeness 10/16 ($p(\text{change at cued cell} \mid \text{change}) = 10/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.67\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $0.98\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.41 vs A_{H1} (persistence) 0.59 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.

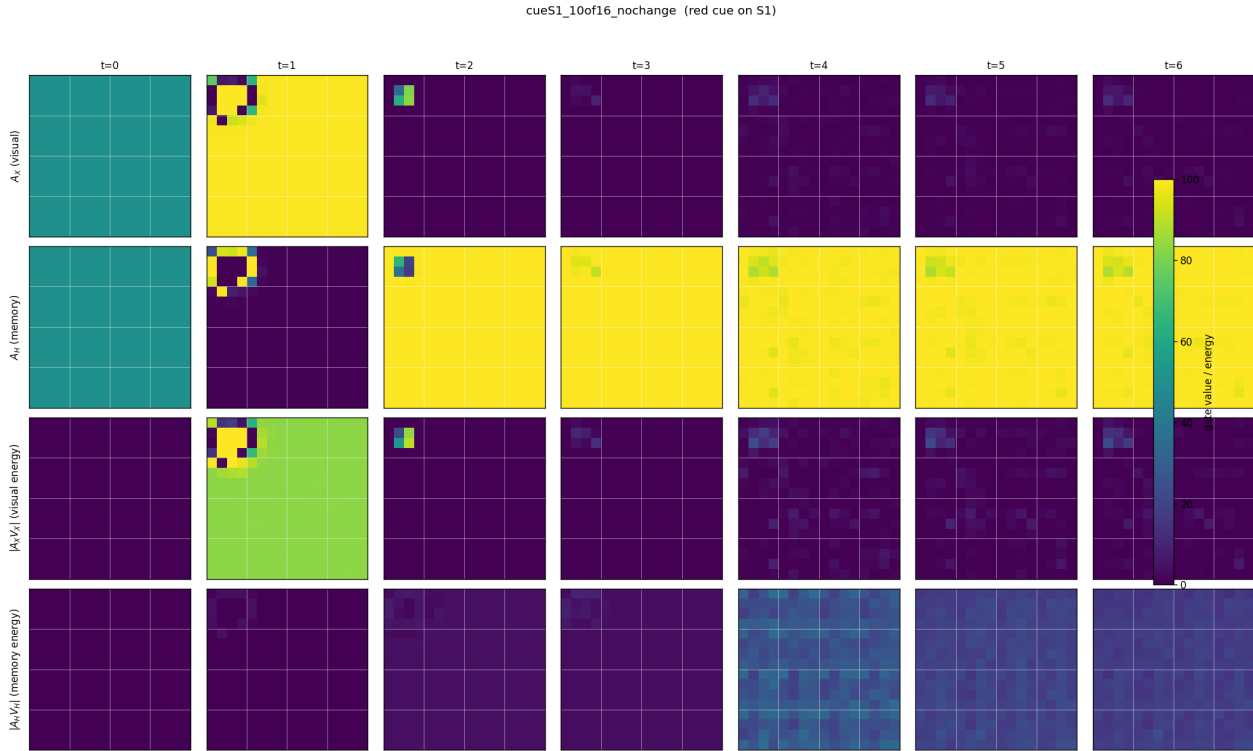


Figure 84: cueS1_10of16_nochange — vis layer.

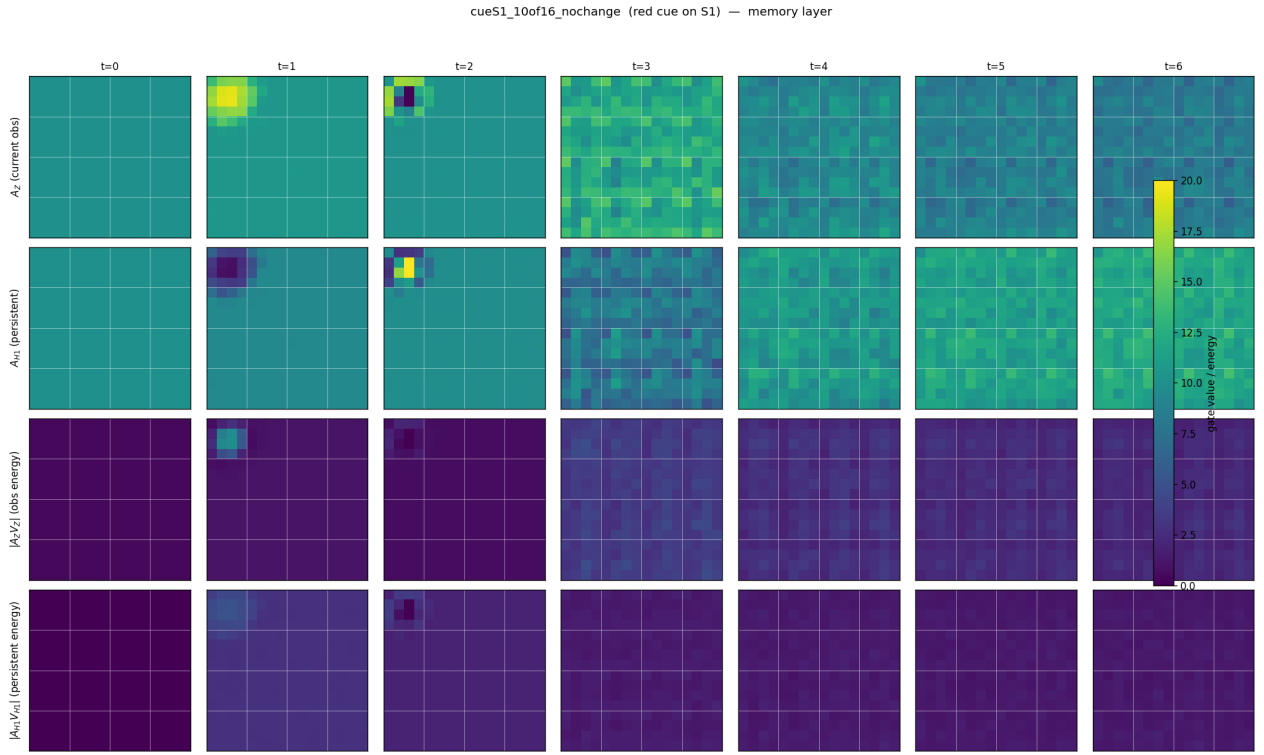


Figure 85: cueS1_10of16_nochange — mem layer.



Figure 86: cueS1_10of16_nochange — acc layer.

3.29 cueS1_10of16_changeS1

Cue fixed on cell S1 (red), ring completeness 10/16 ($p(\text{change at cued cell} \mid \text{change}) = 10/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.67\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.22\times$ the global mean ($t = 6$: $1.24\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.45 vs A_{H1} (persistence) 0.55. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 87: cueS1_10of16_changeS1 — vis layer.

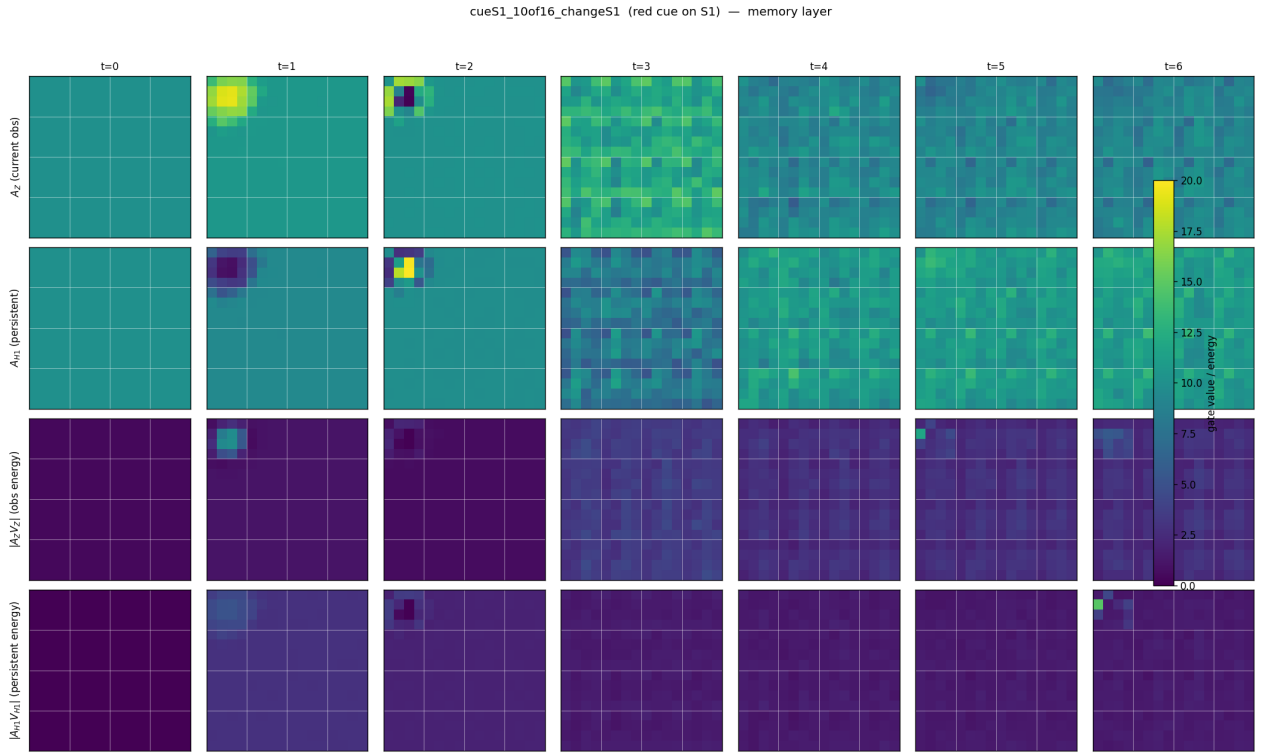


Figure 88: cueS1_10of16_changeS1 — mem layer.

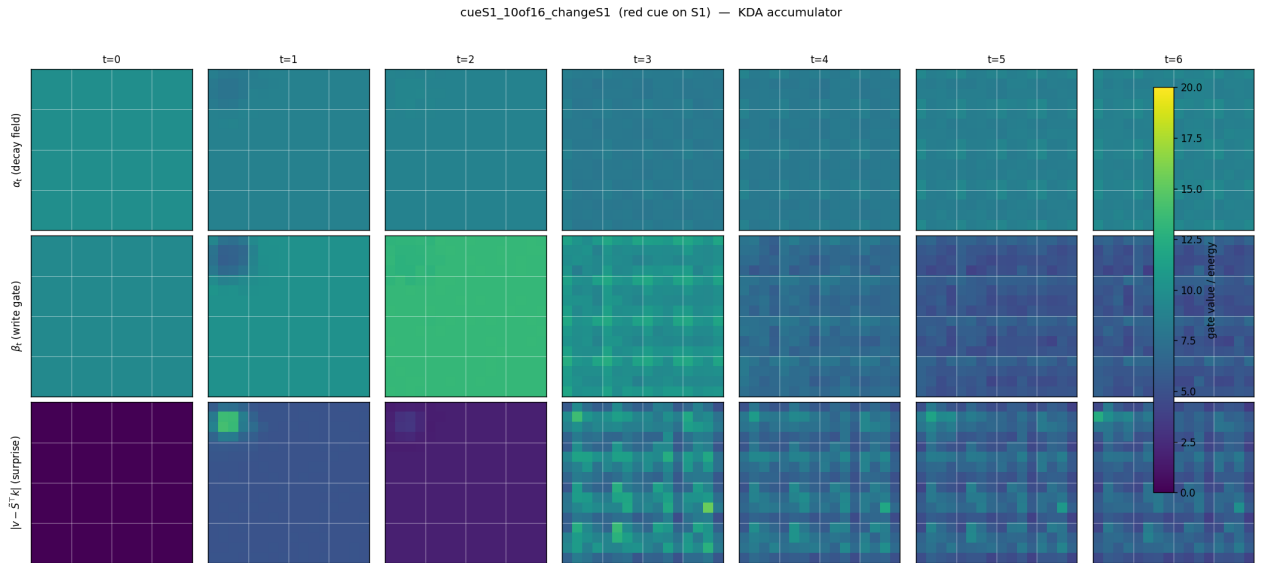


Figure 89: cueS1_10of16_changeS1 — acc layer.

3.30 cueS1_10of16_changeS4

Cue fixed on cell S1 (red), ring completeness 10/16 ($p(\text{change at cued cell} \mid \text{change}) = 10/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.67\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.00\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

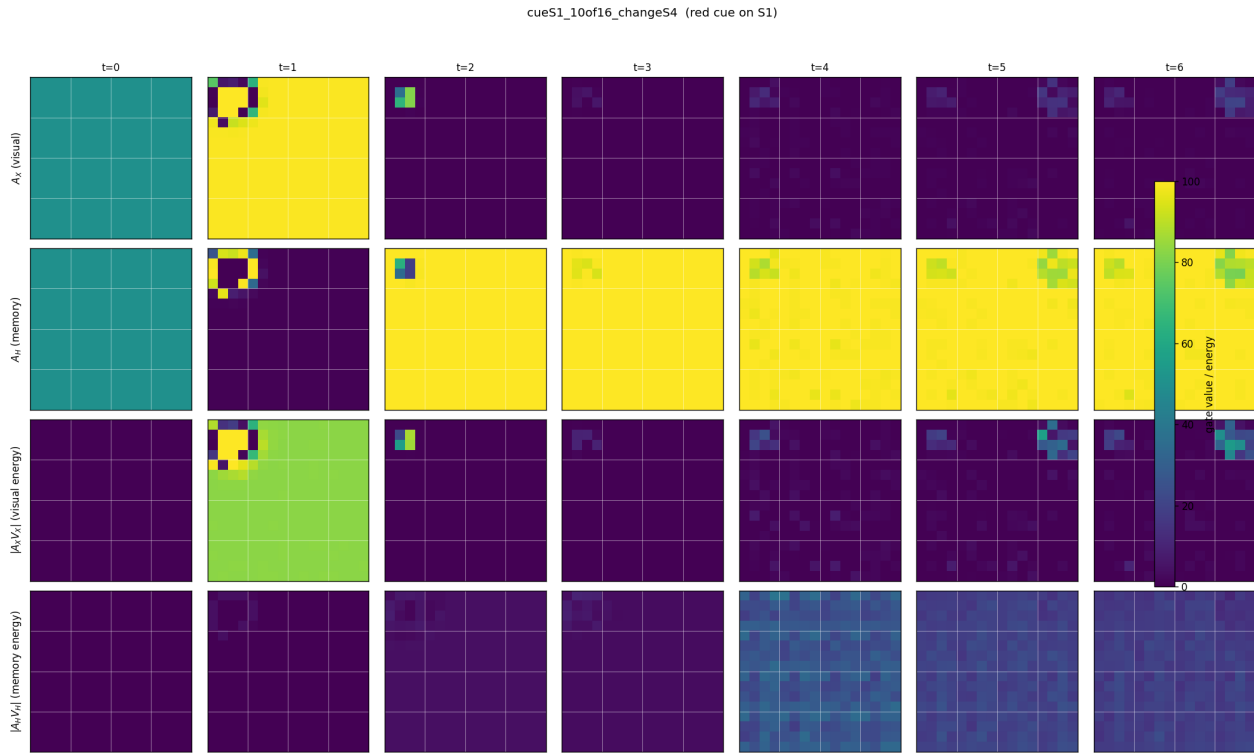


Figure 90: cueS1_10of16_changeS4 — vis layer.

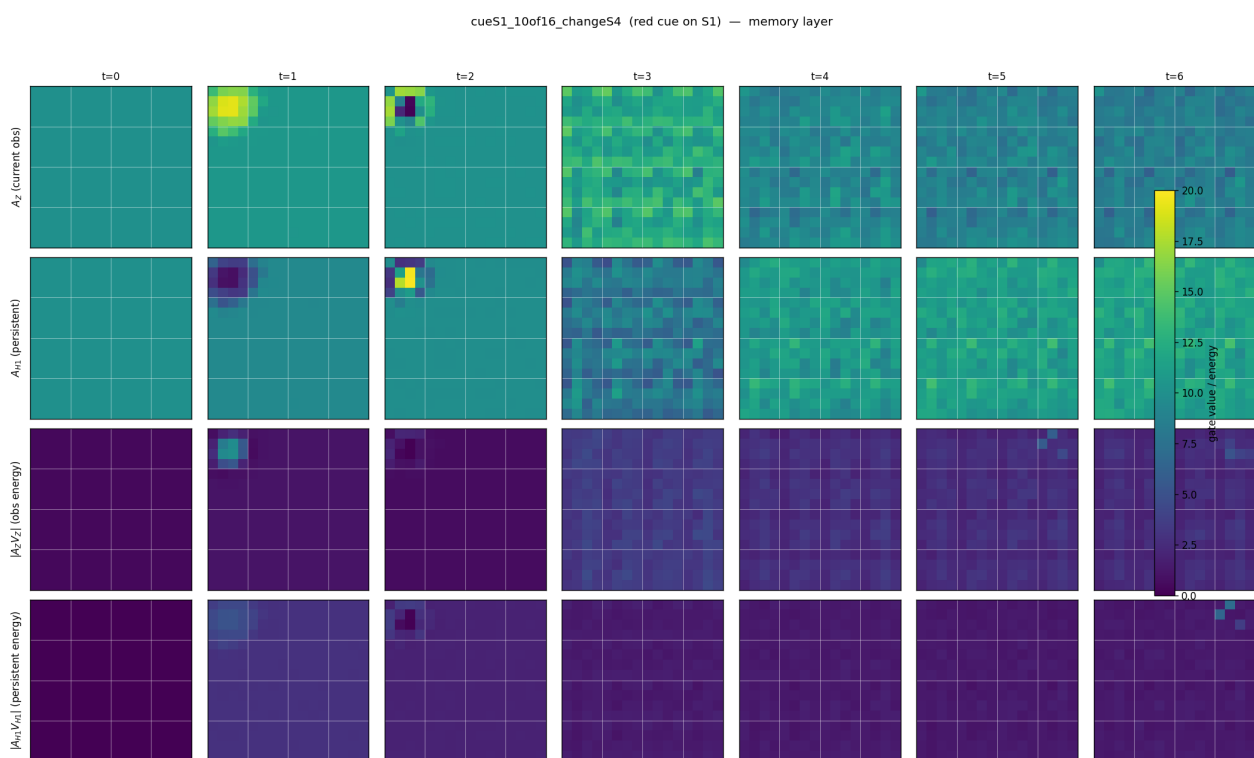


Figure 91: cueS1_10of16_changeS4 — mem layer.

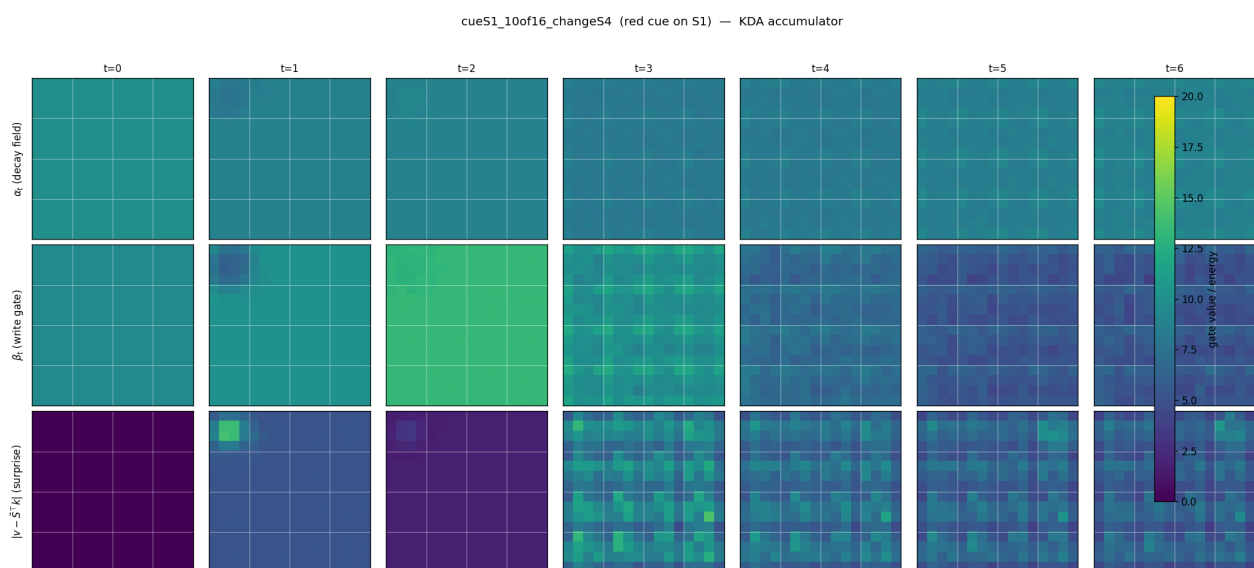


Figure 92: cueS1_10of16_changeS4 — acc layer.

3.31 cueS1_11of16_nochange

Cue fixed on cell S1 (red), ring completeness 11/16 ($p(\text{change at cued cell} \mid \text{change}) = 11/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.69\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.03\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.26 ; spatially flat — localization lives in the error term, not the gates.

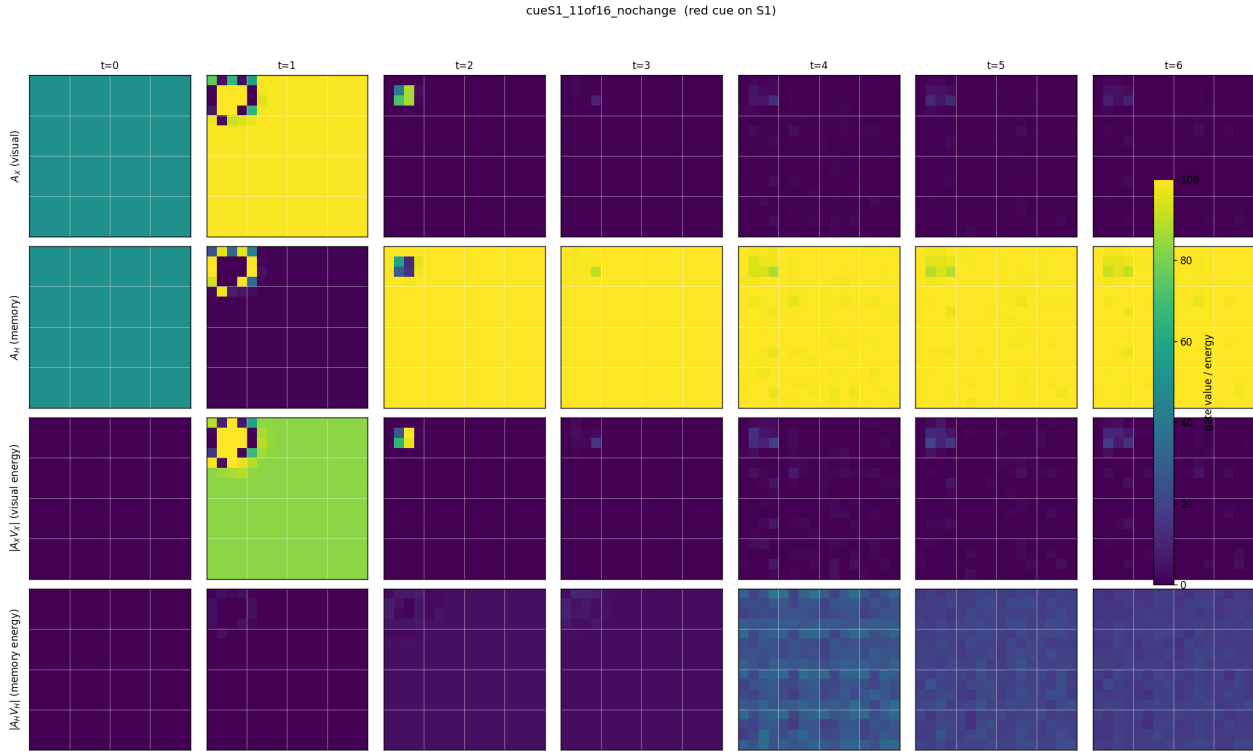


Figure 93: cueS1_11of16_nochange — vis layer.

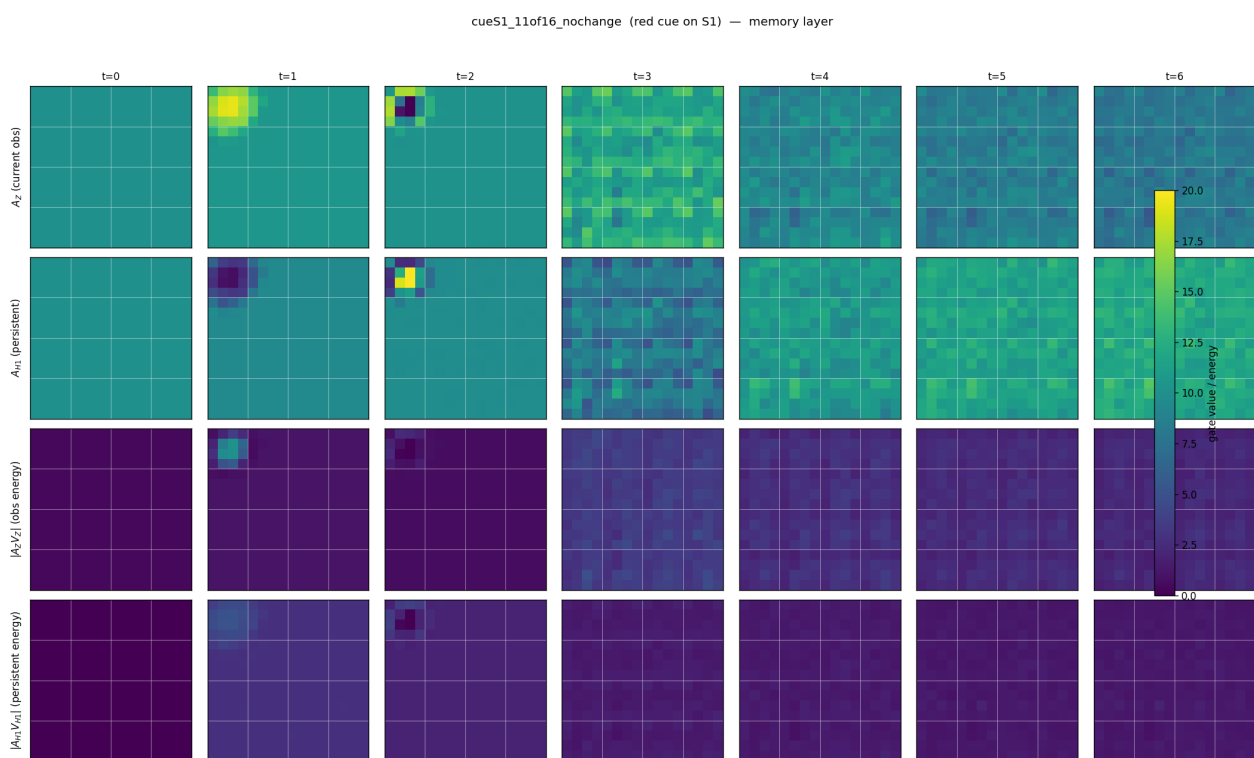


Figure 94: cueS1_11of16_nochange — mem layer.



Figure 95: cueS1_11of16_nochange — acc layer.

3.32 cueS1_11of16_changeS1

Cue fixed on cell S1 (red), ring completeness 11/16 ($p(\text{change at cued cell} \mid \text{change}) = 11/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.69\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.21\times$ the global mean ($t = 6$: $1.17\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.44 vs A_{H1} (persistence) 0.56. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 96: cueS1_11of16_changeS1 — vis layer.

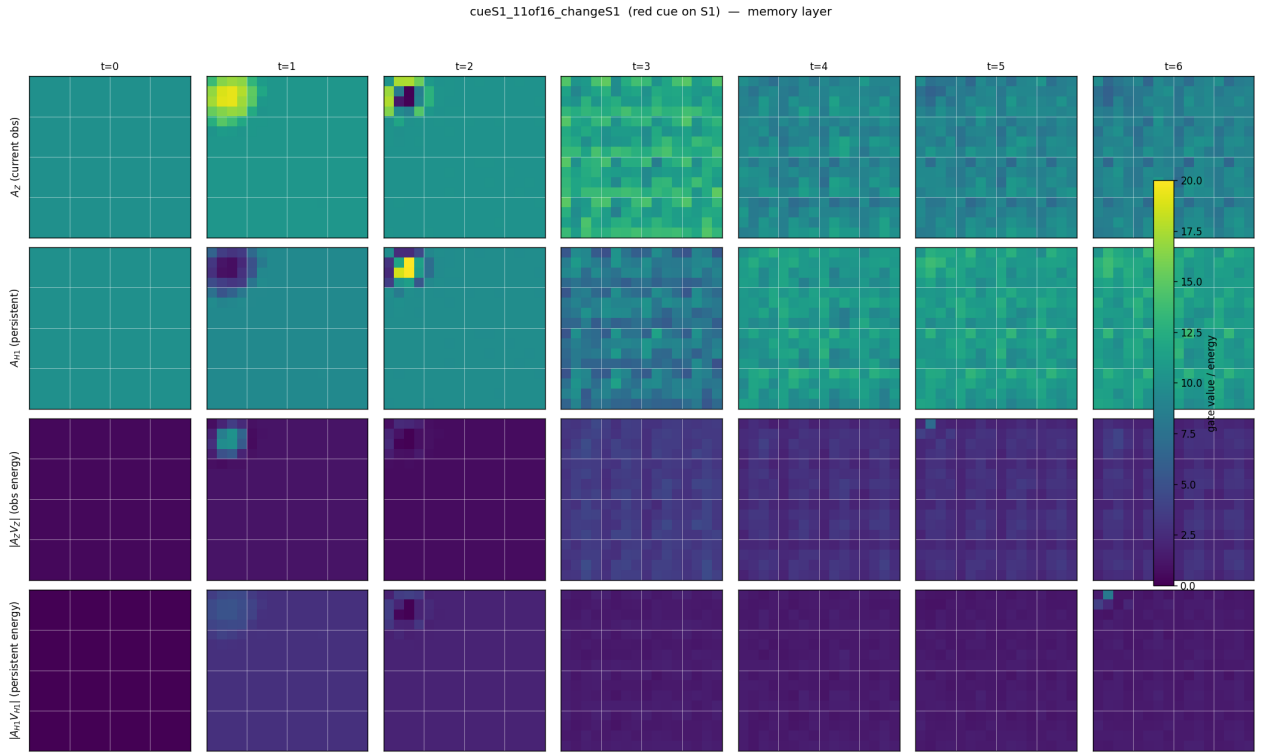


Figure 97: cueS1_11of16_changeS1 — mem layer.

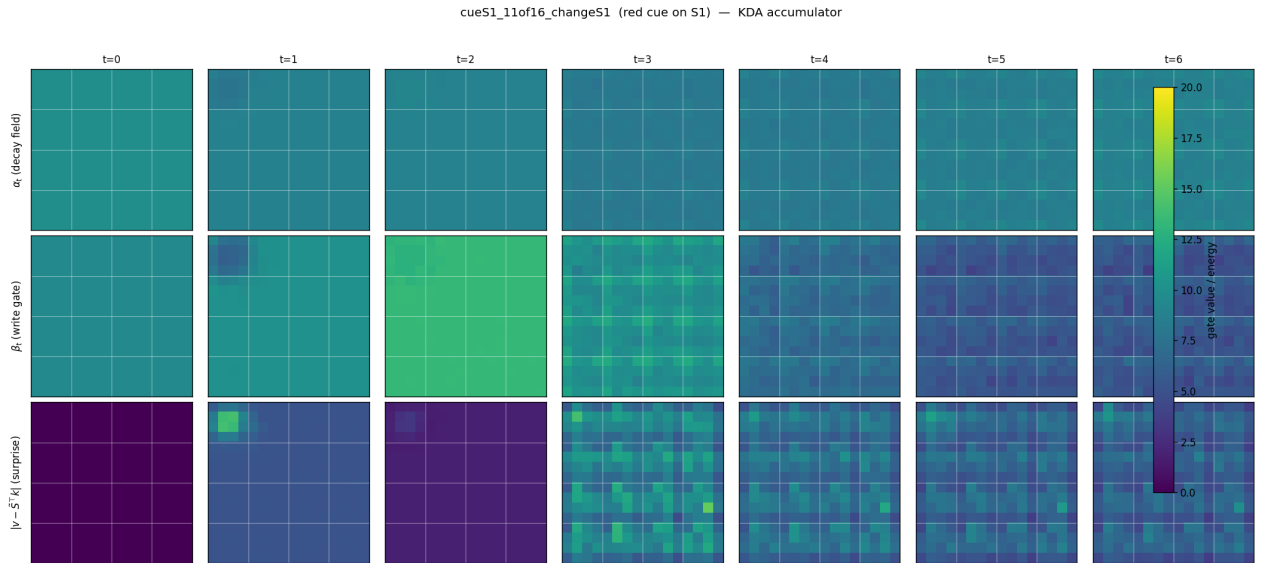


Figure 98: cueS1_11of16_changeS1 — acc layer.

3.33 cueS1_11of16_changeS4

Cue fixed on cell S1 (red), ring completeness 11/16 ($p(\text{change at cued cell} \mid \text{change}) = 11/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.69\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.00\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.



Figure 99: cueS1_11of16_changeS4 — vis layer.

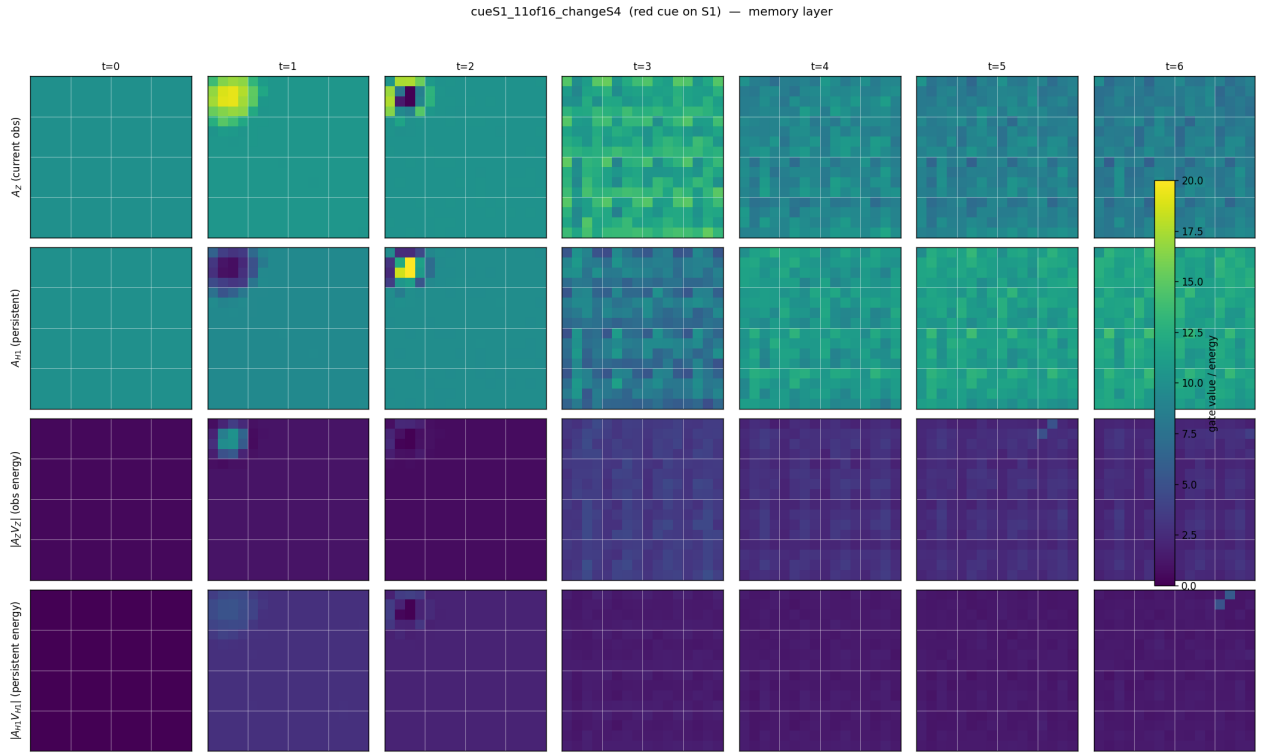


Figure 100: cueS1_11of16_changeS4 — mem layer.



Figure 101: cueS1_11of16_changeS4 — acc layer.

3.34 cueS1_12of16_nochange

Cue fixed on cell S1 (red), ring completeness 12/16 ($p(\text{change at cued cell} \mid \text{change}) = 12/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.72\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.04\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.

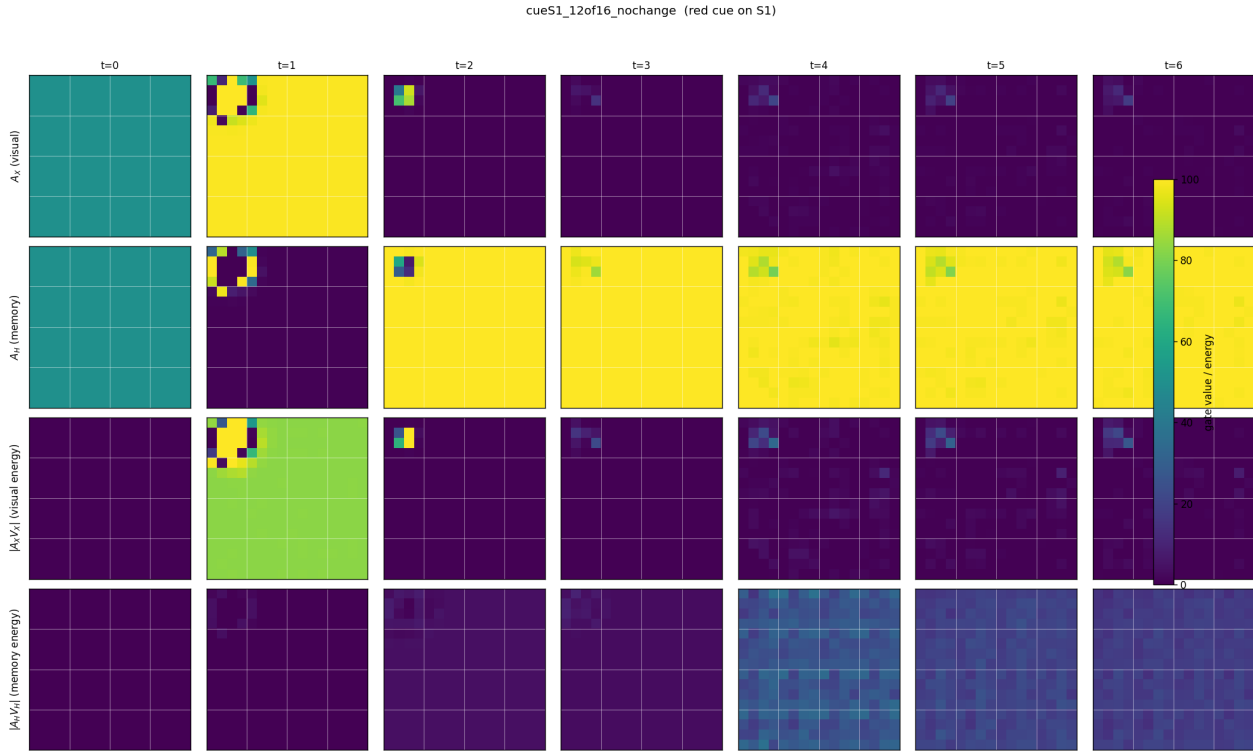


Figure 102: cueS1_12of16_nochange — vis layer.

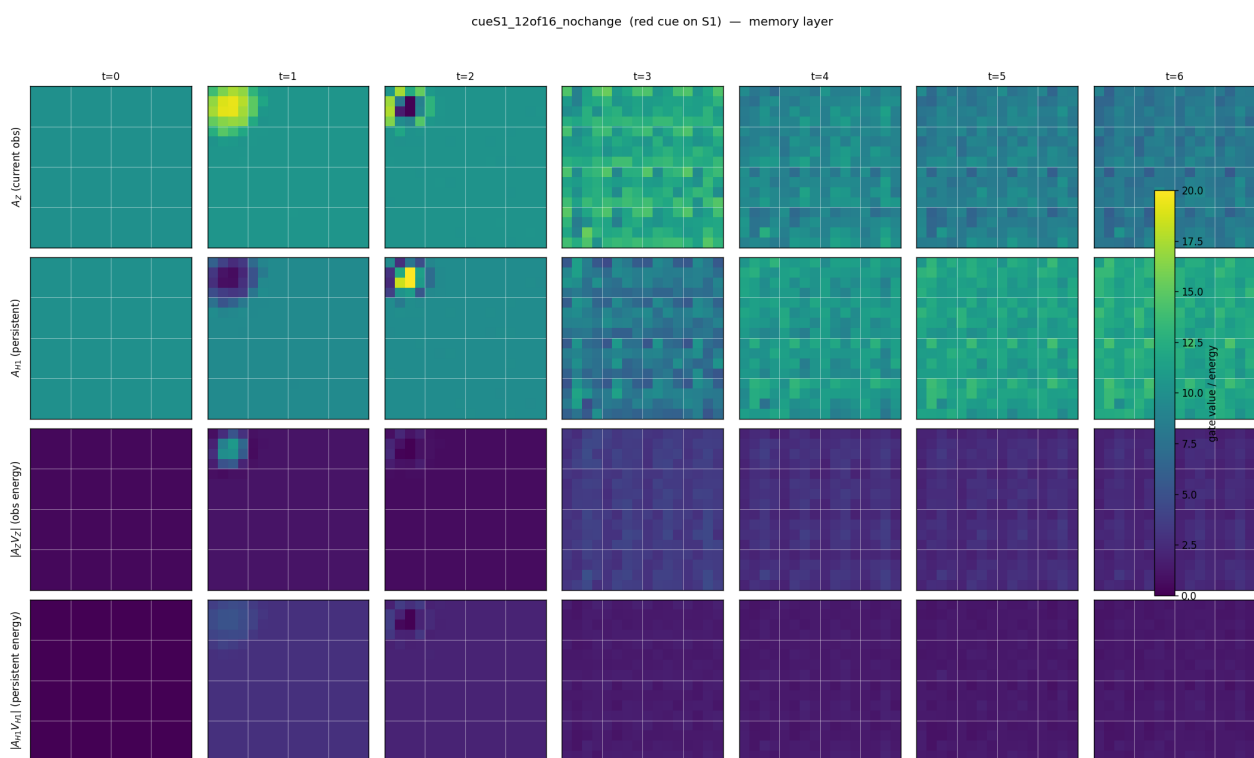


Figure 103: cueS1_12of16_nochange — mem layer.



Figure 104: cueS1_12of16_nochange — acc layer.

3.35 cueS1_12of16_changeS1

Cue fixed on cell S1 (red), ring completeness 12/16 ($p(\text{change at cued cell} \mid \text{change}) = 12/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.72\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.22\times$ the global mean ($t = 6$: $1.19\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.45 vs A_{H1} (persistence) 0.55. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 105: cueS1_12of16_changeS1 — vis layer.

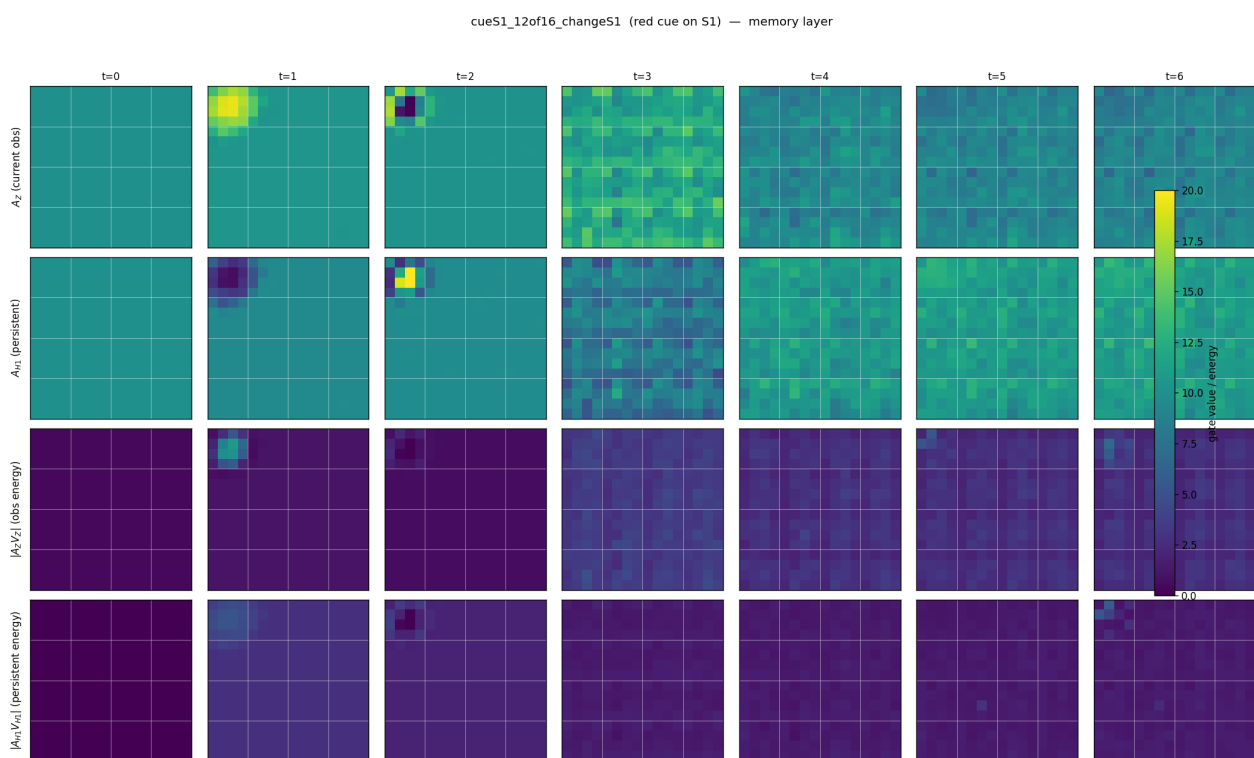


Figure 106: cueS1_12of16_changeS1 — mem layer.



Figure 107: cueS1_12of16_changeS1 — acc layer.

3.36 cueS1_12of16_changeS4

Cue fixed on cell S1 (red), ring completeness 12/16 ($p(\text{change at cued cell} \mid \text{change}) = 12/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.72\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.00\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

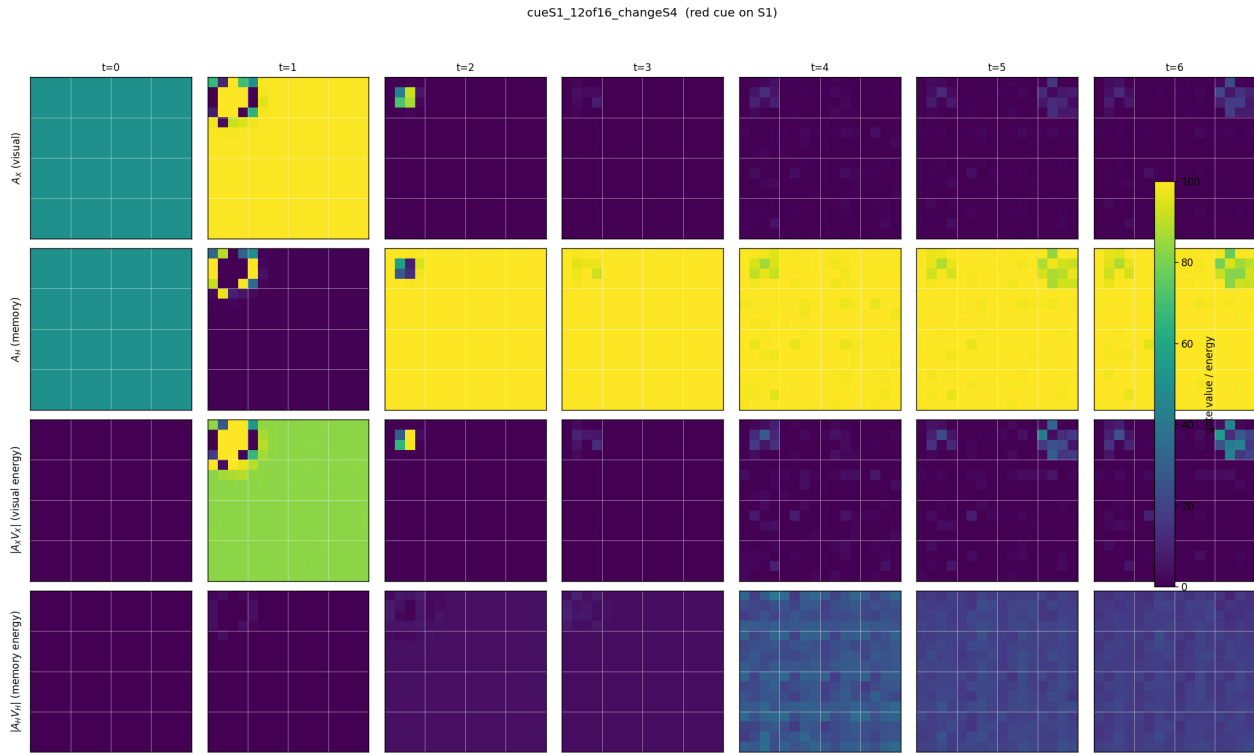


Figure 108: cueS1_12of16_changeS4 — vis layer.

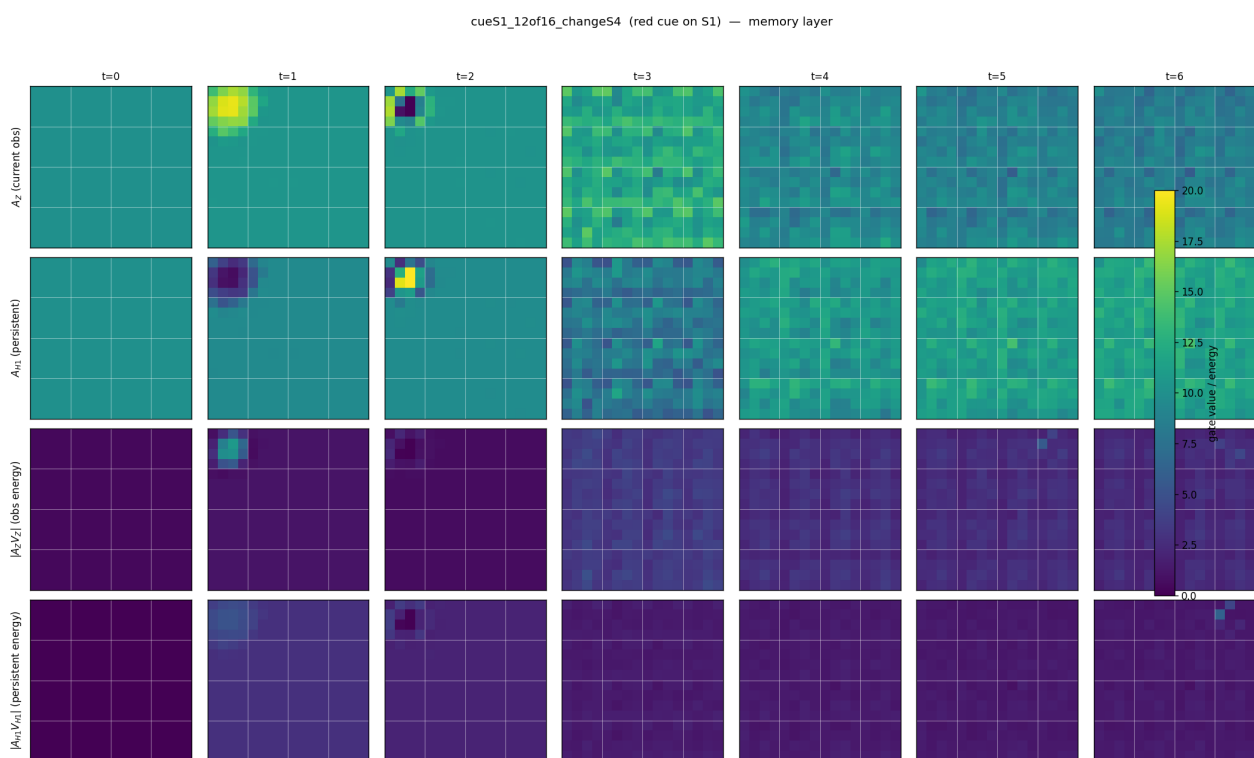


Figure 109: cueS1_12of16_changeS4 — mem layer.



Figure 110: cueS1_12of16_changeS4 — acc layer.

3.37 cueS1_13of16_nochange

Cue fixed on cell S1 (red), ring completeness 13/16 ($p(\text{change at cued cell} \mid \text{change}) = 13/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.74\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.00\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.

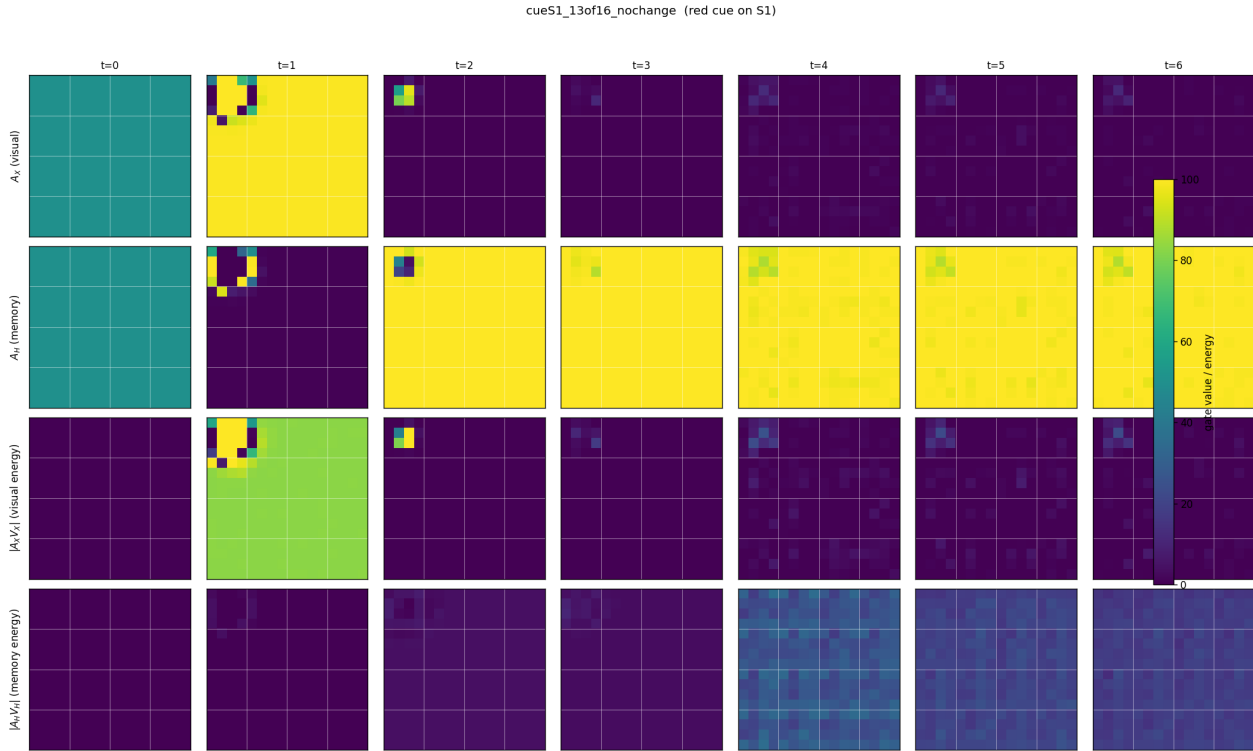


Figure 111: cueS1_13of16_nochange — vis layer.

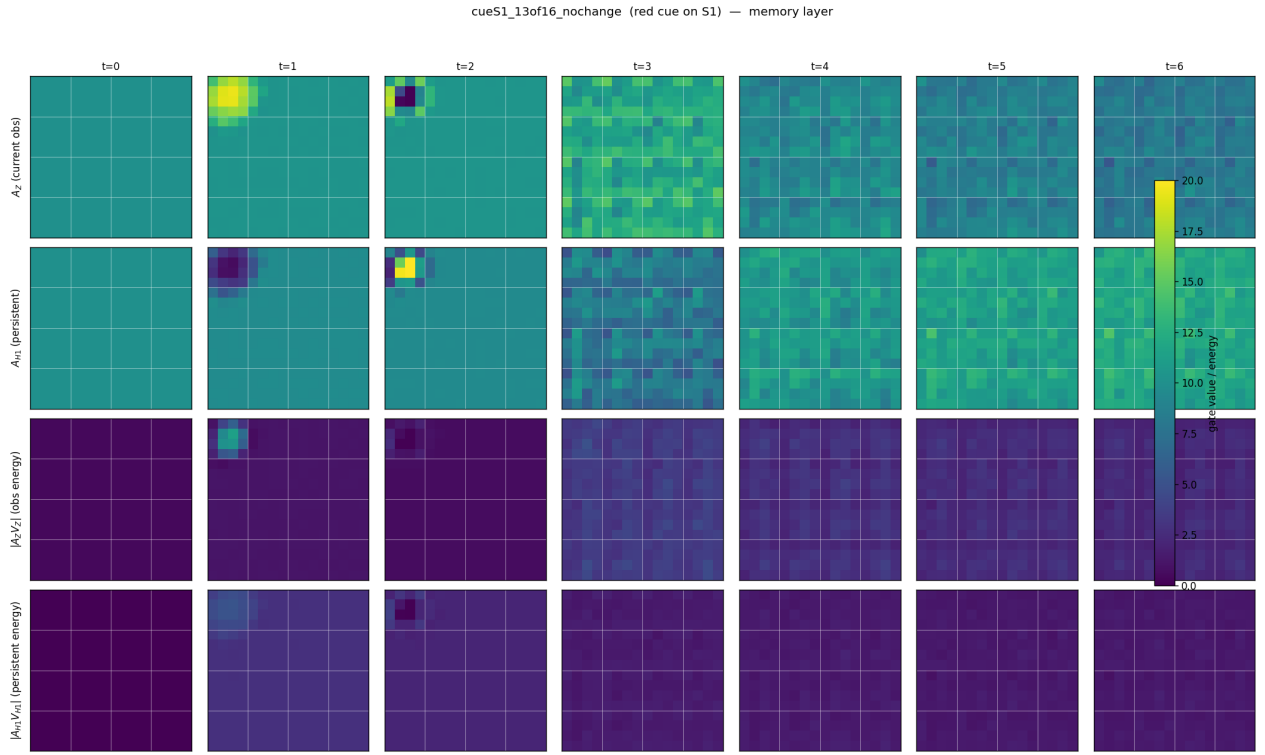


Figure 112: cueS1_13of16_nochange — mem layer.



Figure 113: cueS1_13of16_nochange — acc layer.

3.38 cueS1_13of16_changeS1

Cue fixed on cell S1 (red), ring completeness 13/16 ($p(\text{change at cued cell} \mid \text{change}) = 13/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.74\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.25\times$ the global mean ($t = 6$: $1.25\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.46 vs A_{H1} (persistence) 0.54. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.28; spatially flat — localization lives in the error term, not the gates.



Figure 114: cueS1_13of16_changeS1 — vis layer.

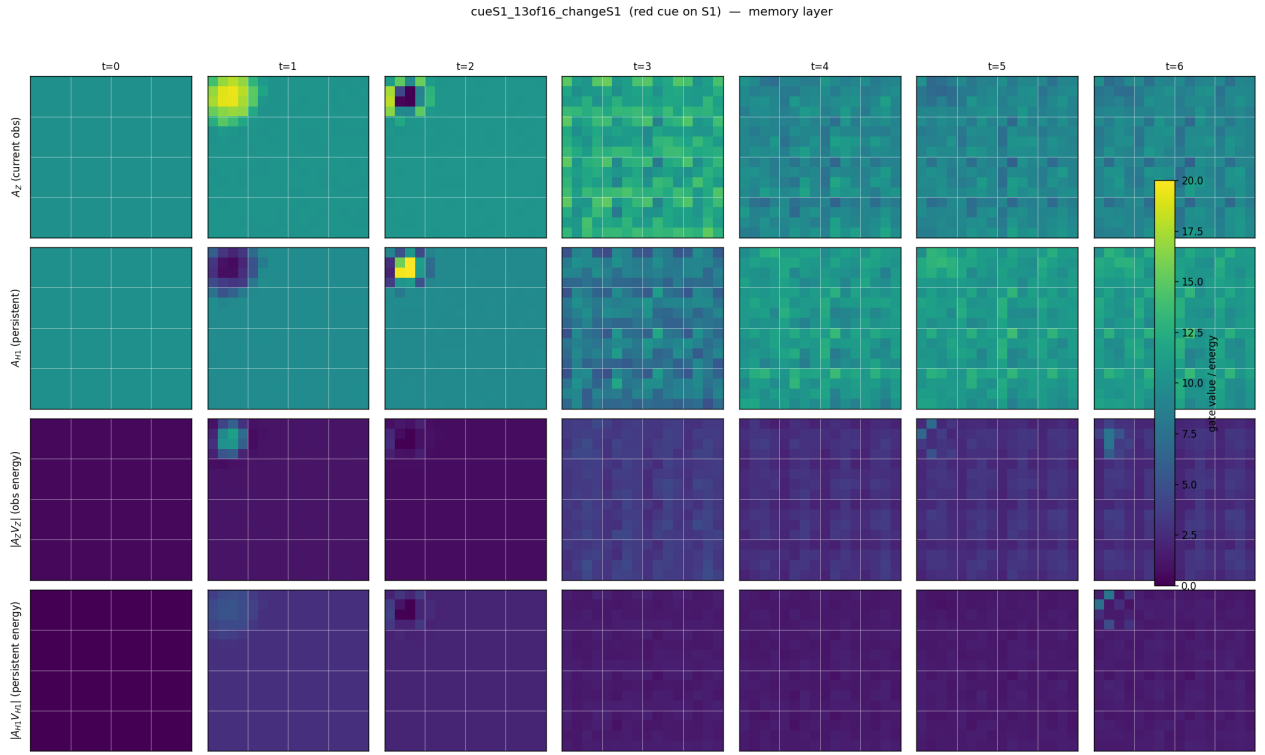


Figure 115: cueS1_13of16_changeS1 — mem layer.

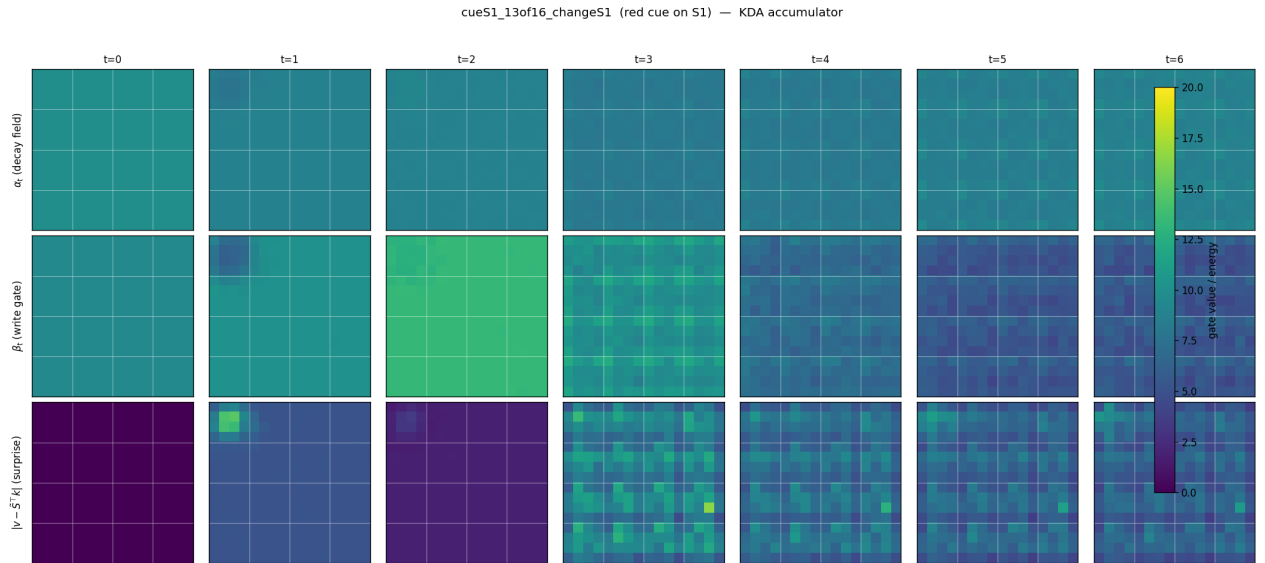


Figure 116: cueS1_13of16_changeS1 — acc layer.

3.39 cueS1_13of16_changeS4

Cue fixed on cell S1 (red), ring completeness 13/16 ($p(\text{change at cued cell} \mid \text{change}) = 13/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.74\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $0.99\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

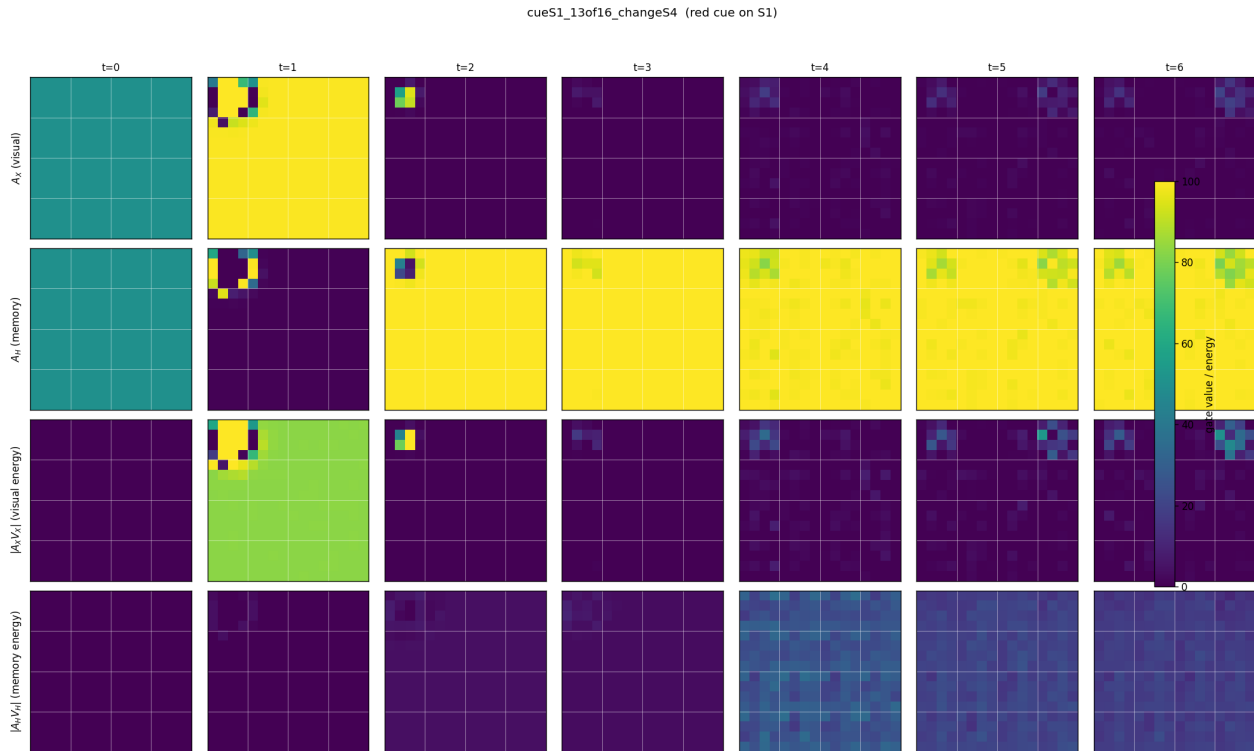


Figure 117: cueS1_13of16_changeS4 — vis layer.



Figure 118: cueS1_13of16_changeS4 — mem layer.

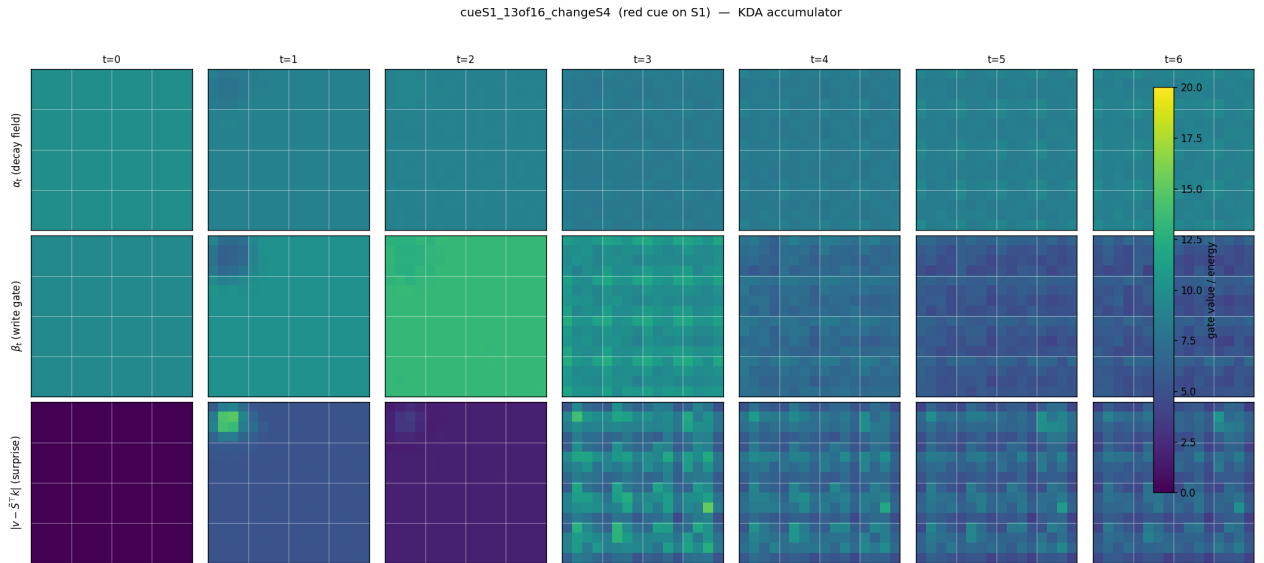


Figure 119: cueS1_13of16_changeS4 — acc layer.

3.40 cueS1_14of16_nochange

Cue fixed on cell S1 (red), ring completeness 14/16 ($p(\text{change at cued cell} \mid \text{change}) = 14/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.77\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.00\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.26 ; spatially flat — localization lives in the error term, not the gates.

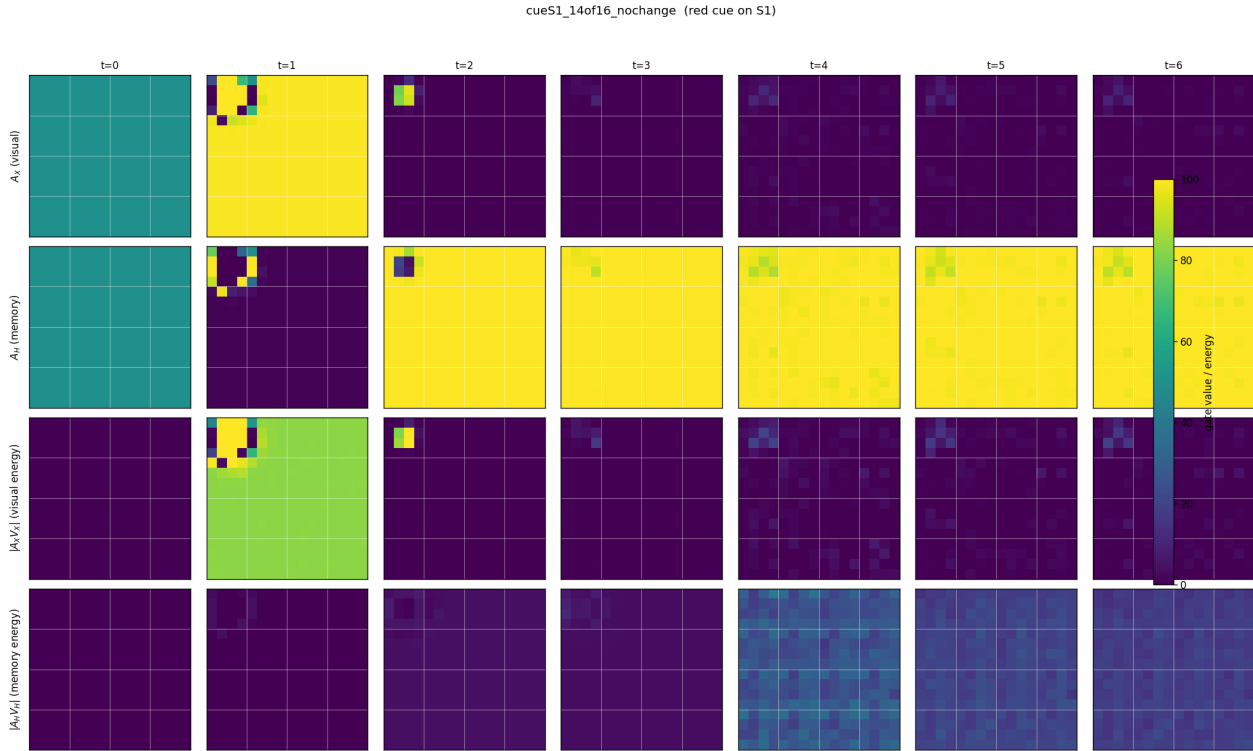


Figure 120: cueS1_14of16_nochange — vis layer.



Figure 121: cueS1_14of16_nochange — mem layer.



Figure 122: cueS1_14of16_nochange — acc layer.

3.41 cueS1_14of16_changeS1

Cue fixed on cell S1 (red), ring completeness 14/16 ($p(\text{change at cued cell} \mid \text{change}) = 14/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.77\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.25\times$ the global mean ($t = 6$: $1.27\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.45 vs A_{H1} (persistence) 0.55. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 123: cueS1_14of16_changeS1 — vis layer.

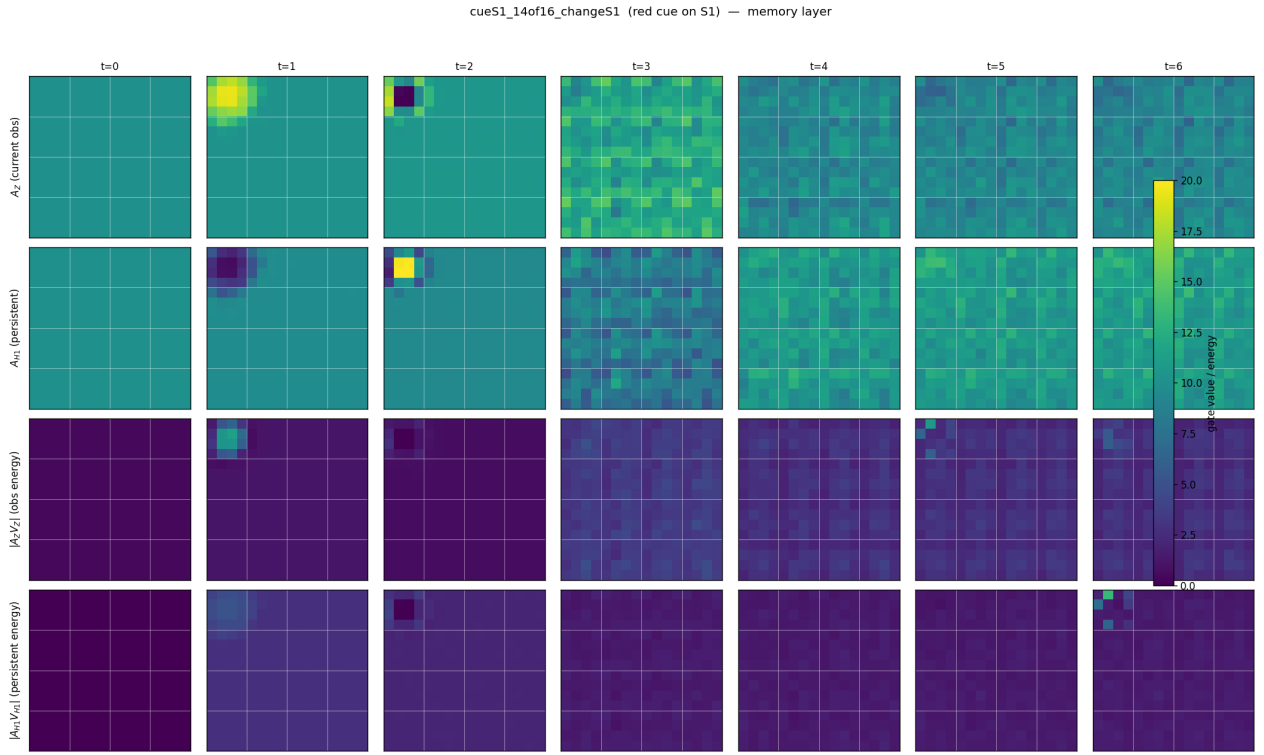


Figure 124: cueS1_14of16_changeS1 — mem layer.



Figure 125: cueS1_14of16_changeS1 — acc layer.

3.42 cueS1_14of16_changeS4

Cue fixed on cell S1 (red), ring completeness 14/16 ($p(\text{change at cued cell} \mid \text{change}) = 14/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.77\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $1.01\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean 0.01/0.99 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

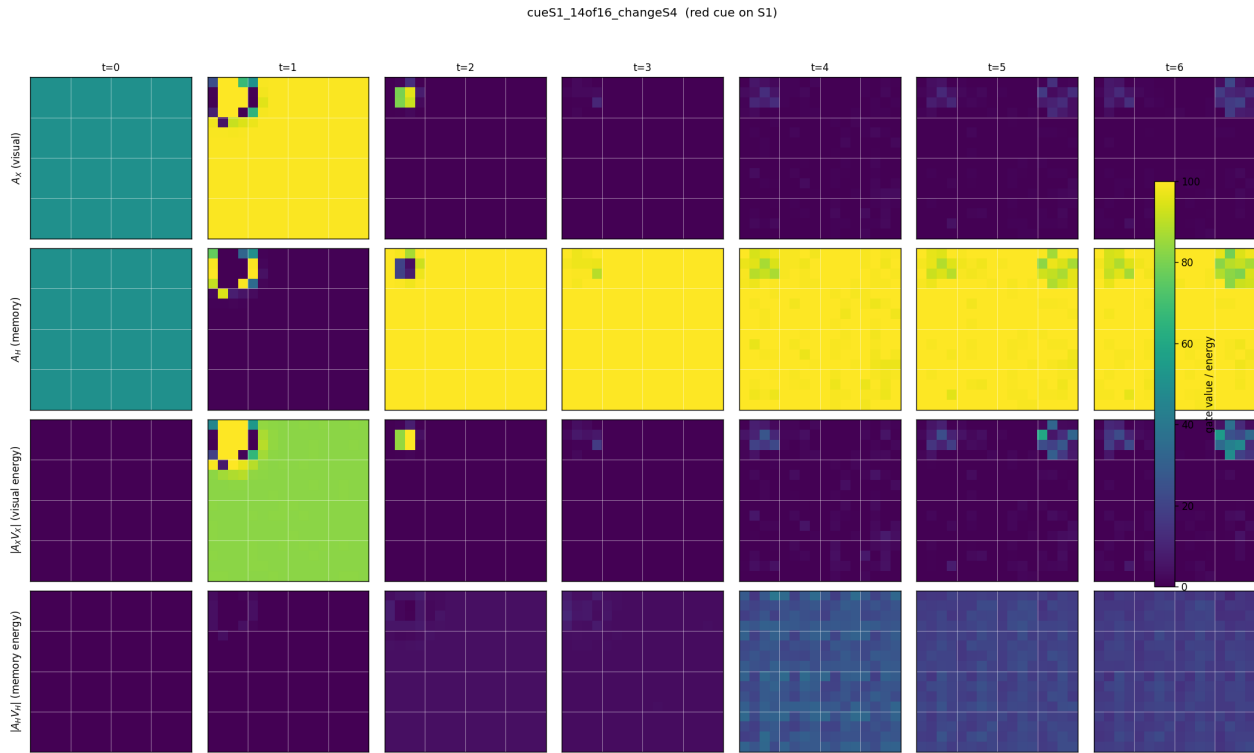


Figure 126: cueS1_14of16_changeS4 — vis layer.



Figure 127: cueS1_14of16_changeS4 — mem layer.

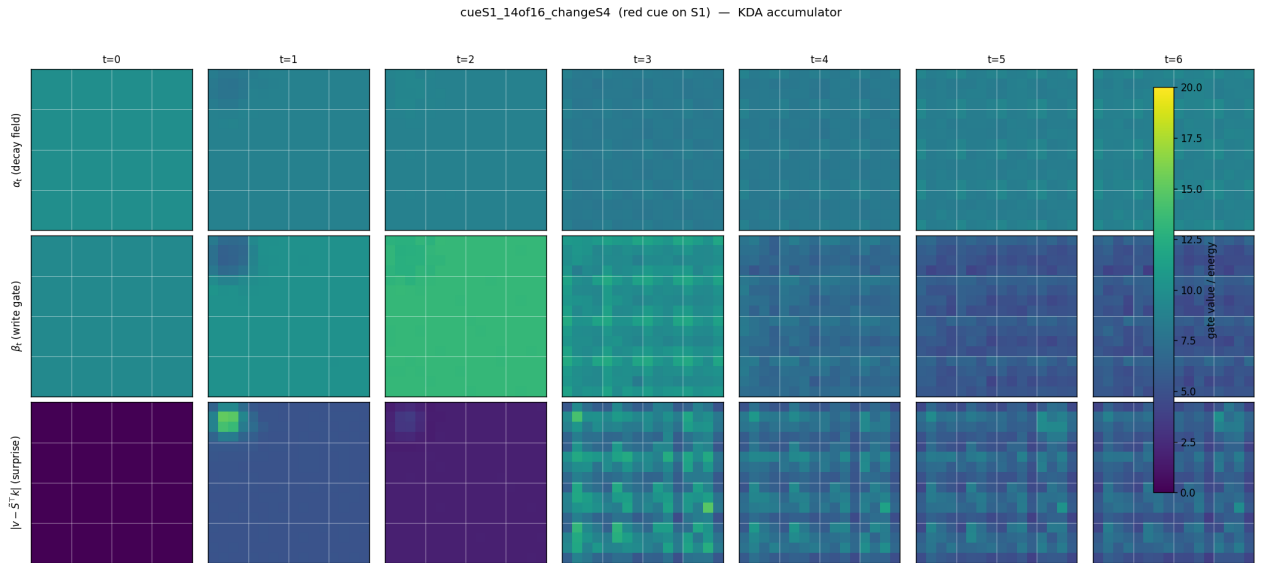


Figure 128: cueS1_14of16_changeS4 — acc layer.

3.43 cueS1_15of16_nochange

Cue fixed on cell S1 (red), ring completeness 15/16 ($p(\text{change at cued cell} \mid \text{change}) = 15/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.79\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.00\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.26 ; spatially flat — localization lives in the error term, not the gates.

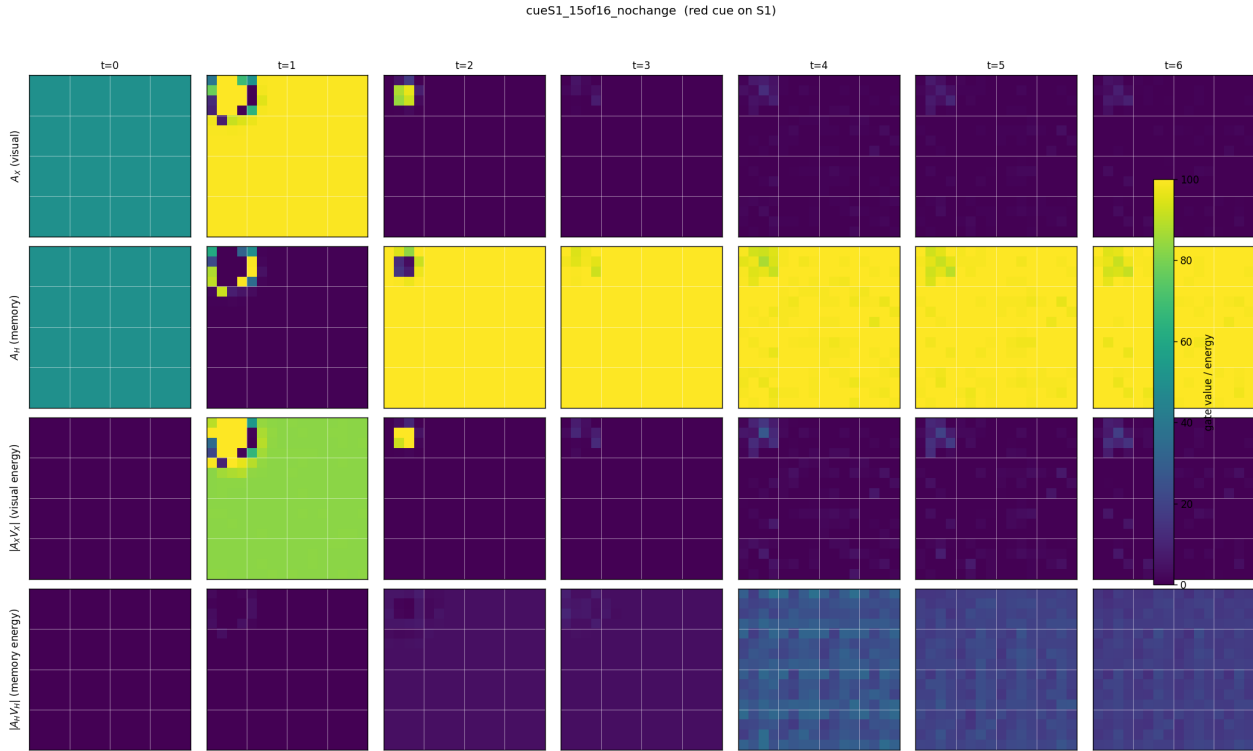


Figure 129: cueS1_15of16_nochange — vis layer.

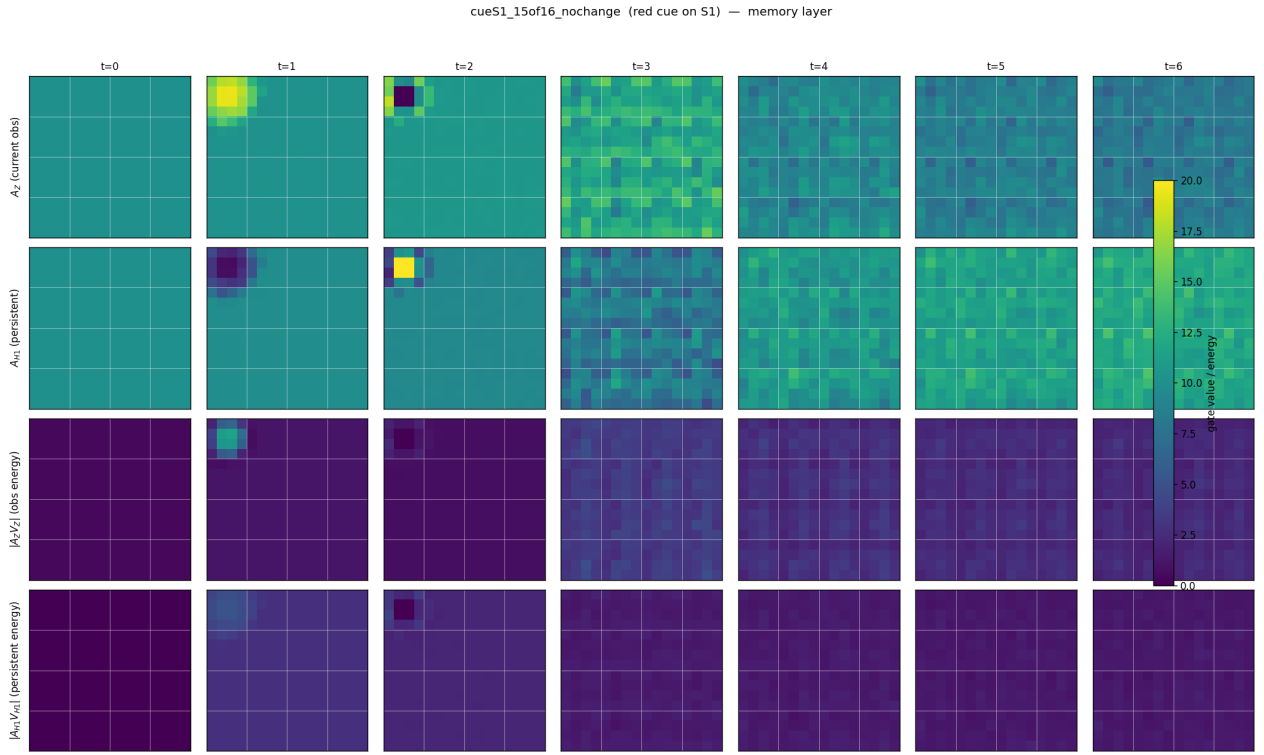


Figure 130: cueS1_15of16_nochange — mem layer.



Figure 131: cueS1_15of16_nochange — acc layer.

3.44 cueS1_15of16_changeS1

Cue fixed on cell S1 (red), ring completeness 15/16 ($p(\text{change at cued cell} \mid \text{change}) = 15/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.79\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.21\times$ the global mean ($t = 6$: $1.22\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.46 vs A_{H1} (persistence) 0.54. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.27; spatially flat — localization lives in the error term, not the gates.



Figure 132: cueS1_15of16_changeS1 — vis layer.

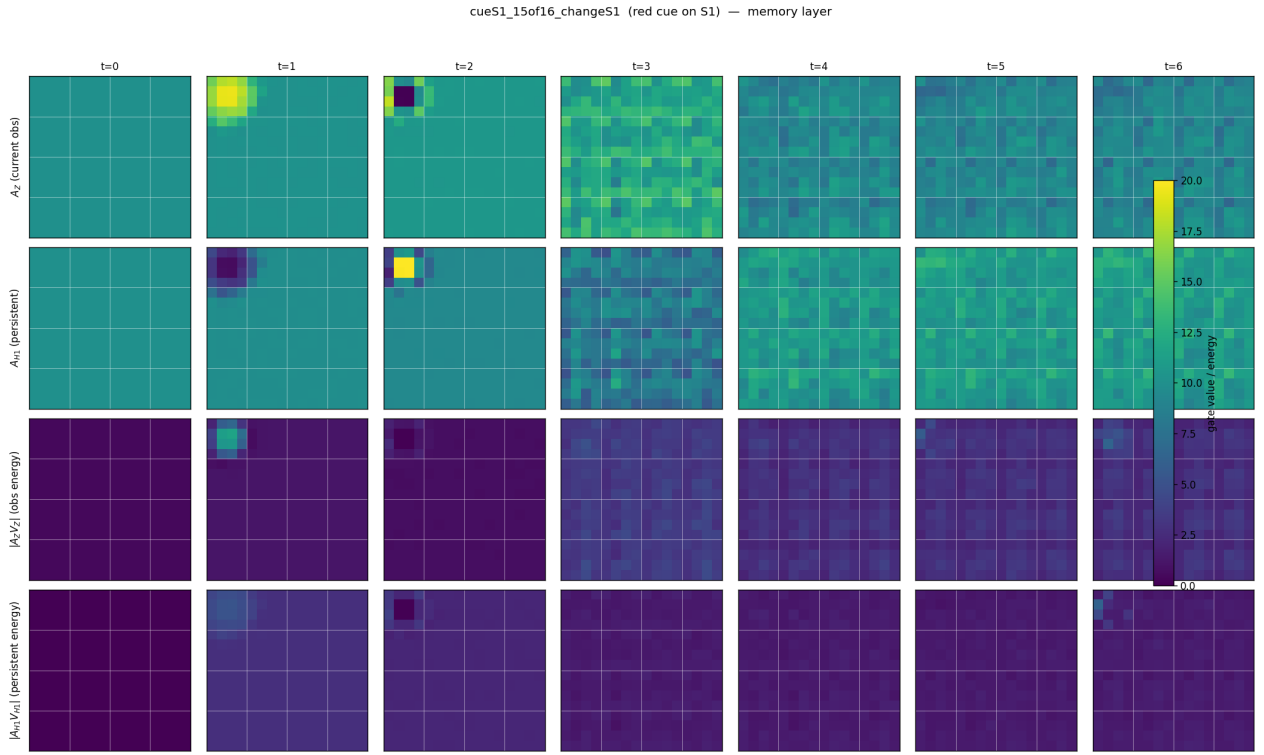


Figure 133: cueS1_15of16_changeS1 — mem layer.

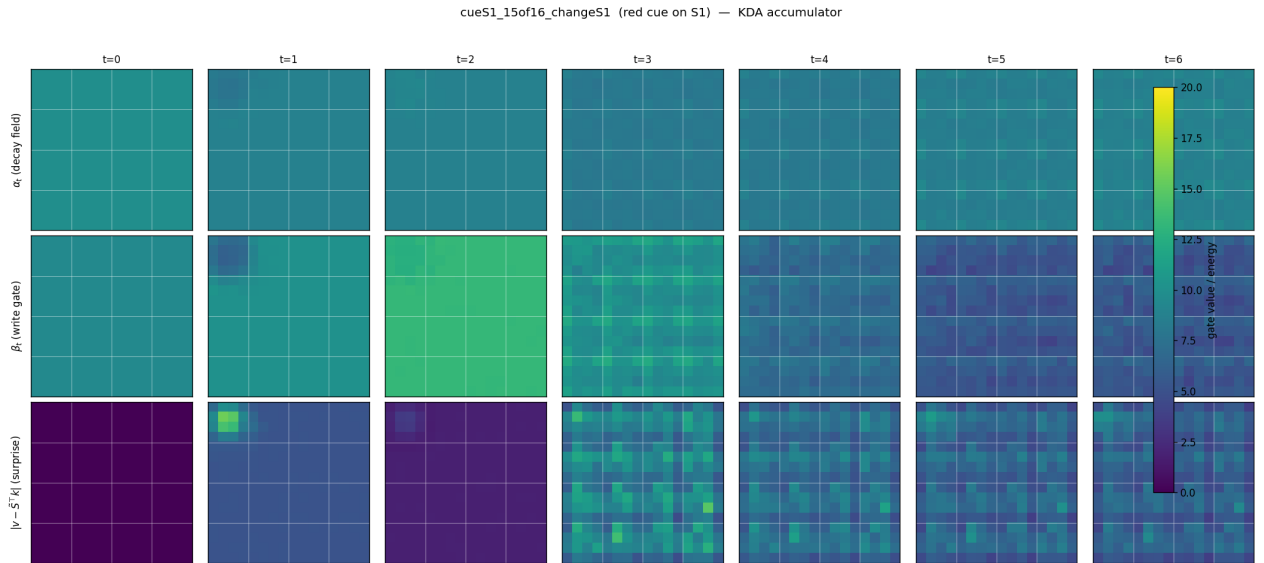


Figure 134: cueS1_15of16_changeS1 — acc layer.

3.45 cueS1_15of16_changeS4

Cue fixed on cell S1 (red), ring completeness 15/16 ($p(\text{change at cued cell} \mid \text{change}) = 15/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.79\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $0.99\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.42 vs A_{H1} (persistence) 0.58 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

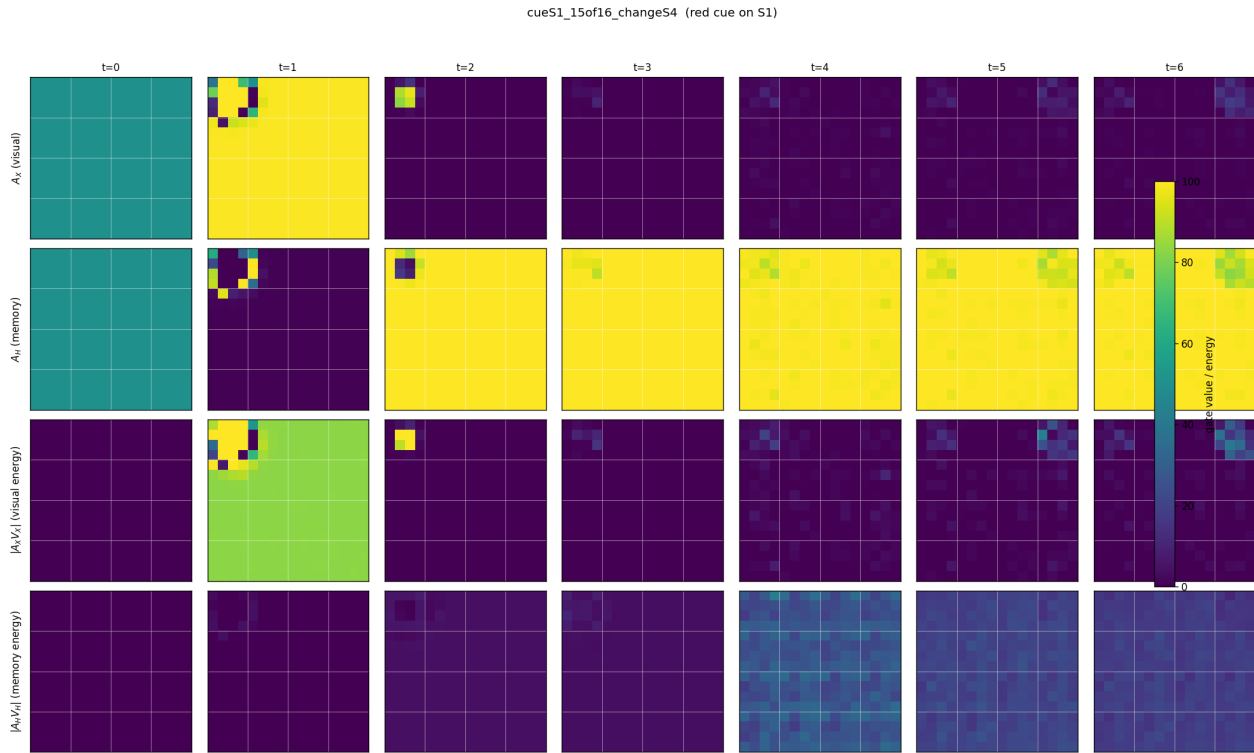


Figure 135: cueS1_15of16_changeS4 — vis layer.



Figure 136: cueS1_15of16_changeS4 — mem layer.

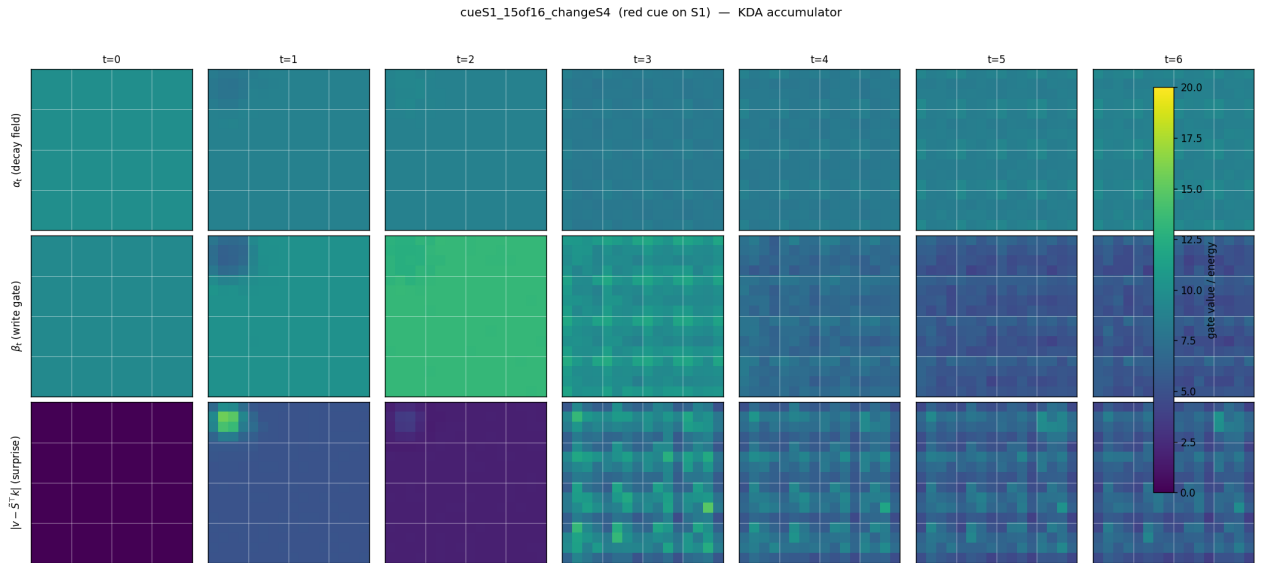


Figure 137: cueS1_15of16_changeS4 — acc layer.

3.46 cueS1_16of16_nochange

Cue fixed on cell S1 (red), ring completeness 16/16 ($p(\text{change at cued cell} \mid \text{change}) = 16/16$). No-change trials: no orientation change ever occurs. **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.81\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** With no physical change, surprise at $t = 5$ stays flat (S1 ratio $1.00\times$), confirming the accumulator is quiet absent a stimulus change. **Vision gates.** A_X/A_H at $t = 6$: mean $0.00/1.00$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.40 vs A_{H1} (persistence) 0.60 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially flat — localization lives in the error term, not the gates.

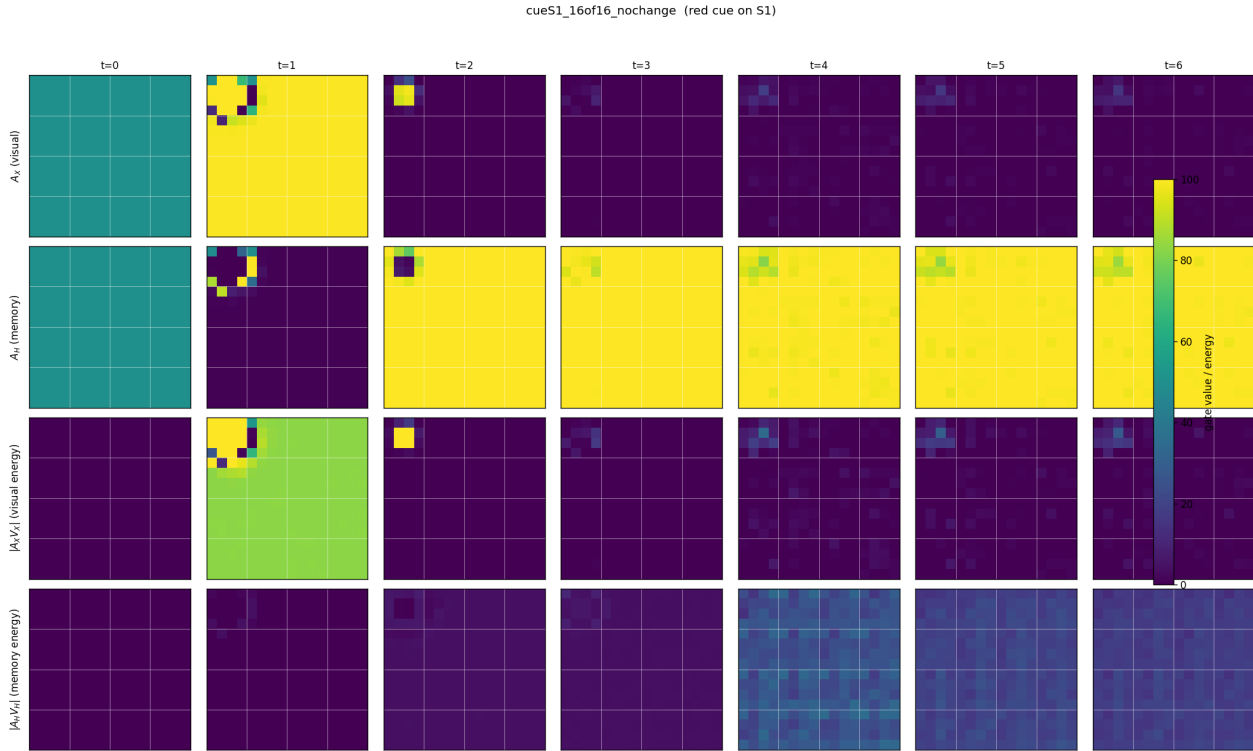


Figure 138: cueS1_16of16_nochange — vis layer.



Figure 139: cueS1_16of16_nochange — mem layer.



Figure 140: cueS1_16of16_nochange — acc layer.

3.47 cueS1_16of16_changeS1

Cue fixed on cell S1 (red), ring completeness 16/16 ($p(\text{change at cued cell} \mid \text{change}) = 16/16$). Change trials with the change pinned at S1 (the cued, valid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.81\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S1), surprise at the changed cell is $1.21\times$ the global mean ($t = 6$: $1.23\times$), while the other tracked cell sits at $\text{nan}\times$. The delta-rule error localizes the change. **Vision gates.** A_X/A_H at $t = 6$: mean 0.02/0.98 (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.49 vs A_{H1} (persistence) 0.51. **Accumulator gates.** α_t (decay) mean 0.44, β_t (write) mean 0.28; spatially flat — localization lives in the error term, not the gates.



Figure 141: cueS1_16of16_changeS1 — vis layer.

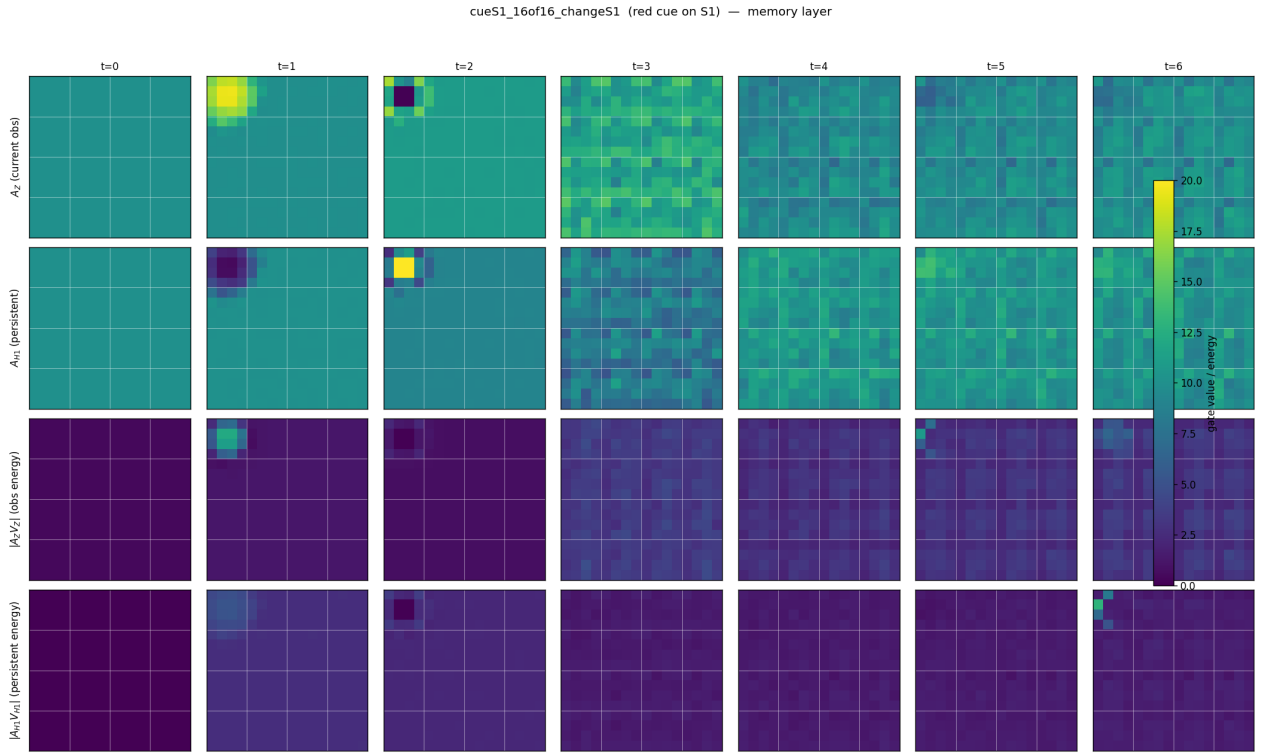


Figure 142: cueS1_16of16_changeS1 — mem layer.

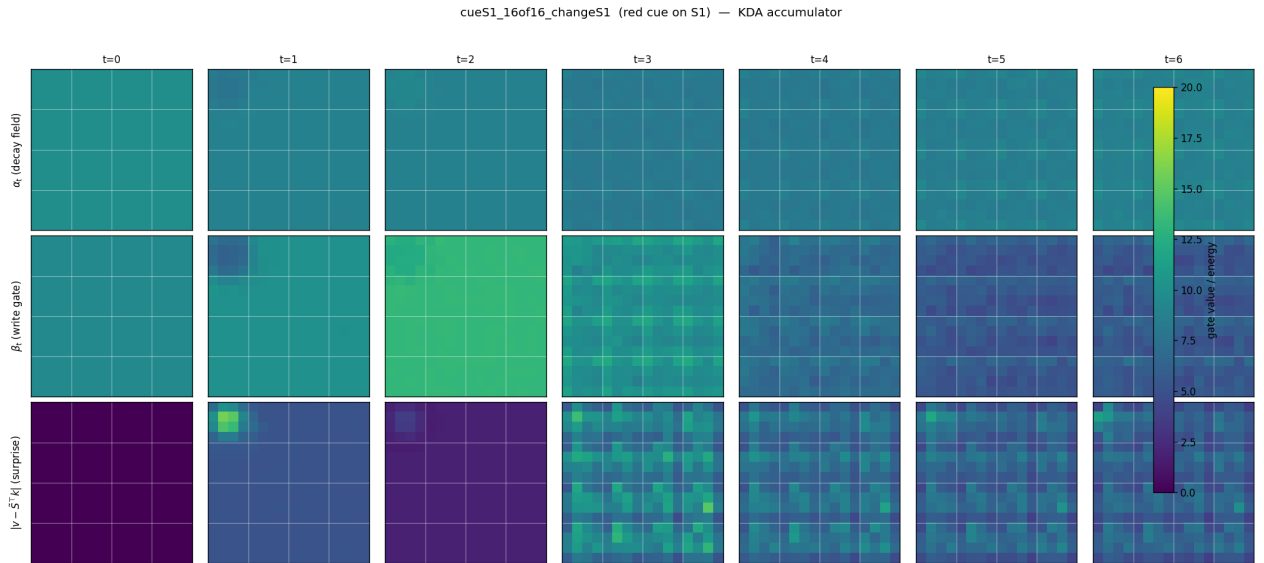


Figure 143: cueS1_16of16_changeS1 — acc layer.

3.48 cueS1_16of16_changeS4

Cue fixed on cell S1 (red), ring completeness 16/16 ($p(\text{change at cued cell} \mid \text{change}) = 16/16$). Change trials with the change pinned at S4 (an uncued, invalid location). **Cue transient.** At $t = 1$ the accumulator surprise at the cued cell is $1.81\times$ the map-wide mean — a strong localized prediction error to cue onset. **Change epoch.** At $t = 5$ (change at S4), surprise at the changed cell is $\text{nan}\times$ the global mean ($t = 6$: $\text{nan}\times$), while the other tracked cell sits at $0.97\times$. Localization is weak in this condition. **Vision gates.** A_X/A_H at $t = 6$: mean $0.01/0.99$ (memory-dominant routing). **Memory gates.** A_Z (current-observation write) at $t = 6$: 0.43 vs A_{H1} (persistence) 0.57 . **Accumulator gates.** α_t (decay) mean 0.44 , β_t (write) mean 0.27 ; spatially modulated ($\text{nan}\times$ at the tracked cell) — localization lives in the error term, not the gates.

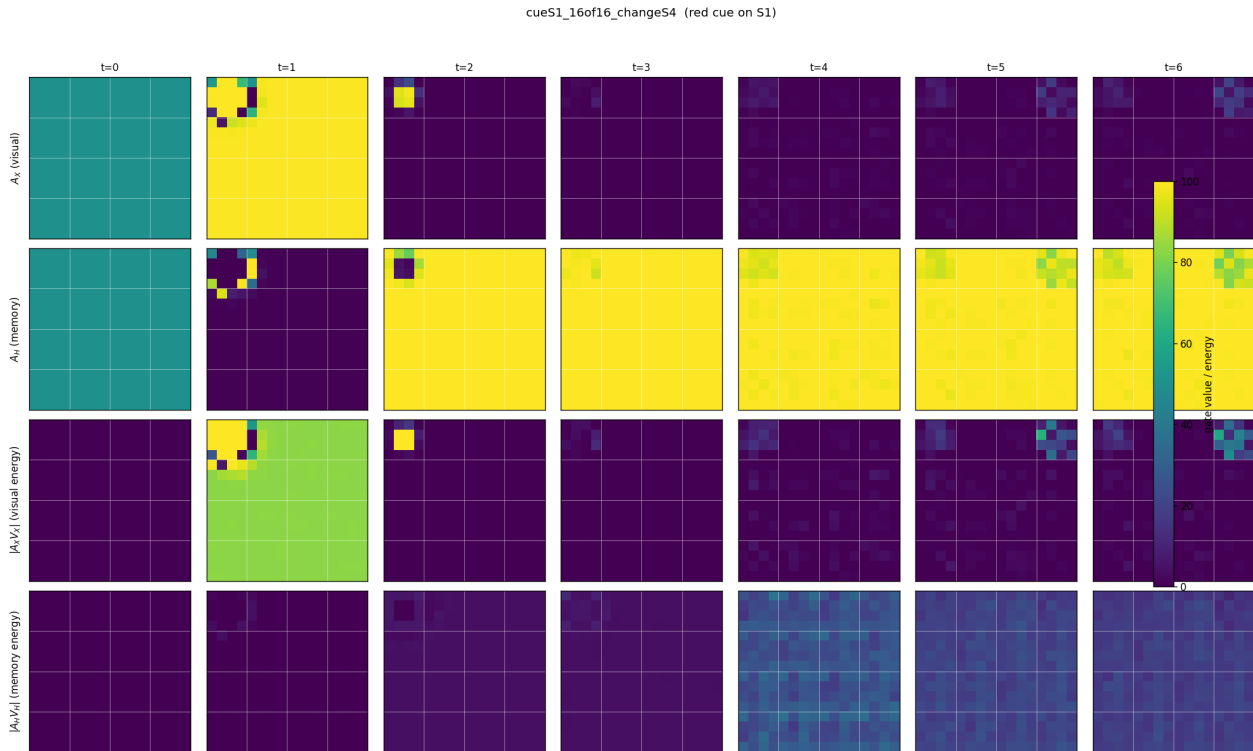


Figure 144: cueS1_16of16_changeS4 — vis layer.

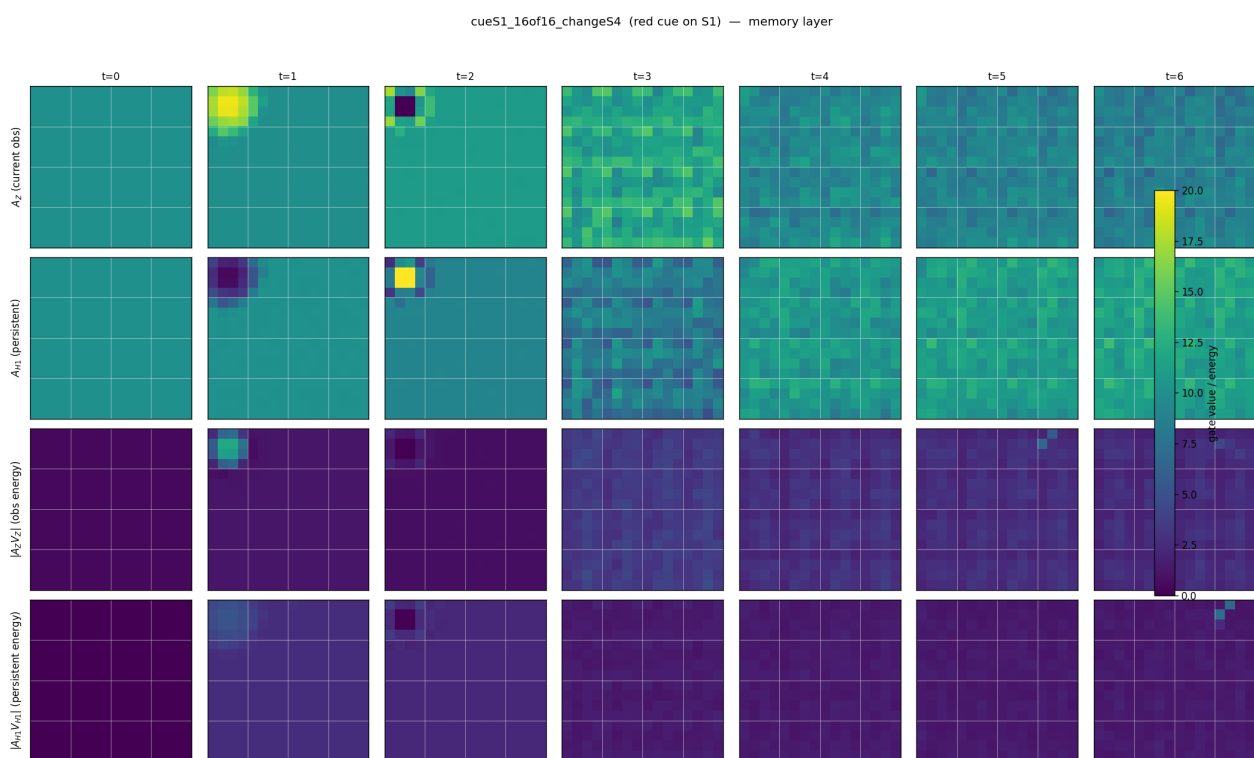


Figure 145: cueS1_16of16_changeS4 — mem layer.

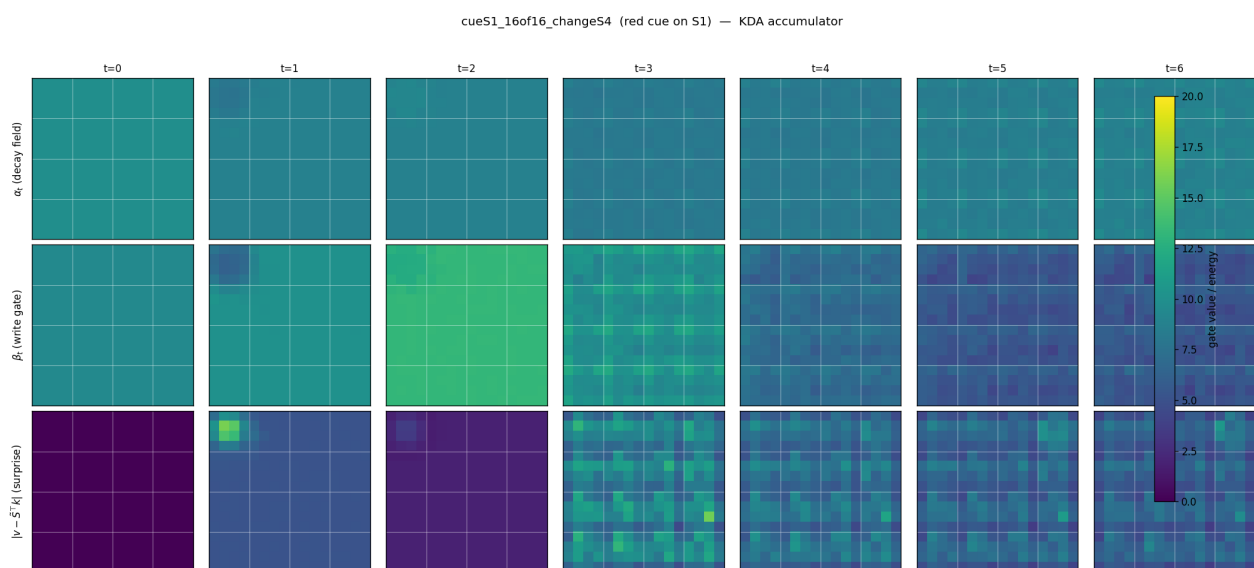


Figure 146: cueS1_16of16_changeS4 — acc layer.